

Confidential

Field Guide to Scientific Entrepreneurship (Draft Manuscript)

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Chapter 1: Introduction

Why a Field Guide?

A biological century led by start ups

It is trite, now, to say that we have entered the biological century, but it is true. The new technologies and measurement tools that let us read the underlying code of what makes us human and peer into the machinery of our cells provide endless new ways to turn those insights into medicines. Every day that we get deeper into the fundamental workings of biology we uncover the errors in our molecular instruction manual and come up with new ways to fix them.

We've found therapies for diseases that were once our biggest killers and rendered chronic what were once death sentences. We have developed drugs to cure hepatitis C, control HIV, and eradicate certain leukemias. We have unlocked the power of the immune system to treat cancer with checkpoint inhibitors and immune-activating vaccines. We have hyper-activated antibodies by adding new binding sites and toxic payloads. We have figured out how to safely harness viruses to bring corrective new genes into diseased cells and edit mistaken genomes. We have cracked the manufacturing of cells to allow for patient-specific cell therapies for everything from cancer to neurological disorders. We have sped up the timelines for discovering and making specific and protective vaccines for emerging pathogens.

And the pace of innovation only accelerates. We are learning to trigger neurons with laser light to induce new memories or erase old fears in mice. We are learning to connect the composition of the microbes in our gut to our response to food, our susceptibility to inflammation and allergy, and even to our mood and behavior. We can do amazing things and will continue to turn the once unimaginable into reality.

A new feature of this biological age is the rise of the start-up, which is how most of these innovations actually become products. The start-up has emerged as the most efficient way for smart scientists and visionary leaders to come up with and implement creative solutions. These aren't yesterday's start-ups that simply translated ideas from a lab and developed single assets. Instead, the new start-ups are ambitious and innovative. They dream up new technologies, make discoveries, and create entire new drug modalities. They raise large sums of capital to build cutting-edge platforms and develop multiple therapeutic products, with a goal of sustainable success and lasting innovation.

Success begets success, and as start-ups have successfully grown into large and profitable biotech companies and had above-market returns, the private dollars have poured in, fueling new companies and new innovations for years to come.

Accompanying the rise of the start-up is an entire new class of professional careers. The start-up scientist, for example, who excels in the uncertainty of building new technologies and loves wearing many hats in service of creating drugs. Or the

professional entrepreneur who starts company after company. Or the venture capitalist turned venture creator, who doesn't just invest in existing companies but seeks to build new ones from the ground up. Or the translationally focused academic professor who cares about the products that will come from the lab's research as much as the publications.

But innovation is a cottage industry and expertise is experiential

But despite the flourishing of start-ups and their associated careers, it is still very much a cottage industry. Whatever expertise has been built up, experientially, over the years is held close by the individuals and firms that have it; the know-how for how to do it well is not shared. Even those of us in the field who have the expertise, lack the language to describe what we're trying to do. We don't have an instruction manual. Our instincts are based on experience – have I seen this movie before and did I like it? – rather than a rigorous framework, which makes it really hard to teach the next generation of innovators how to do it.

The old playbooks don't really apply. The instructions for creating a single asset start-up – a checklist of activities, clear milestones, and known value inflection points – aren't relevant to today's new class of ambitious, innovative, multi-product platform start-up. The tech model of rapid A/B testing, quick failures, and lean methodology doesn't really work in biotech where costs are high, experiments take time, and you can't give people versions A and B of your drug to ask which they prefer.

How is the start-up scientist supposed to know how to do her job well? How is the new entrepreneur supposed to perceive the right paths if he doesn't have a half dozen of these experiences under his belt already? How will the venture capitalist who is trying to get "founder economics" learn that origination is not just earlier diligence on more tenuous science? How will the entrepreneurial professor figure out what questions to ask and experiments to run to launch the most impactful companies?

In place of a framework and a language, myths arise. Myths like that of the lone genius with a magic touch. (You *do* have to be smart, but it's not simple genius, it's not done alone, and it can certainly be taught.) Myths like it's possible to pick the winners. (The "winners" aren't predestined – it's up to you to lay the groundwork for massive value and then to execute to that vision.)

More problematic is the time lost to re-discovering the basic lessons of value creation, and the energy wasted in getting mired in the old muck of asset-centric drug development. And meanwhile, we all suffer because the innovations that will herald the biological century don't make it to us as quickly, as readily, or as thoroughly as they could.

A new framework and language for creating value in biotech

Into this void, we step. In this book we hope to provide a language for describing value in the new multi-product platform start-up. Here, we offer a framework for identifying, creating, and conveying value, and processes for building innovative and valuable

companies. We developed this framework over multiple years and multiple companies. We've practiced and honed these processes as scientists, entrepreneurs, investors, and executives. We've taught these methods to our junior colleagues, associates, employees, and teammates.

The framework and processes were developed at Flagship Pioneering, which was founded by Noubar Afeyan in 2000 as a hybrid venture capital/venture creation firm. Flagship and Noubar were the first to affirmatively choose to create new biotech companies through internal innovation rather than just invest in existing ideas. Within its walls, Flagship scientists, inventors and entrepreneurs systematically explore new areas of science, come up with novel and transformative platform hypotheses, and create companies to test and realize them. Flagship has earned a reputation in our industry as a deeply creative institution, willing to imagine, invent, and invest in technologies that sound completely unreasonable, and to launch companies to make these technologies reality. Once unprecedented but now common fields like mRNA medicines (Moderna), microbiome therapies (Seres), and genome editing (Editas), were pioneered by Flagship companies. Outlandish ideas have become accepted reality through the companies' successes.

At Flagship, we (Jordi, a protein engineer, and Avak, an RNA biochemist) have together created, launched, and led multiple valuable public and private companies in fields as disparate as DNA sequencing, RNA medicines, genome editing, enucleated cell therapies, neurobiology, gene therapy, and artificial intelligence in medicine. We have ideated new concepts for companies, designed and run experiments, written and filed intellectual property, hired scientists and built teams, led day-to-day operations from the most focused scientific lab meeting to the highest-level strategic planning, raised funds from private and public investors, brought first-in-category therapies to the clinic and sold first-in-category instruments.

The ideas we present here were forged by our experiences, but also deeply shaped by our partners at Flagship and our colleagues across the biotech and entrepreneurial ecosystem.

Because every company is different and every technology is different, because market conditions may change and individuals are unique, our goal isn't to simply describe our successes and failures or offer a step-by-step how-to guide. Instead, here we'll outline our framework and processes – the questions we ask, structures we apply, and methods we use – that lead us to instinct and to habit. This is how we see the world.

About the Field Guide

How it's structured

The crux of this book is that we must understand value in the life science industry – what it is, how we create it, how we realize it, and how we convey it. In particular, the book will explore how to wield your understanding of value along the entire life-cycle of an innovative multi-product platform company, from the earliest stages of ideation through product development, and from inward-facing team-building to outward-facing selling, so that you create the most successful, impactful, and valuable company possible.

In the coming chapters we will:

- Introduce the Multi-Product Platform company as the model of a biotechnology start-up best suited to maximize the power of innovation (Chapter 2)
- Define Value: what it means, how to describe it, and how to build it in a Multi-Product Platform company (Chapter 3)
- Describe how to come up with a valuable idea so that you know you're working on the right thing (Chapter 4)
- Explore how to create value when you enter the lab by generating results that maximize your value (Chapter 5)
- Discuss product development strategies unique to Multi-Product Platform companies so that you don't squander your potential by collapsing behind a single asset in the clinic (Chapter 6)
- Explain how to create value in intellectual property so that you can protect your innovations from competitors (Chapter 7)
- Explore how to sell your idea to ensure that you convey the value and get the support you need (Chapter 8)
- Examine how to maximize the value of people on your team by building a culture of innovation (Chapter 9)

At the end of each chapter we've highlighted key takeaways, so you can quickly reference them as needed as you ideate, innovate and build your company.

Who it's for

We began this book with a clear audience in mind — our junior colleagues at Flagship just beginning their entrepreneurial journeys for whom we didn't have a good way to convey all the collected wisdom of years of experiential learning. But as we've written the book and formalized the concepts within, and as we've exposed them to other entrepreneurs, investors, and academics in our ecosystem, we're learning that the audience might be quite a bit larger.

We expect this book will be useful for the entrepreneur hoping to build valuable multi-product platform life sciences companies. Thus, the narrative generally follows the path of creating and building a company. But we also expect that this framework may help the many venture capitalists who are trying to become venture creators and get involved

in companies at earlier stages. We think it will help guide start-up scientists who want to understand what's happening at their companies, and who are interested in jumping from the bench to strategy, business, or investing – since really, it's all just solving the same optimization equation for value. We think this book will also be useful to our academic friends, including the professors looking to do impactful work and the students considering spinning out their projects, to navigate this very different world.

There is no one right way to innovate in the life sciences. There are many paths to building successful biotech companies. We hope our perspective is helpful to you and a useful contribution to the professionalizing of our field. We hope this begins a conversation and we look forward to hearing from you.

Onward!

Chapter 2: Setting the Stage

This is a book about starting innovative companies. A book about identifying, creating, and conveying value. A book about navigating the frontiers of science and building products that transform human health. Before we set off on this journey, though, we need to cover some basics.

Life science start-ups exist in a rich ecosystem of Pharma, biotech, universities, venture capital financiers, entrepreneurs, and scientists – all working together to discover, invent, and create new medicines. Start-ups germinate in this ecosystem, grow in it, and permanently shape it.

To navigate your start-up to success, you need to understand the steps of drug development, the players in the ecosystem, and the ways they interact.

In this chapter, we'll start with a high-level review of the process of turning a molecule into a medicine, and we'll learn how the key players in the biotech ecosystem work together to make this happen. We will explore how start-ups fit into this puzzle, how they raise money to fund their growth, and how, in an inherently uncertain business, they mitigate the risks of drug development. We will introduce an important modern class of start-up – the Multi-Product Platform Company – that we've found to be the best model for creating maximum value from sustainable innovation.

Ch2.1 Making a Medicine 101

A molecule's journey from discovery to drug happens in three stages: Discovery, Preclinical Development, and Clinical Development. This high-level review will touch on the key scientific and regulatory milestones of each stage and discuss where the funding comes from, since the costs tend to increase by an order of magnitude with each stage.

During Discovery, we pinpoint how to intervene in a disease. We identify and validate targets that we think matter in pathophysiology. We then identify the molecule itself – through screening or by design – and show that it acts on the target in the way we want.

No regulatory check-ins are required during Discovery, though if you're inventing a new modality or tackling a disease in an unusual way, it can be useful to get an early opinion from the FDA about your clinical development plans.

Discovery was once the purview of Pharma, but today's inventions increasingly originate in enterprising academic labs and start-ups. Discovery typically costs anywhere from hundreds of thousands to a few million dollars, with funding from venture capital (seed or Series A), private angel investment, or government grants.

In Preclinical Development, we refine the molecule for potency, efficacy, and toxicity and build the case for evaluating the molecule in patients. For medicines that work in new ways, we develop biomarkers that tell us whether the molecule acts in the body in the way we hypothesize, which is invaluable information for selecting the right patients and proving the mechanism of action in clinical testing. During Preclinical Development, we also establish how to manufacture the molecule, purify it, and store it stably.

The capstone of Preclinical Development is the filing and approval of an Investigational New Drug (IND) application with the FDA. The IND describes the molecule's animal pharmacology and toxicology data, the assessment of Chemistry, Manufacturing, and Controls (CMC), and the clinical plan for initial human studies.

The costs of Preclinical Development are typically \$5-10M, driven largely by the costs of refining the molecule, conducting animal testing, and manufacturing the medicine. Funding for start-ups at this stage typically comes from venture capital (Series A, B, or beyond) or partnerships with Pharma.

During Clinical Development, we assess the molecule in patients, first for safety and then for efficacy. In early trials, we adjust the dose to find the optimal therapeutic quantities; in later trials, we compare the molecule head-to-head with placebo and existing treatments, if they exist.

Success in Clinical Development is defined as the submission and approval of a New Drug Application (NDA for a small molecule drug) or Biologics License Application (BLA

for a biologic therapy), which gives you license to market the drug and defines the conditions you can treat and the doses you can provide.

Clinical Development can cost \$50-100M, depending on the nature of the disease, the number of patients you need to test, and the time it takes to see a benefit. Funding comes from venture capital and other large private investors, Pharma partnerships, or public markets (if the company has gone public).

Many start-ups take molecules into the clinic, but by the time they reach late-stage clinical development, they no longer look much like start-ups. The regulatory and financial requirements often force the team to expand to dozens if not hundreds of employees, and the start-up joins the newest crop of mid-sized biotechs.

Ch2.2 The Changing Role of Start-ups in Making Medicines

Most of us are familiar with the start-up's traditional role in the biotech ecosystem. Start-ups translate academic discoveries into the commercial realm. Start-ups shepherd new targets, molecules, and hypotheses until they look enough like products that they can be transferred to Pharma.

The explosion of innovation and funding and the cultural lionizing of entrepreneurship, however, have reshaped the ecosystem. Today's start-ups play a new and more meaningful role.

Start-ups as engines of innovation

In the past decade, Pharma has retreated from innovation. New molecules used to come from the legions of Pharma scientists at sprawling university-like campuses, but no more. Over the past five years, nearly 60% of all new drugs approved by the FDA were initially developed by small biotech companies and start-ups, and that share appears to be increasing.ⁱ Pharma has handed innovation off to biotech and start-ups, relinquishing the risk of discovering new targets and creating new modalities and instead concentrating on clinical development, manufacturing, and sales.

Start-ups have stepped up to this opportunity by becoming engines of innovation. No longer do they simply translate others' discoveries – instead, they increasingly make the discoveries themselves. No longer do they limit themselves to single assets; instead, they increasingly develop new modalities and technologies that are broadly applicable. If you're looking for the cutting edge in biotech today, you'll likely find it at a start-up.

With massive funding comes massive ambition

Biotech has enjoyed an explosion of money. The prolonged economic growth since 2008 that sent investors searching for higher returns created a biotech boomlet in the mid- to late 2010s. Start-ups can now access a glut of private capital, select from multiple investors, and even go public with no more than preclinical data. In recent years, the number of start-ups funded per year (>250), the money raised per round (\$25M), and the total VC investment in the sector (\$16B/yr) have soared.ⁱⁱ

As a consequence of living in a world of surplus capital, start-ups have become more ambitious. Entrepreneurs can chase opportunities that are bigger and more transformative than just the next drug or the next target hypothesis. With sufficient resources, start-ups can be first in a class of medicines, first in a disease area, or first with a new discovery platform. We've seen recent examples of this ambition in the multiple start-ups built around CRISPR as a therapeutic (Editas, Intellia, Crispr Therapeutics) and as a discovery tool (KSQ, Tango); around gene therapy (Avexis, Spark, and others); around CAR-T and other cell therapies (Kite, Juno, and many more); and around the many flavors of RNA (Moderna, Alnylam, and beyond).

We can't predict whether the macroeconomic trends will hold into the new decade and beyond, but we've learned that start-ups can be very successful when they tackle big problems. This experience has changed the scale of many start-ups' ambition and has shown the innovative power that small groups of motivated, incentivized scientists can harness.

Changing dynamics between start-ups and Pharma

The interactions of start-ups and Pharma used to be one-dimensional and one-way: Pharma provided the exit for molecules developed by start-ups. While these interactions still occur, the flavors and types of interactions have expanded along with the scope of the problems that start-ups now tackle. We now see much more equal and collaborative relationships between Pharma and start-ups. Start-ups that pioneer new modalities or new discovery technologies now have substantial leverage to tap Pharma resources, including extensive molecule libraries, patient samples, and preclinical expertise.

On the buy side, start-ups have always sought to license molecules from Pharma to seed their development work. But increasingly, thanks to their expanded coffers, start-ups are grabbing discarded Pharma assets and jumping directly into clinical development of multiple therapeutic programs. Companies like Tessaro and Myovant have shown that start-ups can leverage smart licensing and aggressive clinical development to create full pipelines with multiple approvals within just 3-5 years of company founding.

On the sell side, start-ups continue to license later-stage molecules back to Pharma or to split the cost of clinical development in exchange for a share of royalties. But Pharma's increasing hunger for innovation means that we're seeing these deals happen with earlier-stage molecules, with new modalities where the start-up gives the Pharma options to multiple future products in a therapeutic area, and with creative corporate structuring that makes it possible for the start-up to continue to develop multiple new products instead of concentrating solely on clinical development of the lead asset.

New opportunities for success

Start-ups eventually stop being start-ups. The three traditional exits still exist. You can succeed in clinical development and be acquired by a Pharma; you can fail trying and close up shop, auctioning off assets and letting your employees seek their fortunes at new companies; or you can persist, neither striking it big nor flaming out, long enough to stop counting as a "start-up."

But with today's newly ambitious start-ups and changing market dynamics, it is no longer crazy to dream big. Start-ups can go public earlier in their lifecycles, as demonstrated by multiple examples of successful preclinical IPOs, and tap into the resources needed for long-term success. It is now viable for start-ups to aim, from the get-go, to become fully integrated biotech companies with multiple approved products. It is now reasonable to try from day one to become an enduring, sustainable, and

profitable company that creates lasting innovations and brings new ways of thinking into existence.

The explosion of start-ups has also created new opportunities for individual success. Creative scientists – those who crave the thrill of discovery, who hunger for flat organizations that allow leaps in responsibility, who live to make consequential choices and aspire to invent new drugs – now find their skills in high demand in early-stage companies. They can jump from one start-up to the next, applying their insights and judgment to new technologies and the latest unsolved problems. Entirely new industries have cropped up around company origination and incubation to professionalize the cottage industry of entrepreneurship. Venture firms can invest their dollars earlier and own more of the companies in return. Serial innovators can now surround themselves with dedicated investment capital and in-house logistical and operational support to speed and simplify the building of the next generation of transformative start-ups.

Ch2.3 A Quick Primer on Money

Developing a drug costs money – a lot of money. Reliable estimates pin the cost of developing a single drug at over one billion dollars. Early science experiments run you tens of thousands a pop, and the numbers increase tenfold at each stage of development. All of this takes place before the drug is approved, before there is any revenue to offset expenditures.

You can't do biotech on the cheap. Start-ups need to continually raise money as they progress to ever more expensive stages of development. So where does that funding come from, and what strings are attached?

Rounds and runway

Early start-ups hemorrhage cash. The majority of this money comes from private investors – venture capitalists and individual angel investors. Companies raise money in discrete rounds. The first bolus is called the seed round, and subsequent raises are lettered as Series A, B, C, and on.

The money you raise affects your runway, the amount of time before you run out. It's an important metric but an imperfect one, because runway is a function of both your cash reserves and your burn rate. You can certainly blow your money too soon, reaching the end of your runway without having created significant value, and be forced to raise new money at a reduced valuation. But extending the runway by spending too little and inching along creates its own risk – the risk of no progress, of being overtaken by the competition and failing to achieve any real value. An airplane can avoid overrunning the runway by slowing to a crawl, but to get where it's going, it has to take off.

The money you raise has to be spent to create value; you must balance profligacy and parsimony. You must use your limited resources to create as much value as you can before you need to go ask for more.

What do you give up in exchange for the money?

My late Grandpa Jerry, a child of the Great Depression and gourmand of the delicatessen, loved to say that there is no free lunch. The advice is certainly true for fundraising in biotech.

When you get money – whether from VC, other investors, or Pharma – you give up something in return. You exchange participation in your future success for cash today. This participation usually takes the form of ownership or rights: you either sell your investor shares in your company (ownership) or grant the investor a portion of future profits, for example by splitting future product sales (rights).

When you give up ownership, you relinquish control of the company. Choose partners who share your vision for what the company should become, but beware that even perfect alignment can waver over time. Investors pressured to return funds to limited

partners may push you to make business decisions with short-term returns rather than long-term value in mind.

When you give up rights, you relinquish future profits. Because it is difficult to accurately assess your future value today, it can be tempting to discount it if you're desperate for money or just eager to get started. It can seem reasonable to treat shares like scrip when they are worth pennies and illiquid to boot; to treat royalties like toy money when sales are a decade off; to treat market geographies like territories in Risk. But you must be careful. At this early date, your future discoveries may seem like cheap bargaining chips, but when they're worth substantially more, you don't want to regret having given that value away.

One exception to the free lunch maxim is certain government grants – SBIR and similar – that are non-dilutive, meaning that they don't require that you give up any shares or “dilute your ownership” in exchange for the funding. The downside to these grants is that they offer relatively (for biotech) small amounts and have relatively onerous requirements for reporting progress. You may find that the effort you spend maintaining the funds outweighs their value to you.

Pre-money and post-money valuation

How much of your company do you give up in exchange for an investment? The answer depends on how valuable your company is today and therefore the percentage of the total value that is due to the new investment. In finance jargon, we call the company's worth today the “pre-money” valuation and the company's worth after the new investment the “post-money” valuation.

For example, assume you need to raise five million dollars to run your next set of pivotal clinical trials. If your company is worth \$20 million today pre-money, then the \$5 million you raise will increase the value to \$25M and account for 20% of the post-money value ($5M/25M = 20\%$). In exchange for the investment, you'll have to give up 20% of your company. If instead your company is worth \$5M today, then that same \$5M investment costs you 50% of your company. And if your company is worth only \$1M today, the same money costs you 83% of your company!

Clearly the pre-money valuation is important. How do you determine what you're worth? Perhaps surprisingly to those of us coming from the sciences, where important numbers tend to mean something real, valuation is subjective, negotiated between you and your investor.

Your valuation is measurable – you can predict the size of the market for your drug and use historical data for failure rates of similar drugs at your stage of development to calculate a risk-adjusted net present value of the asset. It is, however, still subjective. Those market size projections might be too rosy if patients are routinely misdiagnosed or if payers don't reimburse you at the rate you seek. That historical likelihood of clinical success might be wildly low for your drug because you've found a biomarker to select

the patients most likely to respond. Or any of a million other plausible caveats may apply.

Value is particularly subjective at the earliest stages of a start-up, especially for innovative companies creating new drug modalities and new discovery platforms, where there may not even exist a specific drug that we can put through the wringer of a net present value calculation. That's why it is essential that you learn how to conceive of, create, and convey the maximum value in your company, even from the earliest days.

Ch2.4 Single Assets and Portfolios

All the single assets (All the single assets)

Developing a drug is hard, and most attempts fail. It's hard because new biology can be difficult to understand and manipulate; it's hard because the human body is so complex that predictive models rarely predict what actually happens; it's hard because even if your drug hits its target, it may hit it too hard or not hard enough, or react with a similar molecule somewhere else in the body, or be otherwise toxic; it's hard because even if the molecule works well, it may not be absorbed well enough to achieve an effective dose in people.

These difficulties manifest as low probabilities of success across the ecosystem, from academic findings that can't be reproduced to start-ups that struggle to translate promising leads and to the endemic attrition in Pharma pipelines, where fewer than 10% of drugs that enter Phase 1 of human testing get approved.

For start-ups built around single assets, this risk is acute. If you're right – if the molecule works – the results are lucrative. But these rare true positives are buried in a haystack of good ideas that don't pan out.

Portfolios to the rescue?

If the probability of success is a Dirac delta function, zero everywhere except for the rare massive spike, then one solution is to build a portfolio. VC firms build a portfolio of investments, each of which is unlikely to succeed, but that on aggregate will have enough “wins” to provide a return. Pharma builds a portfolio of drug candidates across multiple therapeutic areas that balances low-risk incremental improvements with transformative new mechanisms.

But portfolios don't work for the entrepreneur who is all-in at one company, for whom the risk is concentrated in the single asset. To the entrepreneur, what the investor chooses to do across the portfolio is immaterial. Even if the start-up builds a nominal portfolio, a “bag o' assets” that are loosely (if at all) related, the high costs of clinical development mean that assets are usually tested one at a time in the clinic. The unrelatedness of the molecules means that the risk of failure is independent and remains equally high for all of them.

Portfolios also do a disservice to the promise of innovation. Risk-averse companies often choose the least-challenging most-likely-to-work applications for their assets. They may pursue marginally better derivative treatments instead of truly transformative breakthroughs. They may tackle diseases with easy clinical trials instead of those that present the greatest unmet need. They may chase targets that others are chasing, reasoning “If we all go down, we go down together,” instead of forging into new biology.

These are all rational choices made by rational actors, but collectively they limit what we can achieve as an industry. It is difficult to tap the true potential of innovation if we are limited by the risk profiles of single assets and portfolios in start-ups.

Fortunately, over the past few years, a new model of start-up has emerged. A start-up that can take big chances, that can chase ambitious new ideas, and that is built to withstand the hurdles that accompany early innovation. Introducing: the Multi-Product Platform Company.

Ch2.5 Introducing the Multi-Product Platform Company

A Multi-Product Platform Company (MPP) is a company built around a central, unifying platform. The platform can be a technology, a suite of technologies, or even a core biological hypothesis that, when applied, creates not just one but (as the name suggests) *multiple* therapeutic products. The multiple products that emerge from an MPP's platform may look similar to one another or they may not, but the products share common features inherent to their origin. Their phenotypes are different, but their genotypes are the same.

Because of the shared genotype across the products of an MPP, the making of the first product suggests that subsequent products can be made with the same process, and success with the first product makes the success of all subsequent products more likely. Each new product that emerges from an MPP informs, validates, and reduces the cost and risk of those that follow. Wash, rinse, repeat.

MPPs are different from what were naively called “Platform Companies” in the early 2010s. MPPs explicitly emphasize the multiple products that emerge from the platform rather than assuming that the platform has inherent value by itself. It may sound pedantic to distinguish between platform companies and *multi-product* platform companies, but the name becomes prophetic. If the products are absent from your description of the company, they will be absent in real life too. Yesterday's “Platform Companies” didn't think of products as inherent to their platform, so they didn't prioritize making products. Consequently, they often didn't make products and therefore rarely succeeded.

MPPs overcome the challenges of single assets and portfolios

MPPs change the risk profile of drug development. Because the platform produces multiple therapeutic products, an MPP doesn't put all of its eggs in one clinical basket. You can build value in an MPP by investing in and creating your platform, thereby laying the groundwork for future products, instead of focusing exclusively on the first molecule. The existence of an underlying platform also makes it possible to capture broad intellectual property early, allowing you to fend off would-be followers and build a foundation for lasting and sustained value well into the future.

MPPs let you maintain optionality. You can choose which of your multiple shots to take and in what order, deploying them strategically to maximize the company's chance of success. You can monetize your products separately to different partners. You can develop your platform in concert with your products to unlock new therapeutic areas and learn early on which paths are hard and which are easier.

MPPs also incentivize the big leaps in technology and innovation that are often missing from traditional start-ups. By emphasizing common genotypes across products and loosening the vise of single-asset thinking, MPPs free entrepreneurs to think creatively beyond the strictures of what is feasible today or even in the adjacent possible, opening

the door to whole new classes of medicines. By increasing the size of the future opportunity with multiple progressive therapeutic products, MPPs whet the appetite of the deep-pocketed investors who are needed to fuel the endeavor.

Three types of MPPs

MPPs come in three primary flavors, defined by the nature of their platforms: those that develop new modalities, those that create new technologies, and those that map new biological insights. Each creates value in a slightly different way.

New Modality MPPs create new therapeutic modalities – new classes of molecules – that enable new forms of treatment. Oncolytic viruses, gene therapy vectors, CAR-T cells, other engineered immune cell therapies, cyclic peptides, nucleic acid drugs, multi-specific antibodies, and protein degraders are examples of new modalities. The multiple products that emerge from the New Modality MPP are molecules of the same form that, with only moderate tweaking, can be developed for different diseases. The platform is based on synthesis and manufacture of the modality. Because the modalities can be readily described physically, New Modality MPPs are often buttressed by strong intellectual property around the composition of the molecules and the proprietary know-how involved in making them.

New Modality MPPs are household names in biotech. Alnylam (siRNA), Moderna (mRNA), Juno and Kite (CAR-T), Editas, Intellia and Crispr Therapeutics (CRISPR), and many others are known for pioneering new therapeutic modalities. Even the old guard, Genentech and Biogen, were once MPPs that pioneered recombinant protein therapeutics based on a technology platform of molecular cloning, expression of recombinant proteins, and purification.

Until they are proven to work, new modalities seem far-fetched. (It can be hard to remember how unreasonable it once seemed to make recombinant proteins as medicines.) The initial evidence – that you can make the modality and that it works as you expect – is deeply validating of the platform and of all future products of that modality. Given the inherent technology risk, New Modality MPPs typically focus initially on indications where the biology is well understood and for which it's easiest to make the molecule, and then strategically expand into new indications as the platform develops.

New Technology MPPs build a technology that identifies new targets or creates molecules in new ways. The platform is the mode of discovery or the method of making, and the multiple products are the drugs (of whatever modality) that emerge.

New Technology MPPs use their technology platform to find lower-risk likely-to-work interventions by more accurately reflecting disease states (CRISPR screening, primary immunophenotyping), exposing new facets of biology (ChIP sequencing, single-cell multi-'omic analyses, molecular dynamics), or making better-quality molecules (computational chemical synthesis, displayed antibody libraries, or primary B cell

isolation). Most artificial intelligence applications in biology qualify as New Technology MPPs.

The multi-product expectation is that when we point the technology to new diseases, it will produce additional therapeutic opportunities. New Technology MPPs often protect their position with strong know-how and proprietary methods for applying their technology, but without broad foundational compositional IP.

New Insight MPPs start from a unifying biological hypothesis to find new pathways and targets that are relevant to disease in previously unappreciated ways. The platform is often supported by robust target identification and biological validation, often with specialized (though not necessarily proprietary) technology. The multiple products – sometimes targeting a related family of proteins, sometimes built around a common set of pathways – can be in one or more modalities. Most New Insight MPPs prefer known modalities, because they are already taking on substantial biology risk.

New Insight MPPs often apply modern tools to make sense of long-observed phenomena and create therapeutic interventions. Agios Pharmaceuticals applied modern metabolomics to identify drug targets related to the aberrant metabolism of tumors, a phenomenon described in the 1890s. Neo-antigen companies like Neon and Gritstone turned a hypothesis about cancer mutations spurring the immune system into personalized vaccines. Microbiome companies like Seres, Vedanta, and others quantified the flora in various human tissues to develop hypotheses for microbial replacements to restore health.

New Insight MPPs benefit from concentrated expertise in a disease area, set of pathways, or target families, which improves the odds of each additional program in that sphere. It can be challenging to protect an insight with IP, but New Insight MPPs fend off competition by arriving first in a given space and developing deep expertise in specialized technologies.

Building successful MPPs

Examples like Genentech and Alnylam show that MPPs have existed for some time. But only recently have start-ups been able to access the requisite capital and begun to think boldly enough to consistently move beyond an asset- or portfolio-centric strategy.

We have seen, built, and run multiple MPPs over the past decade, of all three flavors. Our MPPs created sequencers that could read single DNA molecules, therapeutic gene-modified cells with no genetic material, gene therapy vectors from endogenous unculturable viruses, molecules to stimulate communication between nerves and the immune system, and algorithms to change cell states and generate therapeutic proteins using just a computer. We have conceived of ideas, invented methods, set up labs, led teams, raised funding, taken companies public, sold instruments to customers, and put medicines in people.

Along the way, we've learned a few things. We've learned that being successful in an MPP is not easy. You can't just follow a drug development checklist. We've learned that MPPs are fun! You get to be at the forefront of scientific advances, make real what was once only imagined, and spend your days tackling the world's unsolved problems. We've learned that despite their differences, MPPs share common features and common strategies for creating value. And we've learned that building successful MPPs requires skills that can be learned and taught.

The rest of this book will focus on principles and practices for creating and capturing the maximum value in a Multi-Product Platform Company. We believe that knowing how to build value in an MPP is the essential skill for today's biotechnology entrepreneurs.

Key Takeaways

- Even today's new, ambitious, innovative start-ups need to raise money to function. There is no free lunch. In exchange for the investment, you give up either ownership of your company or rights to future products. Creating value matters because the more valuable you are today, the less you relinquish.
- Multi-Product Platform companies (MPPs) are the best form of start-up to tackle big unsolved problems, build innovative technologies, and create lasting value.
- MPPs are based on a central technology platform from which multiple related products (or families of related products) emerge, where each product is easier to make and likelier to work because of those that have come before it.
- Three common flavors of MPPs are those based on New Modalities, New Technologies, and New Insights.

Chapter 3. Defining Value

Of all the challenges that confront the entrepreneur launching a Multi-Product Platform company, the most difficult *and* the most central to success is understanding value.

If you understand value, you can create value. You can conceive of ideas that are valuable (Chapter 4. Generating a Valuable Idea), maximize value through your actions and decisions (Chapter 5. Creating Value in the Lab, Chapter 6. Creating Value in Product Development, and Chapter 7. Creating Value Through IP), and effectively convey that value to others (Chapter 8 Selling a Valuable Story and Chapter 9 Building Value Through People). Understanding value makes the difference between success and struggle.

When products are bought and sold, value is easy to measure. We can quantify worth from the price at which goods and services are exchanged. But in an early start-up, we don't have those numbers. In an MPP, we don't have a single "future product" around which to build our model. And in a transformative MPP of the type we aspire to create, where the technology is nascent and the innovation futuristic, projecting the full impact with any accuracy can be nearly impossible.

But value is also where the rubber meets the road. Only by defining and describing value, even in an early-stage MPP – *especially* in an early-stage MPP – can you begin to create value.

In this chapter, we will focus on defining value in an early-stage MPP. We'll show how you can conceive of value as the answer to a simple question: "So What?" We'll provide a framework for using that answer to get to effective and testable value propositions that you should place at the center of all your activities. We'll explore value creation in the different types of MPPs and discuss how to maximize the total value by navigating the intersection of platform value and product value. We'll lay out how to find your Platform-Product-Market fit and why it's essential to create value continuously. And we'll end with a reminder that, for all our efforts to make it rational, value is deeply emotional, so you'd better know how to tap into it.

Ch3.1 Value: The Answer to “So What?”

There are a million questions you can ask about an idea. “How does it work?” “What can it do?” “How will you build it?” “What makes it better than other options?” “How fast does it go?” “How much will it cost?” “Who would pay you for it?”

But now we’re going to let you in on a secret. If you want to start talking about value, there is one question above all that you need to ask.

“So What?”

Behind this deceptively simple question – “So What?” – is the secret to all we do. Articulating an answer to So What moves you from the realm of technology, tactics, and timing into a conversation about value. A good answer to So What explains the utility that an innovation brings to the world. Indeed, start-up literature is full of advice to define the customer and identify the customer’s problems. You, the reader of this book, likely don’t have a “customer” yet, but So What gets you the equivalent. For whom are you solving a problem? What benefit does your innovation offer them? Why would they be compelled to pay you? The answers to all these questions follow directly from So What.

“So What” is easy to ask but hard to answer. This is especially true for us as scientists – we are trained to describe “what” we’re trying to do and “how” we’re going to do it, and we usually assume that the answer to So What (if we even think of it) is self-evident. But So What does not follow from What and How, and What and How do not describe value. We must learn to state the impact and implications of our innovations explicitly.

The answer to So What is the value proposition of your idea. With it, you can define the value of your idea, even when the value is not quantifiable. If you can describe a compelling, simple, understandable So What to a potential customer, employee, executive, or partner, you can convince them to buy from you, join you, and support you.

Ch3.2 Features, Advantages, and Benefits

The answer to “So What?” is the value proposition, which defines the value of your innovation to the person who cares about it. Communicating a value proposition is essential to capturing value. But how do you know if you’ve done it right? What makes a good answer to “So What?” How do you cut through the clutter of technical features and platform advantages to get to a real statement of value?

You’ll find all sorts of definitions of *value proposition* in the business literature – a statement of the value you are providing to the customer; the benefits a customer will receive by purchasing your product; the benefit to the customer minus the cost to the customer. The nugget of truth at the core of all these definitions is the word *benefit*. Your value proposition, the answer to So What, states the benefit that your innovation provides and to whom the benefit is provided.

It is useful to emphasize benefits because we scientific innovators tend to get excited by the new features of our technology and the advantages of our platform over existing techniques. In our enthusiasm, we forget to consider the benefits that these features and advantages confer, or we mistakenly think they are self-evident. But if we don’t make the benefits explicit, it’s as if they don’t exist.

A good value proposition gets at the benefits.

The F.A.B. Framework: Features, Advantages, and Benefits

We’ve found that it often helps to explicitly enumerate the features, advantages, and benefits (F.A.B.) of your Multi-Product Platform. The F.A.B. framework forces you to do just this. The simple and accessible exercise breaks our instinctive focus on technology and provides an easy-to-follow guide for stating value propositions.

To set up an F.A.B. framework, list all the technical *Features* of your idea, platform, or technology; the *Advantages* they provide over existing techniques or competing methods; and the *Benefits* that redound to the end-user. (See the example below of an F.A.B. framework for an enucleated cell therapy.)

Features are technical properties specific to your platform. List as many features as you can envision – there will certainly be many. Include features that you have thought of but may not have built yet. Figuring out whether those features are valuable (do they give you advantages and provide benefits?) will help you determine how much effort you should spend on enabling them.

Advantages are capabilities of your platform that are superior to what can be done with conventional methods or competitors’ technology. Couch your advantages relative to an alternative: “Unlike X, I can do Y.” Doing this forces you to know the world you’re trying to disrupt, understand your competition, and speak the language of your end-user.

Benefits are the perks granted to the end-user, the utility provided by the advantages. What can the end-user do now that they couldn't do before? Make better decisions? Discover molecules faster? Treat new diseases? Avoid painful or dangerous side effects? Live longer? You may have several end-users – patients, physicians, drug discoverers, Pharma companies – who perceive different benefits, so try to capture them all.

As a quick aside: don't be overly enamored of novelty. In our scientific training, if we ever ask So What, the typical answer is "because it's new." Novelty is the currency of the academic realm. It justifies the questions that are posed, the proposals that are funded, and the results that are published. But in the commercial world, where utility matters, novelty by itself is not a sufficient value proposition. "Because it's new" or "This couldn't be done before" doesn't cut it. Instead, figure out how your novelty confers benefits. Who is the end-user, and why do they care if the feature is new? Does the novelty let you get meaningful data at higher resolution so that you can make a decision earlier? Does the novelty let you find targets that would otherwise be hidden using conventional tools? Does the novelty let you treat diseases that would not otherwise be treatable? Find your way to the benefits.

F.A.B. Framework for an Enucleated Cell Therapy Multi-Product Platform

Features	Advantages	Benefits
<i>What are the technical features of the platform?</i>	<i>How is it better than today's tech or other methods?</i>	<i>Who benefits? Why should they care?</i>
4-month circulation	Longer than small molecules (days) or biologics (weeks)	Fewer injections, longer-lasting efficacy
Protects payloads from the immune system	Fewer adverse events, prolonged tolerability and activity	Safer and less discomfort for the patient; more efficacious drug
Type O- universal cells	Off-the-shelf relative to other cell therapies	No delay from intent-to-treat to time of dosing

Turning F.A.B.s into Value Propositions

Listing features, advantages, and benefits in a table clarifies the connections between the F.A.B. inputs: the features lead to advantages that manifest as benefits to the end-user. You craft your value propositions by reading from left to right. A value proposition is simply the statement of the benefit derived from the features and advantages.

For the example above, the value propositions of the enucleated cell therapy platform may be:

- Because the cells have a 4-month circulation, substantially longer than other modalities, you can treat diseases with fewer injections without the efficacy of the drug waning.
- Because the cells protect their payload from the immune system, you expect fewer adverse events and less immune-mediated drug clearance, which makes for a safer drug for patients that is more efficacious over time.
- Because the cells are derived from a universal source (type O-), they can be off-the-shelf relative to other cell therapies that must be produced on a one-to-one basis for the patient, which means there is no delay for the physician between deciding to treat a patient and the time of dosing.

Your MPP will never have just one value proposition. Some value propositions speak to the platform, while others speak to a single product or family of related products. Different features lead to different advantages and different benefits for different end-users. Some features may provide multiple advantages and benefits. Different audiences respond to different value propositions; they feel certain benefits more acutely than others. It is useful to have many value propositions because you can tailor your story to present the most impactful version to any given audience.

Here are a few illustrative examples. Use the blank F.A.B. framework on the next page to explore the features, advantages, and benefits of these concepts and turn them into value propositions. Or work through your own ideas or your company's platform.

Example 1: Imagine a computational platform that quickly models and accurately predicts protein-protein interactions. You intend to use the platform to design new functional binding proteins like antibodies. Because it is an algorithm, the platform doesn't require trial-and-error screening of antibodies in the laboratory. One value proposition is that because the platform doesn't require laboratory screening, it is substantially faster than traditional methods, which lets you get to your lead molecule sooner than you would otherwise. A second value proposition (from the same feature) is that because the platform doesn't require laboratory screening, you can design antibodies to protein targets that are difficult to synthesize, like very large proteins or proteins embedded in membranes, which means you can now develop drugs to important disease targets that would otherwise be inaccessible. What other features, advantages, and benefits can you come up with, and how would you frame them as value propositions?

Example 2: Imagine a modality platform of chemically cyclized peptides that are orally bioavailable and do not degrade in serum. Because they are peptides, they have a larger surface area for interacting with targets than typical small molecules do. One value proposition is that because of their small molecule-like uptake and protein-like bioactivity, these cyclized peptides enable dosing by mouth of drugs that today require injection, which is much more convenient for patients and would lead to better compliance. What other benefits derive from these features and advantages?

Idea:

Features	Advantages	Benefits
<i>What are the technical features of the platform?</i>	<i>How is it better than today's tech or other methods?</i>	<i>Who benefits? Why should they care?</i>

Value Propositions:

Because of [Feature] which grants [Advantage], the platform/product provides [Benefit] to [End-user]

- _____

- _____

- _____

- _____

Ch3.3 Testing your Value Propositions

Value propositions are not givens. Your answer to So What is not a fact; it's a hypothesis. It's a hypothesis about what might be valuable in a yet-to-be-built future.

If you can phrase a value proposition as a hypothesis – “We hypothesize that our platform's feature F lets us do advantage A better than others, which provides benefit B to the end user” – congratulations! You now have a testable statement, or a falsifiable prediction, that you can try to invalidate.

How to test your value propositions

How do you go about testing your value hypotheses? One way is to experiment, to build what you want to build and see if it works the way you predict. You can test out features (can you make them?) and advantages (how do you compare to others?), but it is hard to experimentally test a benefit hypothesis. (In Chapter 5. Creating Value in the Lab, we'll cover the best ways to design and conduct experiments to create value.)

Building tends to be the longest and most expensive way to figure out the answer. But there's a faster way to test value. Once you frame your value propositions as hypotheses, you can go talk to people about them.

You are the expert in your platform's features, but others – practitioners of the technology today – will grasp its advantages relative to the existing technologies, and the end-users will recognize the benefits. Go find the practitioners and end-users. Propose to them your value propositions. Ask how they do the process that you think you're advantaged in. Learn where they struggle with the existing technologies and where the technologies work just fine. Figure out whether your proposed benefits are actually useful and beneficial.

Verify your value propositions. Try to disprove them if you can. And, importantly, keep an ear open for new ones. Often these conversations will bring up additional benefits you have yet to consider, and you can map them back to your features and advantages and incorporate them into your next set of value propositions.

If you solicit this feedback and revise your value propositions accordingly, you'll end up with a stronger and more complete set. You will also be more convincing when you describe why your feature leads to an advantage and why this advantage provides a real and valuable benefit to the end-user.

Test your value propositions before embarking

If you're thinking about founding an MPP, you surely expect to take big technological risks. But you cannot accept risk on the value proposition. Imagine working your tail off for years to build a platform, invent new technologies, and surmount all the hurdles that come your way... only to discover that you created the wrong product for the market and no one has a use for it!

A good illustration of this was our experience with Helicos Biosciences, a company we started in 2003. Helicos was a true technological pioneer that conceived of, manufactured, and commercialized the world's first single molecule DNA sequencing machine.

The machine's key feature was the ability to read the sequence of an individual DNA molecule directly, without any amplification. To do this, the machine immobilized billions of single strands of unknown DNA on a surface, synthesized the reverse complement of the strands *in situ*, and imaged each new fluorescent base as it was incorporated into the nascent strand. Helicos combined the cutting edge in optics, image analysis, microfluidics, and molecular biology. The technology was a feat of science and engineering.

However, when it came time to define the benefits of the feature – to define its value propositions – Helicos hastily assumed that the best application of the technology was the sequencing of whole human genomes (otherwise known as WGS). Enthralled by the hype and excitement of the race to the \$1,000 genome, the team didn't stop to question whether this was the ideal fit for the platform and the product. Would a machine that read individual DNA molecules be superior to cheaper amplification technologies, especially when the task required reading a massive 30 billion bases? And since the template (the human genome) was known, did it matter that the machine didn't rely on it?

As the Helicos engineering team was building a massive, \$1M+ instrument to hit the specifications required for WGS, the benefits of single molecule DNA sequencing over amplification-based technologies were gradually coming to light. The obviation of sample preparation, the lack of G/C bias, and the ability to accommodate small sample amounts or highly degraded nucleic acids conferred little to no benefit for those doing WGS, but represented superpowers for scientists working on counting applications (e.g. ChIP-Seq or RNA-Seq), pathologists interested in quickly sequencing a few key cancer genes, biologists working with archival Formalin-Fixed Paraffin Embedded (FFPE) cancer tissues from patients, and genome hunters working to analyze ancient DNA from humans and even dinosaurs.

Alas, none of these better-suited applications required an instrument of the size, cost, and complexity of the WGS machine. It was too late for Helicos to build a new machine. The company had spent tens of millions developing a device with the wrong value proposition, unable to provide a benefit to those users who could actually have used its features and advantages.

In hindsight, the lesson we learned from Helicos is that it is essential to test your value propositions upfront, before embarking on a project. If you know your "So What" at the outset, you can develop your platform in the most valuable directions to produce the most valuable products. You will (perhaps) be spared long and expensive failures, and you will be able to confidently orient everything you do around the value propositions.

Ch3.4 Orienting Around Your Value Propositions

Once you know your value propositions – when you’ve considered them and defined them, pressure-tested them and validated them – you can begin to orient your activities around them.

What does it look like to orient around your value propositions? Your answers to So What inform your choice of experiments, shape your strategy, and guide you toward or away from partnerships. Your answers to So What define the story you tell about your company and the ways you prioritize your time and effort. In fact, everything orbits in a path set by the gravitational pull of your value propositions.

Our solar system is heliocentric; your MPP should be value-centric. Everything, from the highest-level strategic planning to the most nitty-gritty, in-the-weeds, day-to-day decisions, should be viewed in light of your value propositions. Here’s what that means:

On the science front, let your value propositions determine what experiments you do. Know what demonstrations – what data – will validate your greatest value propositions. Know the So What of your experimental results first; then strive to generate this data. This backward-in thinking is the inverse of the way we usually describe scientific innovation, but is essential if you’re oriented around value.

A few years ago, George Whitesides, a Harvard chemist and polymath, published a how-to guide to writing scientific papers in which he suggests that students pre-write outlines of their papers to *guide* their research rather than simply record their results at the end.ⁱⁱⁱ The advice is sound for PhD students and applies to science in early MPPs as well. Define the So What at the outset and use that to distill the purpose of the experiments, prioritize which ones to do, choose how to do them, and decide what you must show to declare victory.

On the strategy front, do the same. Let your value propositions guide your resource allocation, plans for growth, declaration of development candidates, and engagements with partners. Ask: Does this action help me create value or not? When you focus on your value propositions, you know what you need to prove and why it matters, and this knowledge clarifies your strategic decisions.

It may appear that science questions and strategy questions belong in different realms, but they’re made of the same fundamental building blocks of value. We joke that lab meeting *is* strategy meeting, since lab results tell us which of our value propositions hold up. Set your value propositions and shape the strategy around them.

When you sell, let your value propositions guide your storytelling. When you present your idea or pitch your company, lead with the So What. Attention spans are short. The people you need to convince – those with money and influence – need to care about your idea before they choose to spend mental energy deciphering what it is, how

feasible it might be, and whether to support you. When you lead with So What, you are instantly more persuasive.

Furthermore, when you place your value propositions at the center, everything falls into place. The story lays out the value proposition; the data generated supports the value proposition; the strategy sets the path to realizing the value proposition. Once you've defined your value propositions, crafting the narrative becomes easy (though not trivial!).

Orienting around your value propositions also lets you be effective and intentional with your time. Recent research on productivity shows that attention management is the secret to navigating today's hyper-distracted world. Being deliberate about where we choose to devote our attention and thoughtful about where we choose to spend time distinguishes the productive from the merely busy. If you know what uses of your time create the most value, you can marshal your attention accordingly.

Set your value propositions, and your universe comes into focus. Orient everything around your So What – from top to bottom – and you'll know you're pointing in the right direction. The rest is just execution.

Ch3.5 Creating Value in an MPP

Now that you've defined your value propositions, how do you create value in your MPP? And more importantly, how do you capture the *most* possible value?

Value in an MPP is more than just products or platform

As the name *Multi-Product Platform* implies, you can create value in both your platform and your products. To maximize value creation in your MPP, you must create value in both.

Platforms by themselves don't have much value. Technological innovation by itself is difficult to quantify; what good is a platform if useful things don't come from it? You need to describe how the platform allows multiple products to emerge at reduced risk. Then you need to develop those products to pull the platform forward into the realm of real value.

At the same time, while a single drug may project sales of billions of dollars, its worth is steeply discounted in the early days of discovery. The value of an MPP is more than just the sum of the value of the products you're developing today. It also includes the aggregate sum of the multiple future products that will emerge, with progressively less risk, from the platform.

The maximal value of an MPP combines Platform Value and Multi-Product Value. In the early days, it's on you to prove with data – and to argue before the proof is fully in hand – that your platform will produce multiple valuable products, all sharing common beneficial features that reduce risk and accelerate development.

In the next few sections, we'll describe value creation in MPPs in more detail and offer tips for intelligently building value in your platform and products simultaneously, taking advantage of their intersections so that each augments the other.

Value Creation in MPPs

To create value, all MPPs need to validate and develop a platform technology that is broadly applicable to multiple products; demonstrate that the platform is capable of producing multiple products by generating functional prototypes; and generate multiple differentiated products.

These steps manifest differently in the different types of MPPs.

New Modality MPPs create new molecular form factors of medicine from which multiple therapeutic products can be derived.

Platform Value In a New Modality MPP, the modality is the platform. Because all of your products will be of the modality, you create platform value by demonstrating early that the modality works, that you can make it, and that you can protect it with broad IP.

People tend to be skeptical of new modalities, especially those that are unlike drugs that exist today. Well-chosen prototypes – example molecules that *could be* drugs and that demonstrate the benefit of the modality, a differentiated functional advance along some useful axis – help overcome this skepticism. If you choose well, prototype demonstrations can be simple to do and definitive in proof-of-mechanism preclinical studies.

For example, if the skepticism is that messenger RNA can be a drug modality (it's unstable, it's immunogenic, it's impossible to deliver into cells), you may choose to create a prototype mRNA encoding a gain-of-function protein like erythropoietin.^{iv} It's a clever choice: Epo is a plausible drug; when successful, the prototype shows that the mRNA modality is a viable way to produce systemic proteins; the demonstration is easily and definitively measured by counting red blood cells; and the bar is low because only a small amount of Epo is needed to raise the blood cell count in an animal.

Multi-Product Value To capture value, your products need to allow the features and advantages of your modality to clearly and immediately manifest as benefits in the disease.

New Modality MPPs can be selective about developing products. You should choose products to minimize biology risk: products in which the diseases are well characterized and in which the biology of disease aligns with the features of your modality.

Early clinical efforts that show mechanistic validation can create significant value, especially if the applications are apparent once the premise is validated.

New Technology MPPs develop a discovery tool or technology that lets them better identify new targets relevant to disease (target discovery) or make more potent molecules against known targets (molecule discovery).

Platform Value In a New Technology MPP, you create platform value by showing that the platform works (that it gives rise to new and useful targets or molecules), that you've assembled (or can assemble) the necessary technologies, and that it can be applied repeatedly to multiple disease areas or targets. It helps, too, to show that the platform is unique to you and differentiated from others. You should be able to find things that others miss and make things that others can't.

To validate that your platform can find useful things, look for “positive controls” – targets or molecules that we know work and that your method also discovers. Positive controls only take you so far if you don't have a proprietary angle for developing them. Complement positive controls with new hits and molecules to prove that your platform has breadth.

As much as possible, you will want to protect your platform technology. Depending on its nature, you may not want to disclose it in a patent (more on this in Chapter 7).

Creating Value Through IP) but instead keep it as a trade secret. Regardless, you create platform value when you're the only one with the expertise or the rights to use the tool.

Multi-Product Value New targets and molecular hypotheses carry with them a great number of unknowns, so you will want to minimize translational risk. The easiest way to do this is to pick modalities that are understood so you don't layer technology risk onto your development path.

Reducing translational risk also means speeding the time to Proof-of-Concept for the target or molecular hypothesis. If your MPP enables target discovery, seek diseases for which the link between target and effect is strong and the clinical demonstration is achievable in short timeframes. If your MPP enables molecule discovery, for example making molecules for new sites on targets, pursue applications where your mechanistic hypothesis for efficacy is clear and can be shown convincingly.

New Biological Insight MPPs start from a new biological hypothesis, a new lens through which to view disease, or a new set of targets and interactions against which to develop drugs.

Platform Value In a New Insight MPP, you create platform value when you show that the insight is relevant to disease and creates multiple opportunities for intervention. Furthermore, the insight has to be novel, which provides some degree of protection, and you must have the necessary tools to explore and exploit the insight.

Because your platform is essentially an idea, you need to validate that your hypothesis is relevant to diseases with model systems that are as close to the real thing as possible, like human genetics or clinical data. Lab-based systems like patient-derived models or genetically matched animals are less convincing.

Because your platform is an idea instead of a modality or technology, it can be very difficult to protect it with patents. Who cares if you were first to an idea if others can copy it once they recognize the payoff? Therefore, be prepared to move fast and erect walls. The more your insight requires (or appears to require) deep knowledge or specialized expertise, the harder you make it for followers.

New Insight MPPs can strike early partnerships with Pharma for research collaborations or access to molecules, patient samples, and model development expertise. These relationships increase the value of the MPP because they accelerate the identification of products, confirm that "customers" are excited by the insight, and build relationships for future investment.

Multi-Product Value New Insight MPP should focus on conventional modalities – often small molecules and antibodies – to reduce the total translational risk of their new biological hypotheses. New Insight MPPs should get clinical proof-of-concept as quickly as possible, since new disease mechanisms aren't widely believed until we see them

work in humans. There are sometimes opportunities to repurpose pre-existing molecules for the new hypothesis, which may let you get your drug into people sooner, depending on how much is already known about safety, pharmacokinetics, and biodistribution.

A niche indication, like a rare disease or a genetically-defined subset of a common disease, may provide your path to a fast clinical readout. If that's the case, you can increase the value in your multiple products by showing how to expand into larger indications.

New Insight MPPs have a distinct risk of collapsing into single-asset companies. Because biological validation and drug development takes time and suffers attrition, and because even the most promising leads fall by the wayside, you can easily find yourself with only one drug, target, or hypothesis poised for clinical testing. An MPP with a single compound and no proprietary modality or technology looks an awful lot like a single-asset company, one with a single point of failure (and subject to a novel and risky hypothesis to boot!). If the first test of the hypothesis turns out not to work, there isn't much left to the company.

An Analogy to Oil Exploration

To distill the differences between MPPs in value creation based on compositions, technologies, or hypotheses, we like to make an analogy to oil exploration.

In this analogy, a New Modality MPP has discovered a new non-petroleum energy source. To create value, the company must convince people that the new energy source is worthwhile: that people can harvest it, that they can use it to power useful things – like cars, phones, or the electric grid – and that it provides a benefit; perhaps it is cheaper than oil, less polluting, or more easily deployed on passive surfaces. As the founder of the company, you'll want to pick a set of initial products for which the benefit of your new energy source over petroleum is dramatic and the market risk is low.

In the analogy, a New Technology MPP has developed a new drill to better access hidden reserves. To create value, you must convince people that the new drill can reach places that others can't. You might do this by showing how efficiently you can access existing reserves, but for your argument to be persuasive, your audience must be aware of untapped pools of petroleum, inaccessible today, that you're uniquely able to reach.

In the analogy, the New Insight MPP is sitting on land that has yet to be drilled. To create value, the company must convince people that there is petroleum underground. Everyone knows that petroleum has value – you have to make them believe they'll find it under you. You can do this by drilling a few test wells and showing that there is oil beneath the surface or by comparing your geological features to those on known productive fields. But in the absence of an operation to fully mine the territory, much of the value relies on your ability to sell the future potential.

Depending on the nature of the MPP, you will strike a different balance between Platform Value Creation and Multi-Product Value Creation. The full value of an MPP, however, can only be created and captured when you are alert to the value present in both the platform and the products.

Ch3.6 Integrating Platform and Product Value Creation

Integrating platform value creation and product value creation can make your work especially fruitful. To leverage their intersection, you can do three things: define the Platform-Product-Market fit; develop your platform strategically in concert with your products; and simultaneously advance multiple products that validate your platform value propositions.

Define your Platform-Product-Market Fit

You've likely heard of product-market fit, a concept from marketing that says that a product, no matter how innovative it is or how long you worked to make it, isn't worth much unless it meets a need in the market. The product-market fit describes how much benefit the product provides to the end-user, or the "pull" from the market that makes the product desirable. Helicos' single molecule DNA sequencing machine was a case of a product developed without a strong product-market fit.

An MPP, however, has multiple products. What's more, MPPs often have multiple families of related products, each of which shares common benefits that derive from a subset of platform features and advantages. Each individual product must fit a need in the market, but more importantly, each product in a family shares a common product-market fit that comes from the platform. This is your *Platform-Product-Market fit*.

The Platform-Product-Market fit describes how your platform enables a family of products with shared value propositions. When you specify your Platform-Product-Market fit, you clarify how your products relate to one another and how they relate to the platform. A Platform-Product-Market fit can encompass multiple value propositions, and you will likely have different Platform-Product-Market fits for different families of products.

Defining your Platform-Product-Market fit is important for several reasons. First, it helps you figure out what families of products exist for your MPP *and* helps you identify all of the products within each family that share the same Platform-Product-Market fit. The more closely these multiple products share value propositions, the more de-risked one product will be when you succeed with another. Second, once you identify your product families and the products within them, you can predict which product families have the biggest opportunities and therefore which you want to pursue. This prediction will tell you what platform features are most important to enable. Finally, by explicitly tying the market desire for your multiple future products to features of the platform, you can begin to explicitly value your platform investments.

Develop your platform in concert with your products

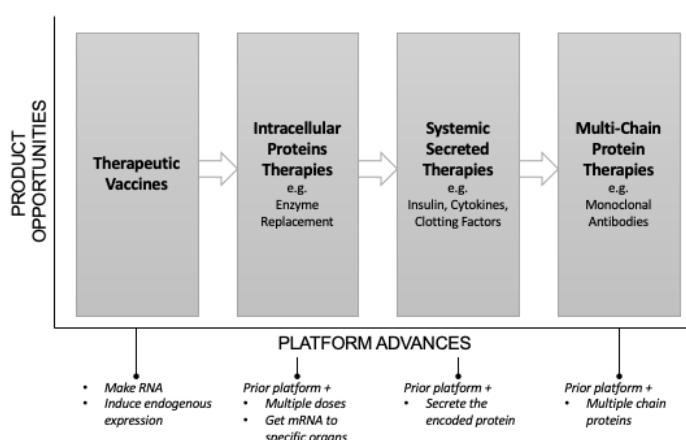
Once you've identified the Platform-Product-Market fit(s), you can begin to develop your platform strategically in support of your multiple products.

Imagine your platform the way a software developer might view a piece of code. If today's platform enables a set of products, certain improvements will let you make the same products better, like an upgrade from v1.1 to v1.2 that runs faster and fixes small bugs. Other improvements are step changes in technology that open the door to new families of products, perhaps new indications or new mechanisms, like software version 2.0 that introduces entirely new features.

Think strategically about how you develop and deploy your platform. It's not a monolith – all features or none, perfect or worthless. You control what elements you build and when. When you know what platform features unlock what opportunities, you can invest in your platform wisely, building the right capabilities to create the right value at the right time.

Moderna is now famous for its rapid development of a coronavirus vaccine candidate in response to the COVID-19 pandemic of 2020. Its readiness in therapeutic vaccines is a result of prioritizing that modality and developing its mRNA platform strategically. Why did Moderna prioritize therapeutic vaccines over other modalities like intracellular protein replacement or systemic secreted therapies? Because it worked with version 1 of its platform. Therapeutic vaccines require fewer doses of mRNA than would be needed for a chronic condition, and they require less protein to be expressed per mRNA molecule because the patient's immune system amplifies the effect. But getting good at version 1 of the platform has benefits beyond vaccines. The multiple candidate vaccines that Moderna put into clinical development taught the company how to manufacture and formulate mRNA drugs, and the COVID-19 pandemic validated the core value proposition that it's possible to instantaneously generate a mRNA medicine from a sequence.

It can help to visualize the relationship between platform and products by plotting your platform advances (x-axis) against your unlocked product opportunities (y-axis). In the Moderna example (*see figure below*), version 1 of the platform, where you can manufacture mRNA, administer a few doses, and get endogenous expression of the encoded protein, unlocks a family of therapeutic vaccines. Version two, where you administer multiple successive doses and formulate the mRNA so that it gets to specific organs, opens the door to intracellular protein replacement, such as treatments for enzyme deficiencies. Version three, all of the above plus secretion of the encoded protein, lets you do systemic therapies. Version four, where you express multi-chain proteins, lets you make monoclonal antibodies.



Build a similar chart for your MPP. You'll see that each family of opportunities shares a common Platform-Product-Market fit, corresponding to the set of platform features currently enabled. Having a Platform-Product-Market map in hand will help you decide what new features to build and how to invest your resources to create maximal value.

Develop multiple products concurrently that validate platform value propositions

We will spend much more time on the nuances of developing products, declaring drug candidates, and prosecuting their clinical development in Chapter 6. **Creating Value in Product Development.** For now, it suffices to say that you should advance multiple products concurrently and make sure that each validates your platform value propositions.

When you develop multiple products at the same time, you show, irrefutably, the multi-product nature of your MPP. This strategy lets you be flexible with your clinical strategy, choosing when and how to trigger trials. It also buffers against an early failure that would leave you in the unenviable position of raising money for a (seemingly) discredited platform without actually giving it a fair shake. Early failure of a single product practically guarantees that your MPP won't survive to reach its full potential.

With multiple product candidates in hand and a Platform-Product-Market strategic map, you can choose how to stage your development activities. You can drive multiple products in a single family and take advantage of synergies between them, and/or pursue orthogonal products that test new platform features.

Whatever you choose, make sure the products you select validate your platform value propositions. Avoid opportunistic "hacks" like repurposed drugs that have quick clinical paths but lack compelling Platform-Product-Market fits. Be skeptical of schemes to generate revenue from an "easier" off-topic product. We occasionally hear companies propose to use proceeds from a first diagnostic product to build a therapeutic pipeline, or to start with a veterinary product and then transition into human therapeutics. But the

toehold products are rarely easy and never create enough revenue without also sucking up all of the company's energy and resources.

If a product doesn't validate your platform value propositions, it isn't worth developing.

Ch3.7 Final Thoughts on Value

Share Price Appreciation vs. Dollars Raised

At discrete moments on your journey, when you raise money, the value of your MPP-in-progress is quantified. As humans, we thrive on certainty. It is natural to latch onto these events and tempting to focus on the top-line figures, like dollars raised or market cap. But beware: these numbers can be misleading.

The value you create is not measured by the dollars you raise or your market cap. Value creation is measured in the appreciation of your share price.^v It doesn't matter whether you raise \$X or \$10X or \$100X; the important questions are: How much of your company did you give up in return? How much more (or less) is a share in the company worth after the fundraiser?

Don't be fooled by breathless press agog over the latest biotech financing. The total amount of money raised is virtually meaningless without knowing the underlying share price before and after. You want to make sure your share price appreciates every time you raise money. If you are creating value in your MPP, this will be reflected in shares that are worth more.

Value is quantified discretely but must be created continuously

The majority of your time running an MPP is spent between discrete valuations. Though value is quantified and recorded only periodically, it is created continuously. You should therefore operate at all times – especially during the unquantified time – with a singular focus on creating value.

The benefit of thinking about value as continuous is that it aligns your internal metronome with the pulse of your MPP. When you have the answer to So What, know your platform value propositions, and orient everything around creating value, you can be confident that you're spending every dollar and every hour on the right things.

Thinking about value as continuous rather than discrete also keeps you from a common pitfall: trying to create all your value in one big swing. You might be tempted to invest all your resources in doing one massive experiment or building the perfect version of the platform, with few intermediate readouts or potential off-ramps, with the expectation that the result will be stunning.

But if it's not? You'll run out of money with little to show for it. You'll have to raise funds at a reduced valuation. So much for share price appreciation.

Also, when you're waiting on the promise of a major payoff, it's psychologically easy to sweep problems under the rug and to avoid the hard work of planning for failures and evaluating alternative paths.

The solution is to think about creating value at many frequent (if smaller) value creation events. Doing so lets you be dynamic, testing hypotheses and nimbly doubling down on those that turn out to be fruitful. It lets you act entrepreneurially, freeing you to be creative in demonstrating the potential of your MPP.

Value Is Emotional

Here we've made a rational case for creating value – do this, count that, make this many of the other. But value isn't only rational; it's also deeply emotional. Value is in the eye of the beholder, even when tangible goods are bought and sold. The value of an item can far exceed the costs of its components. Think of a Ferrari. People don't buy a Ferrari for the quality of its parts. They're not moved by the sum of steel, rubber, and leather. They want the brand, the prestige, the mystique, the unrivaled speed.

In Chapter 8 *Selling a Valuable Story*, we will discuss how to tap into the emotional side of value when selling your MPP. But you must also be aware of this side of value as you *create* it. You have to generate demand. The fewer innovative solutions exist, the more valuable yours will be. The greater the perceived benefit, the higher the price you'll command. The more enticing the data and the more evocative the implications, the more people will hunger for them.

So do the right things on the rational side. Know your So What. List your features, advantages, and benefits. Test your value propositions widely before picking up a pipette, and when you do kick off operations, make sure to build value in your platform *and* in your products in complementary ways.

But prepare the emotional front too. You'll need everything you've got.

Key Takeaways

- Understanding value is *the* essential skill: only then can you conceive of it, create it, and convey it to build a valuable and successful Multi-Product Platform company
- Your value propositions are the answers to “So What?” You can also phrase them (using the F.A.B. framework) as the *Benefits* that you provide as a function of your *Features* and how they manifest as *Advantages*
- Identify and test your value propositions before embarking on a new venture, and orient all your future activities around building and maximizing value
- MPPs create value in their platform and in their multiple products. Define your *Platform-Product-Market fit* to ensure they’re coordinated.
- Create value continuously, even though it’s only quantified discretely.

Chapter 4. Generating a Valuable Idea

All companies start with an idea. Great companies start with a great idea. But how do you find that great idea?

Let's start by pointing out that "find" is the wrong verb for what's required of you. Great ideas for innovations don't just exist, sitting out there in the world, waiting for you to stumble upon them. The skill that's necessary isn't simply to distinguish the transformative idea from the pedestrian one. Instead, what you need is the ability to cultivate creativity, to synthesize useful innovations from the stew of technological progress and unsolved problems, to come up with a vision of a valuable future that is enabled by not-yet-built technologies, to map the most valuable version of that future and figure out how to get there.

It's a big ask. You might wonder how coming up with a great idea can be called a skill. Isn't creativity ephemeral, un-harnessable, like lightning that eludes the bottle?

Our firm, Flagship Pioneering, has built a reputation for creating transformative Multi-Product Platform companies whose technologies and modalities seem outside the realm of reason when the companies are launched. Over the years, we've learned that it is, in fact, possible to institutionalize creativity. It's possible to repeatably come up with farfetched but valuable ideas – beyond the frontiers of what has been done and often beyond what is deemed doable – and turn them into reality. We've refined these methods over the years, across dozens of companies, and taught them to hundreds of students who have participated in Flagship's summer fellowship program.

The Flagship way of creating companies isn't the only way. You don't have to innovate from whole cloth to be successful in biotech. But we think our methods contain lessons worth learning. Whether you're a serial entrepreneur preparing your next venture, an academic scientist hoping to translate your laboratory discoveries into cures, a biotech executive developing your company's strategy, or an investor trying to participate earlier in new company origination, you'll need to imagine a detailed future, identify the valuable destinations within it, and forge a path to them. You'll be better off if you're able to do it well.

In this chapter, we will explore how to come up with a valuable idea, starting with a reminder to stop and think before jumping in. We'll introduce the process of exploration as a way to build a value landscape, a detailed and actionable map of the future. We'll discuss how to explore productively (whether you're starting from scratch or already have an idea in mind), how to construct a good value landscape map, and how to turn it into a detailed and actionable plan for a valuable company. We'll end by taking you through an example exploration that led to one of our companies.

Ch4.1 Picking the Right Idea Matters

There is an opportunity cost to working on the wrong thing

As an entrepreneur, your first great risk is that you will choose to work on the wrong thing. It may be wrong because it doesn't have a path to making a big impact. It may be wrong because it's too hard (if not impossible) to build. It may be wrong because the value propositions don't make sense.

But regardless, once you dive in, the company will require all of your time and attention. Working on it will preclude working on something else that might be better. The outcome of your choice may not become clear for years, but you will pay the opportunity cost upfront.

And if you realize, down the line, that there was a flaw in the original plans, there won't be much you can do. There is no pivoting in biotech. In tech, there is an entire mythology around founders who radically change their product when a first idea runs into headwinds. But in biotech, the costs of radically changing direction are too high, the time needed to go back to the drawing board when something doesn't work is too great, and the expertise needed to get down a path is too specialized. You don't have room or runway to recover from a "holy crap we were entirely wrong about everything!" moment.

It's worth your time, then, to think rigorously, to explore fully, and to identify the most deserving idea before you start.

Good ideas are rare but also abundant

The good news is that good ideas, while rare, are also abundant. Yes, there are many low-impact, derivative, or flawed ideas out there, but transformative, valuable ideas are abundant in every field, with the supply continually renewing itself as we make discoveries and build new technologies.

In this respect, ideas are like the blue jelly beans in the mixed self-serve barrel at our tony neighborhood market. They are my four-year-old's favorite. In our countless scooping adventures, it's become clear that while a random bean is unlikely to be blue, there are literally ten thousand blue beans in the barrel. We just need a way of picking the blue ones alone!

If you shift from a mindset of scarcity to one of abundance, you'll learn not to fall in love with your first idea. Love at first sight strikes many first-time entrepreneurs. The glow of the moment of discovery, however bright, may trap you on a path with technical constraints and value creation challenges that you'll wish you had avoided. You can always generate more ideas – so don't settle for the first that comes along.

Take the time at the beginning to think, read, and talk. Take advantage of this time before the fixed costs of a lab, the burn of a budget, and the drag of expectations weigh you down. Your time will never be as cheap. Use it wisely.

Ch4.2 Exploration and Value Landscape Maps

Okay, you might say, I get it. Don't rush into a new idea. But if I have the agency to come up with a good idea, how do I do that? And I'm supposed to stop and think about *what*, exactly?

Good questions. Let us introduce you to the practice of Exploration.

Exploring a new land – the future – and creating a value landscape

Exploration is a process of inquiry through which you create a detailed mental map of the future. You want to chart the landscape of what the world will look like, should you build the things you might choose to build. Having a value landscape lets you imagine all the possible futures and figure out which are most valuable before you chase them.

The contours of the landscape are value. The (x,y) coordinates describe the *what* – the technologies you might build and their possible applications – and the (z) coordinate describes the *so what*, or the strength of the value proposition. A topology emerges. Some solutions are connected to one another, readily reached with similar technology, while others are far apart. Some valleys are deep and therefore very valuable; others are shallow. (As in a potential energy landscape, the optima of a value landscape exist at the minima.)

From a value landscape you can posit hypotheses for solutions that would create substantial value, and that therefore merit the formation of a venture. In your exploration you will seek to prove or disprove the hypotheses by exposing them to the world. Putting forth hypotheses gives the inquiry direction and deliberateness, as opposed to simply meandering randomly through nature.

We like the term *explore* because the process of creating a value landscape resembles the exploration of a new land. In your case, the new land is the future. Like a classic explorer, you venture into the unknown, knowing only what will be valuable should you chance to find it (a water route to the Pacific, a “lost” civilization, a better way to treat disease). Like a classic explorer, you broadly and open-mindedly catalog what you observe along the way (flora, fauna, geology, weather, unusual phenomena) because you don't know yet what might turn out to be important. Like a classic explorer, you're a hardy navigator and rugged survivalist who makes quick sense of the unexpected or unexplained, synthesizes many inputs at once, and blazes a path around obstacles.

To explore is to ask questions

Philosophically, when you explore, you're building a map of the un-built future, a world created in real time with the power of your imagination. Practically, what you're doing is asking questions and proposing ideas in a perpetual cycle. You ask questions about how things work and why problems remain unsolved. You come up with ideas for solutions, wonder how you might build them, and question how these new inventions

might change the dynamic of the problem. You ask open-ended “what if?” questions to spur creativity, and then challenge your emerging landscape with our favorite “So What?”

You ask these questions not just of yourself, but also of others, of science, and of nature. You talk to people and read scientific literature, you follow hunches and brainstorm, you learn how things are done today and how today’s systems break down, you find what is accepted as true and poke at where the dogma might be outdated, you apply knowledge gleaned from one domain to another, and you figure out where nature may have already tested hypotheses for you.

Answers are rarely clear, easy, or definitive, but every cycle of questioning and ideation deepens your understanding of the problems, the potential solutions, and the contours of the future. Every cycle sharpens your instincts about what ideas will have value and impact. Every cycle clarifies the waypoints and coordinates of your landscape. Ta-Nehisi Coates put it well: “You may not find the answer but every time you ask it *the question is refined*.”^{vi}

Explore. Ask. Refine your questions. Every query gets you closer to a proper value landscape.

A Value Landscape of the future provides clarity

Remember that your landscape reflects not today’s reality, but rather the reality of a future that doesn’t yet exist. You should assume that any new technologies are fully developed and any new platforms are fully realized. Try to identify all possible outcomes, their relative values, and the paths that lead to them from where you stand today.

You may not be right about the global optimum – and unless you experience the multiverse, you won’t ever actually know for sure – but with sufficient work, you can develop a reliable hypothesis about the value of a particular future and a reasonable hunch about alternatives, and you can make sure you’re not getting yourself stuck in a local minimum.

It is important to figure out which destinations are worth going to before you worry about *how* to get there. A valuable enough destination warrants uncommon effort, but no amount of practicality should entice you to pursue a path of little value.

The metaphor of the value landscape has a close cousin: the “idea maze,” popularized by Chris Dixon and Balaji Srinivasan of the venture firm Andreessen Horowitz.^{vii} The concept of the idea maze is that a founder can map the many branching paths and key decision points in their start-up’s future by using the history of similar products and earlier efforts in analogous industries as guides. Walking the idea maze is a way to watch future challenges play out via the past so you can prepare for them and learn which turns of the maze lead to chocolate and which to mousetraps.

In our stage of exploration and ideation, though, you can't rely on the past, since there is no prior. You don't yet have a tangible product in mind. There aren't any analogous companies. Heck, you haven't even formulated what the company will do! But the prescription to develop a model of the future world where your idea will exist is a good one. In essence, you must use your intuition and the data you collect from your exploration to build a future idea maze around each hypothesis of value in your value landscape.

Peer into the crystal ball, inhabit your value landscape, and familiarize yourself with its contours. You'll need to do this before you can answer important questions from investors, partners, and skeptics (like yourself!): What destinations are most valuable? What technologies should you build first to enable the most value? What experiments should you perform first to prove that your vision of the future is attainable? If you start down one path, what other outcomes remain in play? If one of your assumptions crumbles – for example, a technology doesn't work the way you expect – where will you end up, and how will you get to the next nearest most valuable destination?

You'll be confident in your choice of idea to pursue, knowing that you gave full consideration to the alternatives. All this, from creating a value landscape of the future.

We should note that you can do explorations at any stage of a project. Any time you're forging into the unknown – whether in the earliest pre-contemplation of an idea or in a fully formed company – it's useful to build a value landscape of the future so that you can anticipate what is to come and choose the most valuable paths. In this chapter, we focus on *de novo* ideation, but the same principles and practices apply no matter how advanced you are.

Now let's dive into those principles and practices.

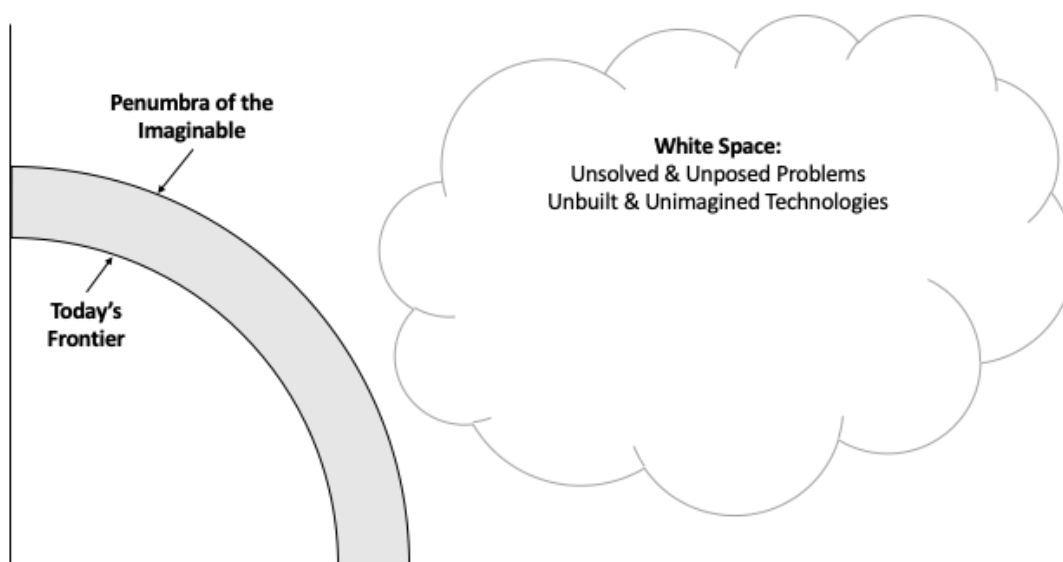
Ch4.3 Thinking Like an Explorer

Before we describe how to think in an exploration, it's worth examining the general shape of knowledge. (See sketch below.)

In any field at any time, there is a frontier of what is known. This frontier is both technological and applied. It encompasses the tools, equipment, and molecules that exist today and the accepted and acceptable uses of those technologies. Beyond today's frontier is the Penumbra of the Imaginable, the shadow cast by what *is* onto what *will become*, the extensions and modifications to technology that will inch the frontier outward, the applications adjacent to the possible, and the problems we can imagine solving. This is where conventional innovation happens, where most start-ups start.

But beyond the penumbra of the imaginable lies uncharted White Space. White Space is where the big unsolved problems reside, alongside the problems that have never been posed because they seem so unreasonable and unsolvable given where we are today. White Space is also home to all our not-yet-built and not-yet-imagined technologies. White Space is where the best ideas start and where the most transformative companies take root – originating either with an unsolved problem or an uncreated technology, and ultimately pairing one together with the other. White Space is well suited to the risk-taking and optionality of a Multi-Product Platform. White Space is where you should be exploring.

The Shape of Knowledge



To end up in White Space, it helps to practice problem-based thinking and to extrapolate the implications of today's discovery and technologies.

Practice problem-based thinking

To practice problem-based thinking means to structure your exploration around unsolved problems, to identify and seek out the biggest and most valuable of them. From the problem, you have free rein to sample various technologies – real or as-yet imagined – as potential solutions. Define the problem first (*so what*), then imagine various ways it might be solved (*what*), and only then work your way backward to the practical building of solutions (*how*).

Problem-based thinking keeps you from two common traps. First, it helps you avoid the “square peg, round hole” problem, where you’re locked into a technology and must find a perfect fit application or otherwise force your solution onto problems that might not be well-suited to it.

Second, problem-based thinking tends to lead to more creative ideas. If you start your exploration from the frontier of knowledge and its rules and limitations, you typically get locked into easily imaginable next steps. If they’re easily imaginable to you, they’re easily imaginable to others too, including to others who have deeper pockets and more resources and will beat you to the finish line, if not the starting line.

It may come as no surprise that research consistently shows that the majority of successful entrepreneurs are in their forties (despite the dominant narrative about young college-dropout founders).^{viii} You need to *know* something to create a useful product and a successful company. By middle-ish age, most of us will have learned something about a set of specific problems faced by our industry. And when we are immersed in problems, we tend to see them clearly enough to name them, craft solutions, and recognize how to apply new technologies to them.

In 2016, we started an exploration around a problem in the world of gene therapy: that viral vectors for delivering therapeutic genes are consistently recognized by our immune system, which limits their efficacy, the types of tissues they can access, and the number of times they can be dosed. Starting from this problem, we imagined that the perfect solution might be a virus (naturally efficient at packaging DNA) that was not visible to the immune system. This thinking led us backward to an emerging field at the frontier of virology, the commensal virome, or the viruses that co-exist with us much the way microbes form our microbiome. Scientists were just beginning to catalog the existence of a commensal virome using snippets of nucleic acids found in human tissue samples. We wondered whether any of these viruses that co-evolved with us might actually be immunologically silent, meaning that our bodies might have learned to ignore them. If so, then they could be the perfect viral vector solution to our problem. Mind you, no one knew how to culture these viruses in the lab, or whether it would even be possible to do so. No one knew whether the viruses could be engineered to carry payloads or travel to specific places in the body. These questions were beyond the penumbra of the

imaginable – you only could get to them if you started, as we had, in the free-wheeling world of problem-based thinking.

Extrapolate the Implications of Discoveries and Technologies

Sometimes, though, you must explore with a discovery or technology in mind, perhaps because it come from your lab, or you're at a later-stage company with an already-built platform, or you've identified a worthy problem and are working backward to consider how you might solve it. If you find yourself grounded in a discovery or technology of today, how do you burst through the frontier, past the penumbra, and out into the unexplored?

The simplest recommendation is to identify the implications of the discovery or technology and extrapolate them as far into the future as possible. This requires suspending disbelief. It requires optimism about not-yet-built features of the technology. It requires putting aside what you think you know about its applications and limitations.

Let's take an example of how you might extrapolate the implications of a discovery when reading the scientific literature. We scientists learn early on how to read papers. We learn to read them in a particular way – skeptically and retrospectively, assuming that assertions are false until we thoroughly vet the data and pore over the historical references. This approach is good for science but bad for exploration; good for building the edifice of knowledge but bad for building new ideas; good for gravitating toward today's truth but bad for launching a new future.

Consider what happens if you read the literature in a believing, prospective way instead. Consider how your mindset changes if you start from the premise that the paper's claims are true, not false. Instead of asking yourself, "Do I believe this?", ask, "If this assertion is true, what are its implications? What other things may it be possible to build? What fundamental features are unlocked and what dogmas upended by these results?" Extrapolate forward from the known into the not-yet-known.

It's a simple change, but it is deeply conducive to creativity. Like the classic "Yes, and!" in improv comedy, it creates a momentum of possibility that can get you from a technology out beyond the penumbra of the imaginable.

You may ask (reasonably): But what if the paper turns out not to be true? Yes, the reproducibility rate of academic literature is low, but that's not of concern now. You're not trying to build a company around this one paper; you're using it as a launchpad to spur your creativity and find the highest value futures. Once you identify the valuable destinations and decide they're worth pursuing, you can ask how best to get there, which will likely require a platform with many technological advances from many sources. If you lead with skepticism and retrospection, you'll never be able to see beyond the limitations of today.

In 2015 we came across literature that described mind-bending experiments in which people given a funny-flavored milkshake along with an immune-suppressive drug would

“learn” to suppress their immune system by sensory recall the next time they drank the milkshake, like Pavlov’s salivating dog. No one really knew how this inverse placebo effect worked, nor what you might do with it... create alternate-reality soda fountains? Instead of focusing on the practicality or the size of the effect, we extrapolated the implications of the observation to the nature of biology. If the results were true, what if, at some fundamental level, the wires of the nervous system and the wires of the immune system were crossing? What if nerves and immune cells were talking to each other in a molecular language? If so, then perhaps we could drug inflammation using molecules and targets from neuroscience, and vice versa. Perhaps, too, since nerves are everywhere in our body, other important cells and tissues like tumors and regenerative stem cells would be responsive to this language as well. Soon we were far past the restrictions of the paper – it no longer mattered what flavor the beverage was, or how many times the patient could recall the immune suppression. We were now launched firmly beyond the frontier and into uncharted White Space, where new neuronal targets and mechanisms might be relevant to otherwise intractable diseases.

You don’t need to be an expert already – you can become one!

Through the process of exploration, you will become an expert in a field; you don’t need to be one already to explore well. We – a protein engineer and an RNA biochemist – have successfully explored, come up with value landscapes, invented new technologies, and (sometimes) launched companies in cell therapy, gene regulation, computational protein design, microbial metabolism, artificial intelligence, chemical catalysis, neuroscience, RNA medicines, single cell sorting, gene therapy, DNA sequencing, cosmetics, and game theory.

What you do need are general skills like scientific acumen, polyvalency across scientific disciplines, curiosity, and intellectual rigor. You need to be able to enter a new field and rapidly come up to speed, perceive the frontiers and speak intelligently to the current experts about them, understand how the underlying system works and how to manipulate it, and draw on multiple domains to synthesize new insights. You need to understand what it will take to build technologies – what will be easy and what will be hard – so that you can accurately plan for and assess the feasibility of your innovations. These skills are not field-limited.

Ironically, we often find that those with deepest domain expertise struggle to explore in their field because they get trapped by dogma. For the already-expert, the limits of the frontier are often so incontrovertible – They’re facts! They’re in the textbook! – that any advantage familiarity may confer disappears in their inability to imagine possibilities beyond the familiar. If you’re on the outside, you face a steeper learning curve, but you’re not constrained by dogma. You have the luxury of seeing dogma as moveable parameters. You are free to challenge those beliefs with “ignorant” but powerful questions that often elude the practitioner.

Most critically, to explore well and map what you find onto a good, predictive value landscape, you need to understand value. You need a sommelier’s nose for value

propositions. You need to hone your ability to ask and answer “So What?” We hope this book will help you develop this all-important skill.

Ch4.4 Accessing your Creativity

Explorations require creativity. We all have the ability to be creative – some of us unleash Zeus-like bolts of inspiration daily, while others summon it with effort. Though we don’t have a failsafe way to guarantee you’ll access yours, we find that preparing your mind for exploration and being aware of common hurdles can help set you up for creative success.

Enter an Exploration State of Mind

The exploration state of mind is a Zen state, an open and absorbing consciousness that gathers and synthesizes torrents of information without slowing to classify or organize or restrict it in real time. The exploring mind freely associates ideas and follows them where they lead. The exploring mind creates and inhabits future worlds that don’t yet exist, incorporating technologies that aren’t yet possible and believing hypotheses not yet known not to work.

As in any creative endeavor, the route of an exploration can be circuitous, and progress is rarely linear with time. You’ll often feel that you’re not getting anywhere. That’s normal. Get used to it. The visual artist Jenny Odell, who teaches art to engineers at Stanford, advises her students to allot twice as much time to a project as they think they’ll need because half the time won’t feel productive. That downtime is actually when a prepared mind wanders, synthesizes information, and generates insights. It is essential to creativity.^{ix}

Once you hit upon an idea and feel the dopamine surge of discovery, it can be difficult to set it aside and continue to explore. With an idea in mind, you shift suddenly from the meandering unknown to purposeful convergence. It is comforting to finally have a clear set of questions to answer, a straightforward to-do list, and an obvious path to follow. It feels like progress, but beware the path of least resistance: narrowing your focus too quickly curtails your creativity.

When you start to feel as though you know what you’re doing on an idea, stop. Set aside those obvious next steps – they’ll wait for you. Force yourself to return to exploration, to being uncomfortable and uncertain, to searching out the next features of your value landscape.

Visual cues can help guide this process. Mind maps – connected diagrams of ideas that record the path of your exploration, whether they are made digitally or assembled with old-fashioned sticky notes and a whiteboard – can be a useful tool for keeping track of the various threads of exploration. Seeing the whole map can help keep you pushing outward rather than narrowing your focus too quickly.

Be Impractical

Often the biggest impediment to exploration is ourselves. What holds us back from freely imagining the future is our own reluctance to seem unreasonable.

Reasonableness sometimes takes the form of practicality, especially when it comes to money. Pro tip: Don't think about money during an exploration. Don't worry about how much things will cost or where you might get funding. Obviously, you'll need to raise money to do the work eventually, but if you constrain your imagination to what you think is practical, you'll never find the most worthwhile destinations.

It is natural to respond to the pressures of our environment. If the only funding you can imagine is a \$100k SBIR grant, you will only imagine problems that can be solved for \$100k. Over time, these limits will become ingrained and the muscles that let you think big will atrophy. If money worries seep into your exploration, force yourself to pretend that you have 100x more and see how your value landscape evolves. You will likely find that you come up with bigger, more transformative, more impactful ideas.

Bigger and more valuable ideas are more compelling to the type of investors you need to build a Multi-Product Platform, so you may find that you've imagined your way into more resources. But if not – if you have to adjust your tactics to correspond to the funds you raise – you will still be working toward solving the biggest and most valuable problems.

Be Unreasonable

At other times, the desire to be seen as reasonable is driven by fear. We are afraid of sounding foolish and of being proven wrong when our imagination takes us to unusual places.

Science selects for us to be reasonable. From early in our training, we learn to stick to verifiable facts and to defend our conjectures with data. Even academic grant proposals, the supposed engine of unfettered inquiry, are judged on preliminary data that has already been generated, all but ensuring reasonable advances in the frontiers of knowledge.

But if we stick to what is reasonable – even if we allow ourselves to tiptoe one or two steps beyond it – we severely limit the output of an exploration. Reasonableness restricts the possible futures we can imagine and marginalizes our creativity. It shrinks the value of what we might find and increases the competition for that value by guaranteeing that many others will have the logical same idea. As entrepreneurs, our advantage lies in our freedom to be unabashed and create the illogical thing that isn't imagined because it isn't thought possible.

If you find that being unreasonable is a challenge and you cringe at making statements that are not yet known not to be true, remember that you only need to suspend disbelief during your exploration. When you speak to the outside world, you don't have to say, "I

think X [unreasonable statement] is true.” Instead, you’re arguing, “*If X were true, it would be incredibly valuable.* And, therefore, it’s worth trying to make it happen.”

Noubar Afeyan, noted serial entrepreneur and inventor and the founder and CEO of Flagship Pioneering, likes to say: How can we expect unreasonable innovation and unreasonable returns from reasonable people doing reasonable things?

Be unreasonable.

Ch4.5 You can do Explorations for pre-formed ideas too

Today, few companies in our industry start with exploration. Many of you will be entering entrepreneurship from a more traditional starting point. Perhaps you’re an investigator or a student in an academic lab looking to commercialize a research project. Maybe you’re an entrepreneur-in-residence evaluating proposals from a local university technology transfer office. Maybe you’re an investor scanning cutting-edge literature and thinking about how to turn these nascent ideas into the next big company.

And you may be skeptical. Sure, it’s nice to do unfettered exploration, but I’m constrained! I have a fixed starting point! I need to find a nail for my hammer, a square hole for my square peg!

It is dangerous to be so limited at the start. You are likely familiar with the framing that characterizes some innovations as pushed to market and others as pulled forward by the market; that says there must be a match between the solution and the problem for an innovation to succeed. Typically, when we start with a pre-formed idea, we search for the problems that are suited to the innovation. Rarely do we then change our frame of reference, stand athwart the problem looking backward, and ask: If this were my problem to solve, what technology would I choose to solve it? And if we’re honest, even more rarely would the answer be the technology we started with.

If you limit yourself early to a single innovation, you won’t uncover the full value landscape, and you will miss valuable opportunities.

Fortunately, the same skills that we use for unfettered exploration can be applied to a set starting point. Doing so requires you to set aside the idea or technology as it exists today. It requires that you deconstruct it into its component parts, ask “So What?” of each one, and be ready to reassemble them into something that barely resembles the original. In other words, you must be willing to open your mind, practice problem-based thinking, extrapolate the implications of your invention, and be unreasonable. If you can do this, you will walk the space with clear eyes and a ready mind.

This approach is embodied in a wonderful piece of immersive contemporary art by Damián Ortega titled “Cosmic Thing.” Upon entering a plain white room, you see a Volkswagen Beetle fully dismantled, suspended on transparent wires, as though a point charge has been detonated at the center and all the pieces are being blown outward.

You walk around and contemplate the different pieces that make up the whole, how they fit together, how they complement each other.

Imagine that the VW bug is your pre-formed idea. It may make sense; it may be a transformative product. But ask yourself: What if we examined all the components of this idea separately? Can we find where the innovation resides? Can we identify the biggest So What that might point us to greater value?

Maybe the innovation is the vulcanized rubber in the tires that provides strength and durability with less weight. Maybe the innovation is a more compact and fuel-efficient internal combustion engine that lets you drive more miles for less gas, or a catalytic converter that simultaneously oxidizes unreacted hydrocarbons and reduces nitrous oxide pollutants. Maybe the innovation is actually the network of human connectivity enabled by the car, which might revolutionize society the way the telegraph, the steamship, and the transcontinental rail line once did. Or maybe the car is a solution to a real problem, just not the *ideal* solution.

See how that happened? From one idea that came in fully formed, you've suddenly built a value landscape with multiple future opportunities, each with a core innovation that is tied to a component of what you're starting with, each with an orthogonal set of applications and value propositions.

It may turn out at the end of this exercise that the most valuable hypothesis is the original one. You may decide that the biggest So What is the full package of a car for the masses. But you will be better off for having done the full exploration and created your value landscape.

This is the process by which you apply exploration principles to even the most entrenched pre-formed ideas. It can be very hard, especially for scientists who have spent years in the lab discovering and advancing technologies. But don't be swayed by sunk costs. Instead, use them to your advantage. You have a real advantage in imagining a future value landscape – you know what emerging new technologies are possible because you've built them!

Ch4.6 Getting from a Value Landscape to a Company

As you explore, your value landscape will emerge. You will have created a mental model of the future, the different technologies you could build and the problems you could solve with them, and the relative value of those solutions. Now what? How do you ensure your value landscape is in good working order? How do you use your value landscape to choose, design, and plan a valuable company?

Make sure your value landscape is accurate and precise

You'd like your value landscape to be correct. But because it's in the future, you can't be certain. The best you can do is make it as accurate and precise as possible, preferably without picking up a pipette. It must be precise about the technologies you

need to build and accurate about how to build them; precise about their applications and accurate about the advantages you have over other possible solutions; precise about the value propositions and accurate about their strength and thus their value.

One way to build accuracy and precision into your value propositions is to read the literature strategically, as a repository of experiments that have already been done. It's remarkable how often you'll find a published result that provides the answer you need from an experiment that originally asked a different question. In the simplest case, in this age of open science and multi-omic comprehensive experiments, you may find your answers in public databases. In more fun sleuthing, you sometimes find that someone performed your experiment without realizing it – that the answer you seek is hiding in the control group of a supplemental figure.

A second way to refine your value landscape is to query nature. Where has nature already performed the experiment for you? Is there evidence from human genetics that a particular genotype protects from a disease? Are there diseases that have surprising side effects that may suggest a particular mechanism? Do observations from different species suggest new ways to think about modalities and diseases? While these observations may not be definitive proof of any hypothesis, they can be useful signposts that your hunches will hold up.

A third way to make your value landscape more accurate and precise is to talk to smart people about it. We don't mean you should seek the blessing of "Key Opinion Leaders" as a talisman against failure, but rather that you should challenge experts to punch holes in your value landscape. Their feedback is infinitely more valuable to you than any imprimatur of legitimacy.

One class of expert is the technical expert who is deeply versed in the domain of science or technology that you're entering. Challenge technical experts to tell you why your ideas will fail, spar with them over whether you can build your imagined platform, learn from them how high the hurdles are between different pools of value. Identify the soft spots in your value landscape. "This will never work because X," says the first expert. The second says, "It will never work because Y." A third tells you "Z." As you revise your idea in response to feedback (or disregard the feedback when you recognize it as dogma), you'll find that the list of objections shrinks. Soon you will have in hand a refined idea with its holes plugged, and with the only remaining unknowns those things that have never been tested.

A second class of expert is the value expert, a practitioner of the art, a doer of what you want to do, a user of existing methods. Value experts will tell you whether the benefits you propose matter, whether the value propositions resonate, and whether the minima in your landscape are shallow or deep. Mine them for the drawbacks of today's technologies to identify the step-change advantages of your platform. Push them to point out the real problems they face and listen for new angles you haven't considered. Tease from them a hint of what demonstrations would leave their jaws on the floor, and therefore what data you should strive to generate.

Probe, listen, and adjust your value landscape accordingly. It will become stronger, and you will be better prepared because of it.

Design a Company on top of your Value Landscape

Notice that we've said very little about companies in this chapter. Explorations give rise to a value landscape of the future whose contours define the many possible problem-solution pairs and whose depth describes their value. Now is finally the time to turn your value landscape into a company.

Look at your value landscape. You can choose what to pursue, confident that you've considered many alternatives and are not walking blind into a low-value future. Your eyes are open to the risks and uncertainties you'll face down any of the paths. There is no "best" or "only" option, but rather a big, compelling, and worthwhile thing (or multiple things!) to devote your time to.

Next, begin to construct a Multi-Product Platform on your value landscape. Make sure the valuable opportunities are amenable to an MPP, not one-off solutions to one-off problems. If they are, what type of MPP fits best? A New Modality MPP, New Technology MPP, or New Insight MPP? What is the value hypothesis for your MPP? (Something of the form: *It would be valuable to create platform X that can create multiple products Y to solve problems Z.*) Can you begin to phrase a compelling Platform-Product-Market fit for different families of products?

One value landscape may have room for multiple companies, each pursuing individual valuable opportunities. Or one company may instead pursue multiple valuable opportunities accessible by a common platform. You may not be sure yet – you may first want to test the various assumptions that went into building the landscape before determining which opportunities make the most sense.

Be careful not to over-specify your company's future path at this stage. While it is essential that you be precise in the problems you aim to solve and the types of products you could develop, there are risks to locking in on specific products too early. For one, you may find yourself chasing opportunities that are addressable with today's version of the platform, and therefore miss out on opportunities that only arise with platform technology development. At the other extreme, you may set yourself an impossible (or maximally difficult) goal that, by pursuing single-mindedly, causes you to miss valuable destinations along the way.

From this foundation, you can begin to articulate a value creation plan for the MPP. What does the platform entail, and in what order should you build its features? From the questions that remained unanswered in your effort to make the landscape accurate, you will have a finite set of testable but as yet untested hypotheses that will become your experiments when you enter the lab. Which are the most pressing? Which, when answered, get you closest to the biggest opportunities? Which might close off certain paths but open others?

What team and expertise do you need to perform these experiments and enable the platform? How will you protect the platform and products from competitors? What resources will you need to build the MPP? What is the next meaningful value inflection, and are you sure that you are creating value continuously?

Through a good exploration, you will generate an accurate and precise value landscape of many possible futures. From the comprehensive mental map, you will identify the relative value of various opportunities and craft clear value propositions for the most prominent. Now, by layering an MPP onto this landscape, you will have a strong rationale for the company that it makes sense to build and a plan for what the company should do to create value. With this map and plan in hand, you are ready to venture forth and build the future.

Ch4.7 An Example Exploration: Rubius Therapeutics

There are many elements in a successful exploration and many permutations of those elements that can lead to success. To show how they came together in one case, we'll share the example of our exploration that led to Rubius Therapeutics, a company developing a new class of cell therapies based on engineered red blood cells.

When we began the exploration that led to Rubius, in 2013, we practiced problem-based thinking. We wondered whether we could make cell therapies safer, recognizing that the inherent risk of emerging cell therapies was that you lose control over the cells once they enter the body, where they can divide uncontrollably, attack healthy cells, or take on new, unplanned functions. A second problem was that an ideal cell therapy should be able to address many diseases, not just blood cancers (for still-new CAR-T cells) and local grafts of skin and cartilage.

We gravitated to circulating cells, like red blood cells and platelets, that travel widely throughout the body. Interestingly, these cells lack a nucleus. We imagined what it would mean to have a genetically engineered cell therapy that didn't carry any genetic material. Would it be safer for patients? Would it let us treat diseases that couldn't be treated today?

We explored widely and with an open mind. We learned about how enucleated cells lose their nuclei in nature and examined whether we could artificially induce other cell types to lose their nuclei. We puzzled over why certain species of birds and reptiles have nucleated blood. We explored what might happen if we combined natural cells and synthetic particles in interesting ways, and considered how we might isolate engineerable cells from people. We sought insights from the frontiers of stem cell biology, learning the latest factors that control differentiation during development, and from cancer biology, learning what makes cells immortal.

We were unreasonable. The notion that we might manufacture red blood cells from stem cell precursors seemed both absurd and already falsified. Governments, academic labs, and companies had been working on protocols for synthetic blood for years and had always run up against physical limits on the production of the quantities needed for transfusion. High-profile review articles argued that manufacturing red blood cells at any scale was both impossible and impossibly expensive.

But we set those limitations aside. What if we could make exceptionally potent blood that didn't require giving transfusions to patients, but instead could be effective as small-volume infusions? And besides, no one had actually tried to industrialize red blood cell culture. (They had no reason to!) Over the past forty years, bioprocess engineers had figured out how to boost the yield of recombinant proteins from CHO cells by 50-fold while reducing the cost 100-fold.^x Why couldn't something similar happen here? Regardless, it wouldn't be worth worrying about *how* to make the cells until we figured out whether the destination was valuable enough.

The question of what could be done with these cells was unanswered and therefore ripe for unreasonableness. What would it mean to make therapies that persist for months? Could we capture and remove toxic molecules from circulation? Might the protein-carrying cells interact with other cells in the body differently than recombinant proteins do? What biology might be newly druggable as a result? Could we put proteins inside the cells to shield them from the immune system? Could we get beneficial synergy from multiple agents on the cells at once? Would the cells get out of the vasculature, and if so in ways that we might manipulate therapeutically? None of these questions had an answer yet, but we imagined what might be possible if we could do it.

We sought answers to our questions where they may have already been addressed – literature, experts, and nature. From the literature, we learned that for decades physician-scientists had mixed small molecule and peptide drugs into red blood cells and platelets and had been able to extend the circulation time of the therapies and provide clinical benefit. This taught us that our hypotheses about the clinical value of these cells might be right.

From technology experts, we learned that red blood cells get rid of all their proteins and protein-making machinery late in maturation in order to load up on hemoglobin. The experts were skeptical that any therapeutic protein would stick around in the cells, but it hadn't yet been tested experimentally. We put that down on the list of essential experiments we would have to do ourselves.

An unwitting value expert, the CEO of an Italian company that was developing drug-cell mixtures, complained, in recorded comments on an investor call, that until they solved the one-patient-one-dose conundrum, there wouldn't be a path to broader use. That taught us that an off-the-shelf, allogeneic, universal donor-based therapy would be highly valuable.

From nature, we discovered a forgotten function of red blood cells mediated by an overlooked receptor in the complement system. It turns out that in addition to delivering oxygen to our tissues, red blood cells act as garbage trucks for blood vessels, grabbing immune complexes from circulation and carrying them to the liver for destruction. In fact, we even found one report of patients being treated for an acute immune complex disease with a blood transfusion. That red blood cells perform complex biological functions naturally made us confident that our conjectures about new functions were less of a leap than they had first seemed.

The value landscape made it clear that this idea could form the basis for a transformative Multi-Product Platform company with multiple families of functional red blood cells, each with a strong Platform-Product-Market fit, enabled by a core technological ability to engineer and manufacture the cells. We began to layer on a plan. If we wanted to produce the cells ourselves from precursors, how would we modify them? If by viral transduction, with what agents? If by chemical modification, how and how controllably? What would an autologous therapy look like compared to an allogeneic therapy? What technologies would we need to implement first? What results

would create the most value, and how should we ensure that we created that value continuously?

Our exploration led to a value landscape for a future where enucleated cell therapies could be used to safely treat many otherwise untreatable diseases. From this landscape, we conceived of a multi-product platform with a clear set of value propositions, a plan for platform development, and testable experimental hypotheses to validate our hunches. We could launch the company confident that we had considered all of the alternatives and were prepared, as we'll explain more in the next chapter, to create value in the lab.

Key Takeaways

- There's a steep cost to working on the wrong thing, and good ideas, though rare, are abundant. Try to imagine all the possible futures to figure out which are most valuable before you chase them.
- Through the process of exploration, create a value landscape of the unbuilt future. Use the map to generate hypotheses of what would be valuable and what is worth building a company around.
- Explore White Space – beyond today's frontier and the penumbra of the imaginable – even if you're starting from a pre-formed idea.
- Think like an explorer by practicing problem-based thinking, extrapolating the implications of known, and being unabashedly unreasonable.
- Make your value landscape as accurate and precise as possible by reading literature strategically, interrogating nature, and asking smart people to punch holes in it.
- Finally, layer your Multi-Product Platform company onto your value landscape.

Chapter 5. Creating Value in the Lab

Let's take stock of where we are. You've finished your exploration, you've created your value landscape, and you've done all the thought experiments you can think of. You've chosen a valuable multi-product opportunity and structured a multi-product platform company on top of it. You've developed a clear plan for creating value in the company, including a set of testable but as-yet-untested scientific conjectures. Good news – you're finally ready to pick up a pipette and do experiments!

But now what? You can do almost anything – how do you choose which experiments to do to create the most value? The world is big and the possibilities are semi-infinite, so what should you tackle first? What should you try to accomplish? What results should you prioritize? How do you maximize the value of your multi-product platform when you enter the lab?

The answers matter because experiments are expensive. Once you're at the bench, time starts to cost real money. Every experiment must be definitive; every investment should provide the most valuable return possible.

Fortunately, the same value framework we've been using can guide us in the lab. A consistent and cogent understanding of the value propositions of your technology and the landscape of future value unlocked by your platform is the best guide to doing the right experiments. Even if you aren't currently launching a new venture in a dingy basement, the principles can be applied to ongoing research projects to help you maximize innovation in your company or academic lab.

In this chapter, we'll discuss how to think about your experiments through the lens of your value propositions. We'll teach you how to apply the principles of Scientific Marketing to design, carry out, and describe your science for maximum impact. We'll push you to think about early experimentation as a time to prototype your idea and expand its potential, as a cornerstone of the search for value rather than as a capstone of exploration. And we'll explore how these approaches will affect your behavior as an executive, a manager, and a leader.

Ch5.1 You have to do the RIGHT experiments

The early days of a company are busy. You will expend tremendous physical and mental energy. Thermodynamics teaches us that when you expend energy, you can either spin your wheels (metaphorically) and create heat, or turn literal wheels and perform useful work. To make sure you create work, not heat, you have to do the right experiments.

The right experiments are those that create the most value. The right experiments are those that demonstrate your value propositions, answer the biggest open questions about your platform, and unlock the largest pools of value. The best way to make sure you're doing the right experiments is to work backward from your value propositions.

Work backward from your value propositions

The data that you generate in the early days matters: the early results will guide what type of company you become, what problems you try to tackle, and how readily you can recruit talent and investment to your cause. But the data doesn't speak for itself; it can only make sense in the context of your value propositions. Therefore, you should preview your data – anticipate your future results – through the lens of your value propositions.

Recall from Chapter 3 that your value propositions are the answers to that deceptively simple question, “*So What?*” Your value propositions are the benefits that your innovation provides to an end user, as a function of its features and advantages.

So work backward from your value propositions. Craft the story you want to tell that will compel people with the benefits you provide. Create slides with descriptive titles but blank content. Lay out the case, centered on your value propositions, for why people should care about your platform and its products. Doing this should feel natural by now. Your exploration in Chapter 4 will have made you comfortable running detailed thought experiments in an unbuilt future. And this is just the implementation of the work you began in Chapter 3 to orient everything around your value propositions.

Now – what are the results you need to complete the picture? What are the figures that will prove the declarative titles on your slides? What are the experiments that are essential to demonstrating your value propositions? These are the right experiments. The right experiments are pulled into existence by your value propositions. Do them.

Of course there's room for revision and reversion. If the experiment surprises you or if the effect turns out to be null, you're not locked in. But starting from *So What* and letting the answer guide your experiment is the way to make sure you stay oriented around your value propositions.

Data drives your value, but your value propositions drive your data. Rather than getting the data and then using it to tell a story, let the value propositions define the most compelling version of the story. Then figure out what data you need to complete the arc.

Design experiments to test your value hypotheses

Remember that your value propositions are actually hypotheses: “We hypothesize that our platform’s Features will let us do certain things better than our competitors, and that these Advantages provide a Benefit to the end user.” This is a testable prediction, and the time to test it is now.

Build the essential Features. Show that these Features provide Advantages over today’s methods. Prove that these Advantages provide Benefits to an end user.

Importantly, the best experiments don’t build incrementally toward a single product or long-range goal. Instead, consider the sequence of advances that will be necessary to create the valuable future you’ve mapped. Focus on the results that best demonstrate that you’re on your way, that give you the biggest inflections, that unlock the most important doors.

These results will differ according to the type of MPP. For a new modality MPP, you need to prove that you can make the modality and that it functions; for a new technology MPP, you need to show that the technology lets you find better targets or make better molecules; for a new insight MPP, you need to show the insight is relevant to disease (see Ch3.5 Creating Value in an MPP). But keep your eye on the benefits. If the benefit is transdermal uptake, show transdermal uptake. If the benefit is duration in the body, show duration in the body. If the benefit is delivery to a particular tissue, show delivery to the particular tissue. If the benefit is scalability, show that you can scale. If the benefit is speed, show speed. If the benefit is specificity and therefore safety, show specificity and therefore safety.

The value propositions don’t have to be solely in the realm of science. Nimbus Therapeutics is an excellent example of a company that launched with three fundamental hypotheses, only one of which was scientific.^{xi} The scientific hypothesis was that computational chemistry had advanced sufficiently that Nimbus (with its partners at Schrödinger) could use software to drive the discovery of new molecules, instead of just as a tool to optimize scaffolds from conventional screens. Computational molecules would be faster to make and more specific to their targets.

The other hypotheses were related to operational and business strategy. Nimbus hypothesized that it could build a wholly virtual multi-product platform company with no physical lab space, which would make it quite a bit cheaper to run and more flexible than a conventional lab, and that it could design a nested corporate structure to facilitate monetizing multiple assets without having to sell the whole company.

Nimbus, which recently celebrated its tenth anniversary as a public company, tested these hypotheses early. It developed new methods and pushed multiple first-in-class small molecules to the clinic, all with a highly virtual team, and sealed multiple partnerships, including a sale of a subsidiary and two substantial licenses, all while maintaining control of the core platform and company.

Make sure you are working on the Right Stuff

The early days of your company – especially the part about knowing your goals and letting the goals drive the technology development – have interesting parallels to the earliest phase of space exploration. In the 1950s, the US Navy was developing supersonic aircraft that could fly at the edge of space. Chuck Yeager and other test pilots were the daredevils, the bronco-busting cowboy heroes of the post-war technology age.

But when Sputnik happened, the game changed. Control of the upper atmosphere was not good enough. Now the mission was to get a man into space by any means necessary, and soon. The Navy proposed to develop its jet planes to fly at higher speeds and higher altitudes, which would require rockets that were still three or four years out. A new agency, NASA, suggested a quicker solution: shooting a manned capsule into orbit atop a rocket.

The NASA proposal was a radical departure from what everyone had conceived of as “space flight” up to that point. As Tom Wolfe chronicled in his 1979 book, *The Right Stuff*:

The man would not be a pilot; he would be a human cannonball. He would not be able to alter the course of the capsule in the slightest. The capsule would go up like a cannonball and come down like a cannonball, splashing into the ocean, with a parachute to slow it down and spare the life of the human specimen inside... The Air Force named it the MISS project, for “man in space soonest.”^{xii}

The test pilots were dismissive – this wasn't even real flying! – and the Navy brass was aghast that years of development and breakthroughs would be set aside. But value came from getting a man into orbit. Fast. Period.

So set aside your perfect solution. Set aside a single vision of a final product and a fully realized platform. Find the clarity of purpose that comes from knowing your value propositions and letting them dictate the scientific directions. Capture the urgency of post-Sputnik space exploration. You're under time pressure. Every experiment must be meaningful. You have to create work, not heat. Think of *The Right Stuff* to make sure you work on the right stuff.

And who knows – once you get into space, a whole new world of value may appear.

Ch5.2 Scientific Marketing

You may find it helpful to think of your task in the early days of your company as Scientific Marketing.

In therapeutics, we don't know that the molecule we make is *actually* going to work until very late in the process, often after it has been taken by thousands of patients, even sometimes post-approval. And this uncertainty is especially true in the early days of a multi-product platform company, where you are developing a new technology or modality that has no precedent.

Much of the future you envision is not yet real. Your job is to build the case for this future and convince a smart, skeptical audience that it may come true, and that when it does, the implications will be massive. There is a subtle but important difference between assertions that are not yet known to work and those that are not yet known *not* to work. Scientific Marketing emphasizes the not yet known *not* to work and uses data to elicit excitement around this potential.

Marketing has a dirty reputation in scientific circles, where it's seen as a slimy tactic used to obscure what should be an objective reality. But this reputation is undeserved. Marketing is not only an essential skill, but one that is deeply embedded in the most august academic institutions, practiced by the most famous professors. These masters of Scientific Marketing know how to capture the interest of the public and the trade press; they know how to design experiments and talk about their results in such a way that people "get it" and tell their friends.

Remember that the data may be clear, but the implications are not. The implications of the data are for you to formulate and for you to describe. Scientific Marketing is about knowing which implications are most meaningful, and then generating and describing your data in ways that maximize its impact, its relevance, and its value.

Here are some tips:

You get to decide what your data means

The starting block for Scientific Marketing is that you get to decide what your data means. You get to determine what your experiments show. You have the flexibility and the imperative to choose the implications that are most valuable to your story, and to frame the data accordingly.

A single piece of data can mean many things. Let's say you put your molecule into an cellular assay and you see 2x stronger response than in a control sample. What does this mean? If the control is a molecule used in treatment today and the assay is the state-of-the-art standard, maybe you cast the result as preliminary proof that your molecule is more potent than what is currently used. At the same time, your molecule is not just a one-off: it is the vanguard prototype of a whole class of similar molecules enabled by your platform. So the cellular assay result may suggest that the common

mechanism of your new class of molecules works. Furthermore, if the molecule was hard to make, or generally believed to be impossible to make, the fact of its existence and functionality proves that the modality is makeable. If you suspect that your molecule might be resistant to degradation, performing the same assay in serum for an extended period and showing that the molecule retains potency can prove this value proposition too. All this for the same simple plot of 2x stronger response than a control.

Whether you're producing the first steps to a future drug, a representative of future products, or a demonstration of a key platform advantage – or all at the same time – you have the agency to decide the implications of your data and to set the context for your results accordingly.

You control the goalposts and define success

Along with the agency to choose what the data imply, you get to define the goalposts for success. You should, of course, aspire to make the biggest possible impact – setting the goalposts far from where you are today – but you should foresee the smaller but important victories along the way. If you come up short of your grandest aspirations, you can move the goalposts to an earlier point and legitimately declare victory.

In the early days of Rubius, after an early string of experimental successes, we hit a wall. Suddenly none of our red blood cell prototypes were working; none of the genes we put into the cells were expressing. We soon discovered a limitation in our gene delivery methods that restricted the size of the genes – they had to be smaller than a certain threshold in order to get into the cells efficiently. We called an emergency meeting, gathered the whole team (three people!), and together made a list of all the therapeutically relevant bioactive proteins that were smaller than our threshold. We then ordered the genes and put them into the next batch of red cells. All of them expressed. And all of them worked.

When we marketed that result, we didn't focus on the size limitation – we knew we could solve that problem down the line. We highlighted instead the high success rate with multiple classes of proteins relevant to multiple disease areas. Enzymes, binders, ligands, and antigens. Human proteins, microbial proteins, and entirely synthetic proteins. Proteins in the cytoplasm and proteins on the cell surface. We highlighted the incredible breadth of therapeutics that we could make as red cells and the robustness of our platform for producing all of them.

If you find yourself in this situation, think of it this way: you ordered dessert and expected a whole pie, but the science only gives you a slice. You can view your result as a disappointment – only one piece! – or you can sell it as an incredible opportunity to eat the most exceptional slice of pie in the world. Depending on what you need, the slice of pie can be cherry or pumpkin or rhubarb, plain or with whipped cream or served warm à la mode.

You get to define success. You get to set the goalposts for the implications of your results. So think ahead of time of the many possible outcomes, and be ready to adjust the goalposts to highlight your particular flavor of success.

The transitive property of data

In mathematics, the transitive property states that if $A=B$ and $B=C$, it follows that $A=C$. In a world where your platform is not yet fully realized, the transitive property is a useful tool for conveying future value.

In your early-stage MPP, it is cost- and time-prohibitive to build the full version of the platform upfront. As you choose your experiments, think about how the individual pieces of data can come together transitively to argue for the fullest picture of the future. How do the capabilities and prototypes you build hint at the complete version of the multi-product platform? Can you draw the arrow from $A \rightarrow B$, and separately draw the arrow from $B \rightarrow C$, to argue that you've de-risked your ability to get from A to C?

The individual demonstrations will necessarily be discontinuous, but that's okay. A single experiment may validate multiple small arrows proving different value propositions. Importantly, if some of the steps are known to work already, or are likely to work based on others' experience, it's less important for you to re-prove them, since you can argue that success is likely.

As your company grows and you generate more data, these arrows will connect and you will build the incontrovertible fully-realized arrow from $A \rightarrow Z$. In the meantime, learn how to use the transitive property to your advantage.

Anticipate the Coming Objections

Living in a not-yet-known-not-to-work world has certain benefits – it gives you flexibility and potential. But it also means there are alternative outcomes, many ways you could fail, and many reasons you may never reach your envisioned future.

Everyone you speak with – investors, executives, scientists, corporate partners – will come up with reasons your idea won't work. They will view these objections as insurmountable hurdles, and this perception will drive their assessment of your value. Your job is to persuade them that their objections are surmountable by anticipating their concerns with data. Therefore, to maximize the impact of your data, design experiments to overcome the coming objections.

The objections are subjective and audience-dependent. Ask three people, and you're likely to get at three (and likely many more!) different objections! And the objections will come regardless of how much data you've already generated and regardless of the listener's background or degree of risk aversion. It isn't easy to predict what the specific objections will be, but if we revisit the common archetypes of MPPs from Ch3.5 **Creating Value in an MPP**, the shared key value creation events can serve as guideposts.

All MPPs will face skepticism that the platform technology is developable and broadly applicable to produce multiple products.

- In a New Modality MPP, the objections typically focus on whether the modality will work as predicted, with advantages over the state of the art that provide a therapeutic benefit, and whether you can actually make it, especially if the modality is a complex composition of matter.
- 1. In a New Technology MPP, the skepticism will center on whether the technology spotlights new and useful targets or molecules, and whether there are paths forward with low technology risk modalities.
- In a New Insight MPP, the concerns often revolve around whether the platform is protectable from competitors, whether there are low-risk paths forward to create molecules, and whether you can avoid falling into the single asset trap.

How can you predict what the objections will be? Think about your most unbelievable assertions and focus on proving them, or at least reducing their unbelievability. If you assert that you're going to build a table with a dancing monkey on it, you should probably show that you've taught a monkey to dance before you build the table. If you build the table first, you'll still face disbelief that your fully realized vision is possible.

Remember that not all objections are technical (you can't build it, it won't work that way) and therefore not all of them can be punctured with an experimental result. For certain projects, the objections might stem from assumptions about market penetrance, regulatory or reimbursement strategy, or IP strategy. In these cases, conversations with value experts that pressure-test the benefits you provide and early feedback from regulatory agencies or the patent office may take the place of experimental results. Do your best to predict the objections and strive to disarm them, because your solutions will create value.

Avoid the "More Data" Trap

Scientific Marketing helps you overcome one of the pitfalls that builders face when seeking funds from investors or resources from partners: the trap of more data. A skeptical audience that doesn't yet believe in your future can always fall back on a request for more data. "This is nice, but I'm not fully convinced. What if you do the mouse experiment?" Then after those results come in, "What about another animal model?" "Did you try it in my favorite cell line?" The list is never-ending, because there are always more things you could do.

But rarely do the next incremental experiments generate significant value. They're comforting, but they only marginally increase the overall likelihood of success or the size of the opportunity. Your resources would be better spent on more important value creation events.

So don't get stuck in this trap. Refuse to engage with it. Use the lens of Scientific Marketing to shift the focus of the conversation from incremental improvements to the larger implications of the data. Sell the power of the future. For a not-yet-known-not-to-work hypothesis, your burden of proof is that the future you envision is plausible and valuable, not that it exists already.

If your audience still can't get behind your future – and you're not fooling yourself about the data – then there's a mismatch. Find a more visionary audience or make a more compelling case, but don't get stuck running on the hamster wheel of More Data.

Impact matters.

As with any marketing campaign, impact matters. The data needs to look amazing, jump off the page, and be obviously meaningful, because data doesn't sell itself. Animals treated with your drug should live when the controls die. Experiments in cells are better than on a computer; experiments in animals are better than in cells; and experiments in people are better than in animals. Early evidence of activity in humans, especially for unprecedented hypotheses, can pack a wallop of a punch.

We built a company that is developing a new set of gene therapy vectors based on commensal viruses that have co-evolved to exist within us. One of the value propositions is that these viruses, by virtue of growing with and among humans over eons, are not recognized by the immune system and can be transferred from person to person without eliciting an immune reaction.

Like all value propositions, this is a testable hypothesis. It's bold and, if true, it could revolutionize gene therapy by providing new vectors to access new tissues for new diseases. It does not yet exist, but it's also clearly not yet known *not* to work.

It is testable, yes, but it is testable in many ways, some more impactful than others. You could, for example, compare the sequence of the virus to sequences of other proteins in the human body and predict whether or not it would be immunogenic. Or you could isolate the viruses from human donors and run them through an assay to look for immune cell activation. Or you could coat the viruses with human serum and quantify how much "blocked" virus gets taken up by target cells. Heck, you should do *all* of these, but none of them in isolation is a particularly compelling result.

Instead, the team came up with an ingenious experiment. They hypothesized that these commensal viruses might be regularly transmitted between individuals during blood transfusions. If this were the case, then you might be able to detect the safe transfer of virus from individual to individual simply by sequencing the blood of the donors and recipients. With a simple sequencing experiment, the team was able to track the safe and repeated transfer of the vector across human subjects. Talk about a high-impact demonstration of non-immunogenicity!

Confidential

Coming up with high-impact experiments requires creativity, and it's a skill you can cultivate. Starting from high-impact results will make your job as a Scientific Marketer easier and more successful.

Ch5.3 Prototyping your Company

Getting into the lab can feel like a big transition: the end of exploration and the beginning of the execution phase. But while you may have new resources, new equipment, and a new setting, it's important to keep an exploratory mindset.

To reinforce this behavior, at Flagship we call our early projects – what others would term “Seed-Stage” companies – “Prototype Companies.” The difference may seem small, but the words we use matter. When you plant a seed, you know what species you've planted and therefore what you expect to grow. There's no mystery. You just have to water it and let time and sunshine do their work.

But a Prototype Company is different. A Prototype Company is still interrogating its value landscape. A Prototype Company isn't yet destined to reach one kind of outcome or another, and it hasn't yet decided its future because it doesn't yet know which paths will be possible or most valuable. A Prototype Company recognizes that its value hypotheses are merely initial conditions for what is sure to be a discontinuous and non-linear landscape. Prototype Companies are explicit that they are still prototyping and testing their ideas.

When you enter the lab to build your MPP, it helps to approach it as a prototyping exercise in which you try to efficiently identify and build the best future for your company.

Whether you choose to use Flagship's language or not, here are some tips for using your early experiments to prototype your company.

Be Nimble and Test Multiple Hypotheses

In the last chapter, we described how to construct a value landscape from which you derive multiple orthogonal value creation hypotheses. If you've done your exploration well, you'll have multiple hypotheses. Now you will test them.

Use your experiments to nimbly test multiple value hypotheses and try to validate, invalidate, or evolve them in response to facts. This is your chance to determine which paths are feasible and which are easier or harder, in such a way that your value landscape becomes less hypothetical and more real.

You will test platform hypotheses, figuring out which features are buildable today, which are easily within reach, and which will require substantial investment. You will test product hypotheses, discovering which product opportunities are most tractable, which platform features unlock multiple related products, and how to craft a compelling story using the data.

You will test your value propositions too. As you learn more about your platform's features and converge around products, you should continue to talk to experts – especially Value Experts of the type we described in Ch4.6 Getting from a Value

Landscape to a Company. Probe them with your updated Feature-Advantage-Benefit triads. Explore new benefits that you may not have thought of. Define and refine your value propositions in light of your ongoing findings.

Earlier, we highlighted the importance of not relying on a future “pivot”. Nimbly testing multiple hypotheses simultaneously is what lets you figure out your value landscape in real time and pick the most valuable path. Maintain your options. Prototype many versions of your future to keep from prematurely committing to one vision. If you go too far down the wrong path, you’ll be forced to try to pivot when things don’t pan out.

Seek Failure

If there are key failure nodes, you should seek them out. If you've made assumptions that will cause the whole edifice to crumble if they turn out not to be true, you should do test those assumptions first!

The desire to fail is counterintuitive, especially for most entrepreneurs. You may have told your parents and friends about your new company. You may have printed business cards or updated your online profiles. You’ve put time and money into getting this far, and you’re clearly invested in a successful outcome. But the game is long. It feels as though you’ve come far, but you’re only at the beginning, and the costs and time will only increase. The opportunity costs of working on the wrong thing are still accumulating – but it’s not too late to change course.

Listen to the data. There is no shame in saying “We couldn’t make it work,” “It’s not possible with today’s technology,” or “We were wrong.”

If you are optimistic during an exploration, assuming that all of the premises are true in order to let your mind find the unreasonable but valuable destinations, then once the lab work starts, you should be ruthlessly rigorous in interpreting the data. There is no benefit in fooling yourself. Listen to the data.

Remember, too, that good ideas are rare but also abundant. You’ve just discovered how easy it is to come up with a good new idea through exploration. You can do it again if necessary.

Data Must be Definitive

We’ve talked about doing the right experiments, but you also have to do them right. If you intend to make decisions based on the data, the data must be definitive. You don’t want half measures; you don’t want inconclusive results; you don’t want meandering fishing expeditions. When you fail, you want to learn why you failed and not be left with a mess of uninterpretable data. Design your experiments such that failure teaches you why you failed, and success is incontrovertible.

In particular, you should ensure that all of your experiments in the early days are hypothesis-driven rather than open-ended. This is particularly true for New Technology MPPs and New Insight MPPs, where it can be tempting to apply your tool or insight to a

new space and anticipate that you'll uncover great treasure. It may be easy to say you will search for and find interesting things and then craft a story around them, but such stories are rarely definitive. Instead, you'll likely find hints of "perhaps" and "sometimes maybe." If you don't declare what success will look like at the outset, you'll find yourself chasing after more and more data and never reach a definitive decision.

Decide up front what the definitive data will be for each of your value propositions, and pursue it in a hypothesis-directed manner so that you can make your decisions with confidence.

Do Experiments "Quick & Clean"

To test multiple hypotheses in a short time under resource constraints, you must do your experiments quickly. To ensure that the data is definitive, you must do the experiments cleanly. When you're in prototyping mode, therefore, your science must be Quick & Clean.

Quick & Clean means doing experiments that get right to the heart of the most important value questions and are at the same time definitive, robust, and reproducible. Quick & Clean means you don't optimize for spending less at the expense of time because you recognize that time is your biggest cost. Time drives your burn through salary, rent, overhead, and missed opportunities. Time is worth immeasurably more than any assay or reagent.

Quick & Clean means you don't settle for a Minimum Viable Product as popularized by Lean Startup methodology. Getting a crummy version of a product into the hands of consumers may work in software, where the key unknown is the way a user will engage with different features, but dirty doesn't work here. In biotech, dirty gets you uninterpretable data from which you can't draw conclusions to make decisions.

So how do you do Quick & Clean? You have a good start if you know your value proposition hypotheses and have aligned your experiments to test them. You're in the right place if you feel the urgency of time, have mapped your most important questions, and are laser-focused on answering them. Now, as you plan those definitive experiments, ask yourself: Can I do this more quickly? What is the most direct path to the information I need? Will the data be definitive?

Build a Rapid Prototyping Engine

If you want to nimbly test multiple hypotheses and robustly generate definitive data, you have to find a way to evaluate many things quickly. In our early-stage companies, figuring out what the Rapid Prototyping Engine looks like is one of the first things we do. We build it early and leverage it often to create value.

A Rapid Prototyping Engine will look different depending on your platform, but at its core, it's a process that lets you quickly and repeatedly make exemplars of your products and test them for the properties that matter.

At Rubius, our Rapid Prototyping Engine was a protocol for putting up to 96 different genes into red blood cells, such that a single run through our culture and differentiation process would let us test 96 discrete prototypes of functional red blood cells. At Cygnal, where we explore the interface of the nervous system and cancer, the Rapid Prototyping Engine was a workflow for validating targets that emerged from the platform's target identification assays. At Ring, where we make new gene therapy vectors, the Rapid Prototyping Engine was a process for making different commensal virus vectors carrying multiple payloads to rapidly assess yield, carrying capacity, and infectivity. At Generate, a computational protein design company, the Engine was a pipeline for synthesizing genes and producing and purifying proteins to evaluate their function. At Codiak, an exosome company, the Engine was a protocol for inducing exosomes from different cell types, loading them with different payloads, and purifying them for analysis.

Your specifics will vary, of course. But the common theme is that you have to be able to rapidly make what you're trying to make. A Rapid Prototyping Engine will let you sample your value landscape widely, test multiple hypotheses quickly, and create opportunity-expanding data for Scientific Marketing. A Rapid Prototyping Engine lets you learn what you have under the hood of your MPP.

Prioritize Important, Easy, and Wow!

You'll likely find that you have more experiments in mind than you could possibly accomplish with the time and resources you have. You will have multiple value propositions to validate. How do you prioritize them?

A framework we find helpful is: Important, Easy, and Wow!

Imagine a Venn diagram where the intersecting circles are labeled Important, Easy, and Wow! You will accumulate the most value at the overlap of the three circles. It is nearly impossible to hit all three, but you can often satisfy at least two of these parameters with each of your experiments.

Important means results that are crucial to the whole project. These are testable hypotheses that, if false, could sink the entire idea. Important experiments are rarely sexy, but they are essential to solidifying the foundation of your MPP.

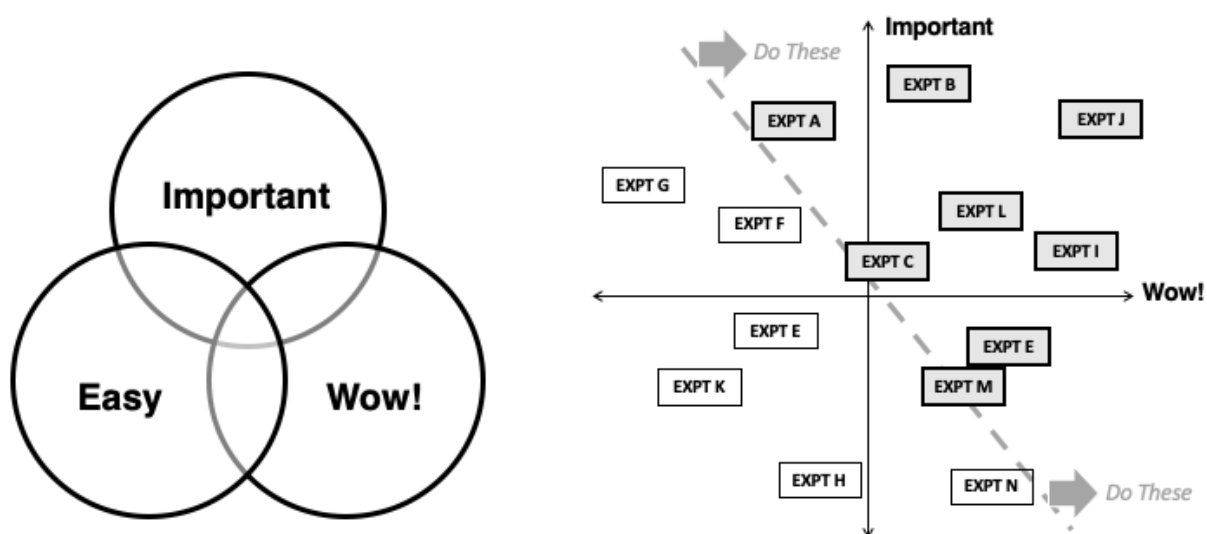
Easy means both easy to try and likely to work. Here, we think about experiments that are readily doable with your existing resources, those whose results are straightforward to generate, and those whose outcomes will be easy wins. The idea is to make sure you aren't only doing really hard things or generating data in the hardest way possible.

Wow! means high-impact. Wow results make people sit up, spark their imagination, tickle their sense of what might be possible, and demonstrate big opportunities. Scientific Marketing relies on Wow! experiments. Wow! experiments are rarely the most basic version of experiments; instead, they rely on creativity.

Earlier we described the experiment that our colleagues at Ring Therapeutics thought up: using blood transfusion donor-recipient pairs to show that commensal viruses could be transferred across people without inducing an immune response. This is a great example of Important, Easy, and Wow! In action. The immune silence of commensal viruses was *important*, an essential demonstration for the whole platform because, if it turned out not to be true, the path to making generalizable gene therapy vectors would be nearly impossible. The experiment itself was *easy to try* – collect blood samples and sequence it for virus. And the outcome – showing safe transfer of the viruses in people and across multiple transfusions – was *jaw-dropping*.

Another good example of Important, Easy, and Wow! is Moderna's early experiments with erythropoietin (Epo) that we described in Ch3.5 Creating Value in an MPP. Recall that the company faced skepticism that mRNA could be a drug because it was thought to be unstable, unable to enter cells, and highly immunogenic. The demonstration that systemic administration of mRNA could encode functional proteins was *important* because it addressed head-on the skepticism of mRNA's therapeutic potential. The experiment was *easy* because the amount of Epo needed to see a biological effect is small – the effect of the protein is catalytic rather than stoichiometric – and the outcome only requires drawing blood from the animals. And because it was performed in animal models with an approved therapeutic molecule, the experiment made it easy to envision a future where any protein could be made as mRNA. *Wow!*

Figure: Experimental Prioritization



For the visual thinkers among you, a useful exercise is to write down all possible experiments on sticky notes. Then, on a large whiteboard with axes labeled *Important* (X) and *Wow!* (Y), place the sticky notes. Look at all your experiments on a single set of

axes; move the sticky notes around relative to each other. (Yes, it would be great to add an “Easy” axis, but in the absence of a 3D wall, rely on your instincts for designing easy experiments.) Take turns with your teammates, debate the relative positioning of the potential experiments until you reach consensus. Once the axes are populated, you have your prioritization: the top quadrant (important and high-impact), or everything to the northeast of the diagonal.

The exercise is qualitative, but it can help make sense of an excess of choices and free you to pursue your experiments with a clear head, knowing you’ve evaluated the alternatives.

Ch5.4 Expand the Opportunity in Addition to De-risking

All this talk about designing and prioritizing experiments, and not a word about de-risking? That’s deliberate. In the early days of an MPP, especially one with an unprecedented platform, there is very little you can do to materially reduce the risk of the endeavor.

Let’s start from the basics. The expected value of your MPP is a product of the probability of success and the size of the opportunity:

$$\text{Value} = p(\text{Success}) * \text{Opportunity}$$

Value is inversely proportional to risk. By de-risking your idea, you increase the likelihood that it will work and therefore increase its value. Unfortunately, we often don’t know that a specific drug will work until very late in clinical development, when it’s been taken by hundreds or thousands of patients. Any pre-clinical advance that improves the likelihood of success 10-fold, 100-fold, or 1000-fold is still an infinitesimal blip that barely registers on the global likelihood of success. Early de-risking does not reduce the overall risk by much, so it doesn’t build much value.

But there is another lever, a more powerful one, that you control in the early days of an MPP. That second lever is the size of the opportunity you can address, and it is a powerful driver of value.

Expanding the Opportunity

You can expand the opportunity by showing that your platform is in fact a multi-product platform, where the fundamental features shared by your products provide real benefits and where success in one product implies success in subsequent ones. You can expand the opportunity by proving that your products are useful in multiple therapeutic areas or disease mechanisms. You can expand of the opportunity by building elements of your platform that unlock broad value propositions, not just specific products. Do the

experiments that open these new opportunities and use Scientific Marketing to sell them.

Expanding the addressable opportunity is an effective way to create value. Imagine that your opportunity is no longer \$50 million, but \$500 million or \$5 billion or more. The risk of any one product, high though it may be, is suddenly tempered by the upside.

Expanding the addressable opportunity will also attract investors willing to put in the time and money required to build an MPP. You want investors who understand the long game, who will help you navigate your Value Landscape, who won't force you to converge on a first asset, and who won't sacrifice the platform and its multi-product potential at the altar of a quick return. These investors are undeterred by risk and tantalized by big opportunities. It will be easier to recruit them if you grow the size of the opportunity, and you want them on your side. Having MPP investors in your corner will make you more likely to succeed.

There is a place for De-Risking

There is a place for de-risking, but it is focused and defined. Save your de-risking for the essential threats, the killers of your technology, the key failure modes and anticipated objections that you must resolve with data.

In the early days of Rubius, we knew that the whole platform hinged upon our ability to grow red blood cells from stem cells, so we made sure to figure out how to do it early. We knew that we would have to manufacture the cells, so we quickly set to work to show that it was possible to produce ever-greater quantities of enucleated red blood cells in bioreactors. We knew that the platform hinged upon our ability to put genes into cells and express functional proteins without harming the cell, so we made sure to prove it was possible to do so. These were all de-risking experiments. They were important, but hardly impactful.

Growing the size of the opportunity, in contrast, meant figuring out just how broadly we could stretch the modality of red blood cells. Could we use them to deliver enzymes protected from the immune system? Could we use them to capture toxins? Could we use them to target diseased tissue? Could we use them to activate signaling cascades in other cells? In the first six months, we produced 150 functional prototypes of red blood cells with different therapeutic payloads. These results taught us that much of our value landscape was realizable – not that it was realized yet, but that a world where red cells are widely used as a genome-free cell therapy for multiple applications in multiple diseases was possible.

De-risking matters, but in the early days of an MPP, remember that expanding the opportunity is a powerful lever for creating value.

Ch5.5 Leading and Managing an early MPP

It's not easy to be at the helm of an early-stage MPP that has entered the lab. The ground constantly shifts beneath you as you test multiple value hypotheses in parallel, learn what predictions hold true and which don't, which paths are harder and which are easier. The opportunity shrinks and grows and changes shape in real time. You, as an executive, manager, and leader, must be ready to respond with equal dynamism. It's not easy, but it can be navigated successfully if you have the right skills.

The Executive

As an executive, you control the allocation of resources. You control what you spend money on, how much, and for how long. When prototyping your company, you must be dynamic with your resource allocation. Titrate your money to different projects – some costs will be high and some low, and the amount will change in response to results.

The most important thing you can do as an executive is know the relative importance of your hypotheses ahead of time. This will tell you how deep you need to go on them. The more important the value proposition, the more effort you should expend.

Sometimes you're grabbing territory, planting your flag, and moving on to the next island. You may have unlocked a feature of your platform and want to show how broadly it can be applied. In this case, do the minimum to prove your capabilities in a space and to enable your intellectual property, but don't worry about crossing every *t* and dotting every *i*. Explore quickly and broadly and don't fret over the few avenues that don't pan out, especially when you're using high-cost venture dollars. Save the incremental bow-tying results for someone else's less expensive capital.

At other times, there is an absolutely essential result that you must achieve no matter what, something required for the success of the whole platform. If you can't build it, solve it, or work around it, the whole edifice will crumble. Hammer at these experiments relentlessly.

Avoid scaling your platform prematurely. You may someday want the ability to run millions of samples in parallel, or read entire genomes, or manufacture kilograms. But that day may not be today. Remember that you control when and how you deploy the various features of your platform (Ch3.6 Integrating Platform and Product Value Creation) – make sure you do so in concert with the product demonstrations and that your resources don't go chasing a distant goal misaligned with today's priorities.

In addition, ensure that the resources you spend return value concretely and continuously (Ch3.7 Final Thoughts on Value). Don't put all of your experimental eggs into one last-minute omelette. Steve Holtzman, veteran start-up executive and perceptive judge of the industry, who reminded us in Chapter 3 to focus on share price appreciation instead of market cap, notes that you can both under-spend and over-spend, and that realizing exceptional value requires walking the tightrope between the two.^{xiii}

All the while, keep an eye on your runway. You want to maximize your shares' appreciation by the next time you need to raise money. Focus like a laser on getting the most valuable results in the time you have with the resources at hand.

The Manager

As a manager, you're in charge of making sure the multiple parallel projects are completed on time with definitive, decisional data. You've worked backward from your value propositions and know which hypotheses must be tested to validate the idea, which demonstrations will best convey the breadth of the platform, and when the money will run out. You need schedules and milestones, but there must be room in your plans for the uncertainty and dynamism that effective prototyping requires.

Your neat Gantt chart, a waterfall with each successive achievement cascading into the next, is a fine place to start, but don't let yourself get locked into a false expectation of linearity. Some milestones are so central to the whole platform that you'll need to keep pursuing them until you reach them or declare defeat. Other achievements are better thought of as a series of sprints: things you may try for a few weeks to see what comes of them. In these cases, you can let the data inform whether and how much farther to go.

Your neat waterfall hides an underlying dynamism. On the ground, be ready for turbulence. Be ready to respond to data, reconsider projects and timelines, set new goals, and respond to changing realities in real time. Don't let your best-laid plans impede new paths to value.

The Leader

As a leader, you must manage your team through the inevitable chaos. You have full visibility, so you know that what looks like chaos is actually responsiveness to data and a shifting value landscape. But imagine what it feels like to the scientists on your team, especially those without prior experience in the crucible of a start-up!

Those coming from academic labs are likely used to multi-year timelines, interminable post-doc fellowships, little accountability, and little oversight. They may feel whiplashed if you keep stepping in to set new goals and directions. Scientists with a career in Pharma may feel micro-managed, and they may believe it's important to let the science run its course until the next annual portfolio review meeting.

How do you keep smart, creative scientists motivated through this dynamic prototyping phase? Transparency. In particular, transparency about your value hypotheses.

The value hypotheses don't change; the way you try to achieve them does. Strategic clarity enables tactical flexibility. Be candid about this with your team. Expose them to the value hypotheses, explain what the experiments are trying to prove and how you'll use that information to tell the most valuable story about the company. But be clear that the specifics are subject to change. The goal is not to "get this experiment to work, no

matter what” but rather to “prove (or disprove) this value hypothesis.” If a new experimental approach is warranted, so be it. Everything is centered on value.

If you alert everyone on the team to the value hypotheses, they'll see how their efforts are aligned with company value creation. They'll come to see chaos the way you do – as the inevitable byproduct of efficient prototyping in the search for value in a not-yet-built future. And you'll be a more effective leader.

Key Takeaways

- Work backwards from your value propositions to make sure you're doing the right experiments.
- Results are for scientific marketing: you control how you design your experiments, what results you generate, and how you interpret them. Remember that it's up to you to convey the "So What" of the data.
- Prototype your company by testing multiple value hypotheses nimbly. Seek failure, generate quick & clean definitive data, and prioritize *Important, Easy, and Wow!*
- In addition to de-risking, you can create substantial value by expanding the size of the opportunity

Chapter 6. Creating Value in Product Development

The value of a Multi-Product Platform lies both in the platform and in the multiple products that emerge from it. For therapeutics, the products are drugs. This chapter covers how to create value when picking your indications, declaring your first products, and developing them into drugs.

For a conventional therapeutics start-up company, one based around a single asset, the choice of indication and product is straightforward. If you have one molecule in hand, that's what you do. First get your asset approved in a disease for which clinical trials are short and definitive, and then look to expand your molecule's use into larger indications with more patients and a bigger market.

In an MPP, you have a choice. You get to choose from a wide menu of diseases, targets, and even modalities. What's more, the choice is highly consequential. The declaration of your first products sets you on the path to creating specific drugs, defines your near- and medium-term development activities (including what elements of your platform you build), and shapes the way you and others will value your company.

In this chapter, we will explore how to maximize value creation in product development in an MPP. We will describe the risks that accompany the selection of your first products. We will discuss the central tension between picking products that will work and picking products that are valuable, and we will show how to resolve that tension by focusing on your platform's value propositions. We will explain why optionality is your friend and how to maintain flexibility by developing multiple products with orthogonal value propositions in parallel.

Ch6.1 Picking your first product matters

What product(s) will you develop first? This choice matters. It defines the diseases you'll tackle, the experiments you'll do, the compositions you'll manufacture, the clinical trials you'll run, and ultimately the drugs you'll make and the patients you'll treat. For the MPP, the choice determines which path among many you'll follow, where you'll focus your energy, and how others will value your company.

The sad truth of Multi-Product Platforms is that many more companies start out as MPPs than end up as them. The majority of MPPs turn into single-product companies, a transformation that happens when the company chooses its first product and begins the strategic narrowing of scope required to develop a therapeutic.

But the dream of an MPP is not just a fantasy of early ideation. It's possible to create large, transformative companies that successfully develop multiple products based on a unified platform and that are active in multiple disease areas. But you have to be alert to the pressures – to focus, to narrow your vision, to concentrate on execution – that accompany product development.

The pressure is exerted by multiple sources: it comes internally, from you and your team, and externally, from the market and your investors. Know the sources so you can push back and make sure your MPP future isn't foreclosed. Go into this phase with your eyes open.

Picking the first product concentrates dollars, expertise, and focus

Developing a drug costs much more than discovery and platform work. You might spend \$50k to perform discovery screens or build the components of your platform, but \$50M to drive a drug forward into development. As the dollar signs grow, so do the consequences of your decisions. Once you commit, it is hard to go back. Very quickly, your option to pursue new products and build the platform becomes secondary to your ability to execute on the lead program.

Product selection also forces you to concentrate your team's expertise, which can be even more damaging to the promise of a multi-product platform. Whereas early work on the platform can be broadly applied to multiple products and early employees are usually polyvalent scientists who can be deployed to various programs, once you begin development around a single product and indication, you will have to go deep. And people who are deep are rarely broad. We experienced this at Editas, one of the first wave of genome editing companies. It was easy to find molecular biologists who could develop nucleases to edit multiple genes in a variety of cell types. But once we committed to specific programs, we knew we would be hard-pressed to find a hematologist who could double as a muscular dystrophy expert or handle retinal degeneration models.

The concentration of money and expertise means that you'll necessarily concentrate your focus.

You get valued on the products you specify

The market will also pressure you to focus when you specify your first products, because the market only knows how to value defined products. You will be valued on your lead asset. When you specify the product, you can put numbers to patients, to the current standard of care and its unmet needs, to the likely rates of adoption and rates you can charge and get reimbursed for. The more defined that first product is, even if it's early in development, the easier it is for the market to assign it a tangible value.

The market knows how to value defined products but has no way to objectively value a platform. Your platform is invaluable in creating massive value but is also un-valuable. Since the market doesn't know how to put a value to your platform, the value ascribed to your platform will be essentially zero.

One of your responsibilities is to tie the value of your company to multiple products. You must be the one to ascribe the value of the product to the platform and to show how the unified platform allows multiple products to emerge, each with reduced risk because of those that came before it. Your job is to convince the world that your success with one product makes it more likely that you will succeed with a second and a third.

You don't have to develop every product to capture its value. Strive to show that the sum total of your opportunity includes both the products you *do* develop and those you *could* develop. Specificity helps. The more detailed and precise you can make your map of future products – what molecules for what indications, and why coming from your platform makes them better than any alternative – the better you can convey the full potential value of your MPP.

But conveying this value requires selling, and this sales pitch will not be easy to make. The infrastructure of biotechnology investing, from the majority of private venture investors to the public markets, is not set up to value multi-product platforms. If you're not prepared to defend the value of your platform, or if you don't know how to make the case convincingly, or if you get sucked into defending the value of your first product alone – you'll end up building a company centered on that first product.

Product investors are not Platform investors

You may find, too, that your investor base begins to change when you begin product development. When you seek funding, you may find yourself pursued by investors who want products. These investors will have seen success with assets pushed rapidly into the clinic and companies that were acquired for their lead molecule. But beware – product investors are not platform investors.

Product investors will not want you to spend their dollars developing the platform, no matter how much you try to explain that it matters for future products. They will want their resources deployed toward pushing forward the first product. If that first product does not succeed, you will still have the many follow-on or orthogonal products that your platform could produce, but product investors won't be interested in funding the

development of a platform. They will instead push to recoup whatever remains of their product-focused investment.

In contrast, platform investors know that innovation takes time and that converging too quickly can be dangerous. Platform investors see value in building a platform as a future multi-product generator. Platform investors are willing to invest in platform development, to be patient while you prototype your value landscape, and to encourage you to fully realize your potential value, so that you're not pushed toward a single asset and eventually left stranded. When you raise money for your MPP, make sure you are screening for investors who share your vision for your company's future.

Ch6.2 The Central Tension in Choosing your First Product: It must work vs. it must be valuable

The choice of your first indication is consequential and carries real risks – but you can't avoid it! If you want to make drugs to treat diseases, you can't dance around the starting line forever. So what should you pick to pursue, and how should you decide?

When you stand at the trailhead with dozens of paths ahead of you, you're likely to feel torn between picking a first product that will work and a first product that is highly valuable.

Your internal argument may go something like this...

It must work

Why must it work? It must work because you're likely only to get this one shot. You're committing a substantial chunk of your capital to move the project forward into development, not to mention the specialized expertise you'll need to bring on board, whether in the disease itself or in the manufacturing of your molecule. Given the costs of development, you won't have the dry powder for a "do-over" if you fail, much less the standing to return to investors and ask for more money without substantially devaluing your share of the company. And after a perceived failure of the platform, even if it is surmountable in a new indication, it will be hard for you to convince new investors to come onboard.

So that first product has to work. Maybe you can search for an indication where the biology is thoroughly understood and where you're confident that your molecule will work, or where you will be able to quickly get a clinical readout, or where you need the least amount of your molecule to have a maximal effect in the patient so that manufacturing is less of a hurdle. The first product has to work, so you set your sights on the lowest-risk opportunities.

It must be valuable

Why must it be valuable? It must be valuable because people are going to start thinking of your company as "Drug X" company. If your drug treats only a few patients, or only works in a niche subset, or might not be fully reimbursed, or faces intense competition from other products on the market or those that might reach the market around the time yours does, or any of a dozen other reasons Drug X may not be a blockbuster, you're going to have a hard time convincing people to invest in your product.

You believe that under your stewardship, the platform will create more than just Drug X. To accomplish this, you're going to need resources, patience, and partners who are willing to let you build the needed underlying technology. But money doesn't usually come with patience. The closer you get to a product, the more you'll be subject to the expectations of public market-facing investors and their standard accounting of value.

You have your arithmetic for the true value of the multiple drugs in multiple diseases, but the market will collapse it to the numbers for your first indication. So these numbers had better be big.

But highly valuable indications are unlikely to be the easiest. They're probably high-value because there's significant unmet need, which means lots of people have seen the same opportunity and haven't been able to act on it. So the science is probably complex, the biology poorly understood, and the clinical trials long. The more you push for high value, the further you may find yourself from "likely to work."

You're not alone in this choice

So how do you choose? How should you pick a first product amid this tension between "it must work" and "it must be valuable"?

The good news is that you're not alone. Let me tell you about a scrappy start-up company that was built around a revolutionary idea for a new modality of therapeutics broadly applicable to multiple diseases and therapeutic areas – a classic New Modality MPP.

The year was 1978. The company was Genentech. The modality: recombinant protein therapeutics. The platform: DNA cloning and bacterial expression.

It's easy to forget how revolutionary Genentech was at its origin – how new the molecular biology techniques we now take for granted were, how artisanal the process of making a protein from a gene sequence. How unreasonable it seemed to propose that human proteins could be made from bacteria, and how vast the opportunity space Genentech opened for protein drugs when the product couldn't be isolated from a donor or an animal.

And in 1978, Genentech faced the same choice between low risk and high value.

In her wonderful oral history *Genentech: The Beginning of Biotech*, Sally Smith Hughes describes this central tension playing out between the scientist Herb Boyer and the investor Bob Swanson.^{xiv}

The team believed they had finally figured out how to produce recombinant therapeutic proteins by cloning DNA into plasmids and transforming them into bacteria. The techniques were still new and pretty shaky – only recently, in a first-of-its-kind make-or-break experiment, the team had figured out that you needed to put a bacterial promoter in front of the synthetic gene to get any protein out at all.

Money was tight and time pressures were real. Competing academic and industry labs on the East Coast – Harvard and IDEC – were fast on Genentech's heels in the race to produce the first recombinant protein therapeutic.

What to make now? What first therapeutic product should the company pursue? The team debated two choices: Somatostatin and Insulin.

Somatostatin was likely to work. The peptide hormone is a short (15 amino acid) linear polypeptide chain. It would likely have been easy to make using the still-primitive bacterial expression platform. But the market would be limited. Only a few thousand patients a year suffer from somatostatin deficiency, so the approval wouldn't provide sales to keep the company afloat for long.

Insulin, in contrast, was valuable. Diabetes wasn't the scourge it has become today, but it was still widespread in the late 1970s, affecting hundreds of thousands of patients. People knew that insulin replacement worked. Physicians had been giving patients insulin from pigs for decades, but a substantial number of patients developed an immune reaction to the xenogeneic product, so a human analog was needed.

But producing insulin would be hard. Unlike somatostatin, insulin is a complex protein. It is more than twice as long as somatostatin and contains two chains that are held together by disulfide bonds. The protein would have to be properly processed by the bacteria and formulated in the lab to make sure it was active.

So which would it be? Ease or value? Genentech faced this choice. You're facing this choice. What to do?

The answer is to reframe the choice by redefining the terms. Your first product has to work and it *also* must be valuable. But valuable doesn't have to mean "makes lots of money." Instead, a valuable first product in an MPP is one that *validates the value propositions of your platform*.

Ch6.3 Valuable Products are those that Validate the Value Propositions of your Platform

This, then, is the answer to the tension between “it must work” and “it must be valuable.” The answer: the first product must work *and* it must be valuable, but it must be valuable because it validates the value propositions of your platform.

Products that validate the value propositions of your platform keep the MPP ball moving forward. By describing the product through the lens of the value propositions of the platform, you validate the multiple subsequent products that will emerge and simultaneously justify your investment in and valuation of the platform itself.

To return to Genentech in 1978: Herb Boyer won the argument and the team pushed forward with somatostatin, the easier of the two products (with a smaller market). But the true value of somatostatin comes from much more than just the product itself. It showed the team’s ability to produce a recombinant human therapeutic protein; it proved (once administered to patients) that bacterially-produced recombinant proteins were safe in people; and it proved that splicing DNA together and coaxing bacteria to produce a protein is a viable and general way to produce *any* therapeutic protein.

The somatostatin product validated the important platform value propositions – human sequences, rapid production that is generalizable to any sequence, an industrialized process more robust than isolating material from biological fluid – that were necessary give Genentech a plausible future as a multi-product platform company.

The choice of that first product doesn’t limit the subsequent products, especially if you choose it to keep the MPP dream viable. Genentech started with somatostatin, yes, but then it made insulin, human growth hormone, and other cytokines and growth factors, and it eventually pioneered monoclonal antibodies in its three-decade run as an innovator and leader in recombinant protein therapeutics.

The total value of somatostatin as a validator of a multi-product future for Genentech far outweighed the specific value of the single product.

Case study: Moderna

Moderna is a case study of a company with a clear sense of its platform value propositions.

Moderna is in the news as we write this because of its central role in the global race to develop a vaccine for SARS-CoV-2. Moderna’s mRNA vaccine candidate was shipped to the NIH 42 days after the sequence of the novel coronavirus was published, and it began Phase 1 study dosing just 20 days later. This near-instantaneous development was one of the original value propositions for messenger RNA as a medicine – the hypothesis was that it was possible to go rapidly from sequence to candidate drug,

independent of protein type, without having to produce and purify the molecules along the way.

But before COVID, Moderna had a choice of what modalities in protein therapeutics to develop first. Should it make mRNA antibodies? mRNA cancer drugs? mRNA enzymes? Of all these options, the company chose to emphasize prophylactic vaccines first. The company developed programs for Cytomegalovirus, Zika virus, Chikungunya virus, and influenza. Few would expect these products to be big moneymakers. Why, then, did Moderna develop them first?

Stefan Bancel, CEO of Moderna, described his attitude to us this way: “The NPV of the first indication can be negative! So long as it works.”

It has to work – and prophylactic vaccines were, for Moderna, the lowest bar to clear. In a vaccine setting, you need only a small amount of virus antigen to trigger an immune response. A little bit of mRNA goes a long way when you can rely on the immune system to amplify the biological effect. You can give a single (or double) dose of the drug, but you don’t have to sustain regular administrations over decades, as you might have to do for a genetic disease or chronic condition.

And yet the prophylactic vaccine programs validated the important platform value propositions about the benefit of mRNA as a medicine and taught the team to manufacture mRNA at scale. They also produced information about the types of formulations that work best, what dosing and toxicities can be expected, how the human body reacts to exogenous mRNA, and so on.

Using the lessons learned from these early programs, the Moderna team expanded to (currently) thirteen additional programs in multiple modalities, including personalized cancer vaccines, systemic immune modulators, and cytoplasmic enzyme replacement therapies.

Moderna, like Genentech, recognized that the first program doesn’t preclude a multi-product future if you choose it well and value it according to the value propositions of the platform that it validates.

Case study: Alnylam

Alnylam offers another study of picking indications that align with the value propositions of a new modality with multi-product potential.

For Alnylam and its silencing RNA (siRNA) platform, the “must work” requirements are that the short nucleotides get efficiently taken up by target cells, that the gene to be silenced is truly the driver of disease pathology, and that a little bit of suppression of the target gene goes a long way.

Today, Alnylam’s pipeline and product portfolio bear these value propositions out. Its products target pathogenic proteins in the liver – an organ that efficiently takes up

particulates (like siRNA nanoparticles) from the bloodstream. The targets are the direct cause of disease, as in the case of transthyretin for amyloidosis, or enzymes that make a pathogenic substance such that a little suppression is catalytically amplified, as in the case of PCSK9 for hypercholesterolemia and heme production enzymes in porphyria.

Successful companies fit their products to what their platform does well. You too should strive to match your development plans to the value propositions of your platform.

Let Product Selection guide your Platform Development

Remember that the likelihood of a product working in a chosen indication is a dynamic assessment, because it depends on what your platform can do today, which is mutable.

In Ch3.6 Integrating Platform and Product Value Creation, we introduced the concept of strategically deploying your platform and understanding how different technological improvements enable different classes of products.

The same analysis is relevant here as you prepare for product development. The indications that “will work” with today's version of the platform may be different from the indications that will work with tomorrow's version of the platform.

Layer platform development on top of your product selection strategy. When you can improve manufacturing yield tenfold, increase affinity by an order of magnitude, produce tri-functional molecules when before only bi-functional were possible, or whatever else your platform might allow you to do, new opportunities will arise. Know what these future product opportunities are. This knowledge will help you choose the most important platform improvements to pursue today.

Ch6.4 How to select your first products

How do you go about prioritizing those first products? How do you boil the ocean down to a manageable gallon of desalinated potable water? You can start with comprehensive lists of diseases, genetic risk factors, or validated targets, but thoroughness doesn't necessarily confer insight.

The only way to efficiently make sense of the world of opportunities is to know what parameters matter to your platform and what parameters don't.

Here are some triage criteria to use in selecting your first products.

It must work

The first product has to work, since so much of your MPP's value hinges upon it. There are three components to a likely-to-work product: technology, mechanism, and biology.

The product must be technically viable. You need to be able to make it with your platform today, or know how to make it with specific improvements to the platform that you will invest in.

The product must be mechanistically sound. When you make it, the product has to do what you think it will do (to the target, the cell, or the body). To illuminate an otherwise black box, you should aim to measure the function, certainly in model systems so that can get an early read on your predictions, and preferably in patients to ease clinical development.

The product must have minimal biological uncertainty. When the product does what you think it will do, it has to matter in disease. We don't presume to understand the human body well enough to make guarantees, but you should do whatever possible to be confident that the intervention will be therapeutic.

It must have a compelling product value proposition

Your product has to have a compelling *product* value proposition. As we described in Ch3.2 Features, Advantages, and Benefits, the value proposition of your product is the answer to "So What?" It describes the tangible benefits to the end user – whether physician, patient, or other – that derive from the features of your product and the advantages those features provide. The benefit may be better safety, better efficacy, convenience, speed, or something else. Regardless of the specifics, the product must provide a benefit, and you should be able to articulate it.

Identifying and articulating the compelling value proposition of your product will help you answer questions of impact (How meaningful to patients is the treatment/cure you provide?), total value (Why will people pay for it?), and competition (How might others try to solve this problem and how are you better?).

In our survey of entrepreneurs and investors, we found a wide range of tolerance for these parameters. Some welcome the validation that comes from group pursuit of the same targets, while others prefer the expanse of opportunity that accompanies an unprecedented high-impact effort. There is often tension between more disruptive hypotheses that face less competition and more reasonable hypotheses that will have to fight for a slice of the market. Regardless of where you fall, you must be able to articulate a compelling value proposition for your product.

It must validate (at least one) platform value propositions

In addition to the product's value propositions, what *platform* value propositions does this particular product validate? Remember: a product will often validate multiple platform value propositions; a single product doesn't have to capture all of your platform's value propositions; and different products can validate different sets of value propositions.

An underlying thesis of your MPP is that your platform lets you make multiple products, each with reduced risk and increased likelihood of success. How does success in this first product read onto other products in your pipeline? What about this product makes your future products easier to make and more likely to work? (It isn't sufficient for a product to simply show that you can make something or do something with your platform. You have to describe why it matters.)

Your platform has an inherent set of "So Whats"; a suite of benefits that any platform-derived product will enjoy as a consequence of originating from your platform. How does this first product manifest those platform-derived benefits? What is the Platform-Product-Market fit for this product, and thus for the family of related products?

Knowing your Platform-Product-Market fits is an essential hedge against the failures you're sure to experience. Some failures will wipe out an entire class of future products. But other failures won't. Failures caused by the specific biology of the product or a particular market dynamic may leave your multi-product platform intact. But only when you recognize the difference and convey the value propositions clearly can you develop your product portfolio with confidence.

It must be provable quickly and definitively

As an early product, you must be able to inspire confidence that it works and that your value propositions matter. How do you do this quickly and definitively? What preclinical results would meaningfully increase the likelihood of success in humans? If the question can only be answered in the clinic, how readily can you get the necessary readouts that build confidence that you're right about the product's value propositions? Not every product or indication cooperates, but those that do can be the most valuable.

The parameters for triaging initial products are similar to those for choosing demonstrations when you prototype your platform in the lab; see Ch5.3 Prototyping your Company. In that context, we talked about the intersection of Important, Easy, and

Wow! – a way to validate platform value propositions in technically and experimentally feasible ways with an eye toward high-impact demonstrations of differentiation.

The similarities between product prioritization and prototyping are no accident, so take advantage of them. Can you pick demonstration projects that have product potential, such that you're simultaneously proving out the platform and developing your products? At Rubius, our first demonstration project for active enzymes encapsulated in red blood cells and cytoplasmic expression of non-human proteins turned into the company's first clinical candidate in phenylketonuria. At Nimbus, the targets chosen as demonstration projects for computational small molecule drug discovery became the company's initial products to reach the clinic.

Prioritize your potential first indications and products early and work back from this future to create a valuable early experimental plan for prototyping your platform.

There may be many right answers

The triaging process gives guideposts, but the answers are subjective and uncertain. The way you weight different parameters will vary depending on your constitution and your biases. That's fine. There is no one right answer.

Take the burgeoning field of gene editing with CRISPR. Three companies launched nearly simultaneously to develop the technology for human therapeutics, all of them firmly committed to building multi-product platforms: Editas Medicine, Intellia Therapeutics, and CRISPR Therapeutics. By looking at their pipelines, we can see how the same technology and initial conditions led to different conclusions about which first products to develop.

All three companies started from the same premises. The first product had to work – it had to be technically feasible and biologically rock-solid. The genetic correction had to be easy to do with CRISPR, using a monogenic target, preferably on a single allele, where knockout (rather than the much harder reintroduction of a new gene) was sufficient to be curative. The link between the genetic edit and the function of the cell had to be unassailable. You won't find much disagreement on these parameters or on the list of targets and diseases that these criteria recommend.

But the drug also had to get to the necessary cells efficiently and safely. And here is where we see differences of opinion emerge. The companies had the option of picking an organ in the body for which viral delivery vectors are well validated (eye, muscle, liver, brain) or where it was possible to confine the molecule to the site of administration (eye, muscle). They could instead have considered editing human cells *ex vivo*, outside the body, where they would have more control over the cell type and could even check to make sure the correction was successful before administering the modified cells back to the patient. Here they might have chosen to modify hematopoietic stem cells to correct genetic blood disorders or alter immune cells to treat cancer.

It is possible to agree on the initial conditions but come to different conclusions. Just look at the pipelines of the three companies. Editas chose to lead with genetic eye diseases and build out subsequent products in hematopoietic and immune cell correction. Intellia instead prioritized stem cell correction for sickle cells with follow-on programs in cancer and liver disease. CRISPR Therapeutics opted for stem cell correction in beta thalassemia as its first product, with follow-on projects in cancer, muscle, and liver disease.

There is no wrong answer if your products are rooted in your value propositions.

Ch6.5 Develop Multiple Orthogonal Products Simultaneously

Modern MPPs Develop multiple “first” products simultaneously

In the CRISPR example in the previous section, you may have noted that all three companies were developing multiple products simultaneously. Furthermore, the products were orthogonal to one another, arrayed strategically across different indications in different therapeutic areas.

This is the modern way to build MPPs. In today’s world, product development does not mean collapsing into a single-asset company. Instead, modern MPPs explicitly embrace their multi-product potential from the outset.

It used to be that the path from start-up to multi-product company with a portfolio of drugs involved getting the first product approved in a first indication, expanding the use of that product into additional indications, and then building out the next generation of molecules. This model still exists and can still be successful.

But today’s MPPs pursue multiple “first” products in parallel. By developing multiple products at the same time, you reduce the concentrated risk of any one program. If you structure your clinical programs such that they’re running nearly simultaneously, you can fend off the market analysts who want to peg your value to a single product or devalue the multi-product generative capability of your platform.

Products should validate *orthogonal* platform value propositions

An MPP has multiple platform value propositions. Different products validate different subsets of platform value propositions. As you select and prioritize your products, try to identify those that validate orthogonal sets of value propositions, that have different Platform-Product-Market fits.

The benefit here is twofold. First, having multiple orthogonal products makes the breadth of your MPP explicit. If you can develop families of products with entirely different platform value propositions, for example multiple products that share each of several mechanisms of action, you can envision a future in which you create multiple multi-product platforms from the same foundation. Second, orthogonality hedges the risk of platform failure because if one product doesn’t work, it doesn’t automatically invalidate all subsequent products. If the failure was caused by a value proposition hypothesis that turned out not to be true, the probability of success of a second product based on different value propositions is not affected.

At Rubius, our first product – the first engineered red blood cell ever administered to patients – was a red cell encapsulating an enzyme for phenylketonuria. The value proposition of that enzyme-carrying red cell was that the long-circulating cells would reduce the dosing frequency and increase the half-life of the enzyme in a way that couldn’t be recapitulated for patients given a recombinant enzyme replacement.

Furthermore, by shielding the enzyme from the immune system, we hypothesized that the red cell product would not cause the severe immune reactions that frequently occurred with the competitor recombinant enzyme.

But a second program, a red cell that presented a combination of immune stimulatory ligands on its surface, had an entirely different set of value propositions. For this second program, the value proposition was that the presentation of multiple ligands on the red cell would mimic the way cell-cell communication happens in nature, thus generating a more potent response than we could get with a recombinant protein drug alone.

These are multiple products emerging from the same platform. They share common features – engineered red blood cells – and as a result benefit from core platform capabilities, but the products themselves are driven by orthogonal value propositions. If one program fails, it is unlikely to affect the other. Having both products meant that Rubius could make a credible case that it was an MPP with the potential to create multiple products in enzyme replacement therapies and multiple products in targeted immune stimulation – two entirely different modalities!

Your multiple simultaneous “first” products should be orthogonal to hedge risk and increase the size of the opportunity, both of which increase your value.

Optionality is your friend

The road to products is long and uncertain; you don’t yet know what will work and what won’t work; you don’t know what the future will look like when you get there. Optionality is your friend.

You may feel unsteady as you navigate the future you’ve only just built, like a musical seventh chord that sounds unstable and open-ended. But just as a seventh chord can resolve to multiple root chords, the uncertain present will resolve to one of several (relatively) fixed states as a consequence of your decisions. Try to remain open to the multiple possible resolutions. Don’t close off paths prematurely or lock yourself into one outcome before you know which will be most valuable.

In an early MPP, strategy is largely about maximizing optionality. In the lab, you’re maximizing optionality as you prototype your value landscape to find the most valuable outcomes. As you develop products, you’re striving to preserve optionality in your platform. Choosing to develop multiple products simultaneously and selecting them to validate orthogonal sets of platform value propositions will help you keep your options open.

Ch6.6 Realizing Value Takes Time... So don't rush it!

You have grand visions of the value you can create in your MPP, but realizing that value takes time. Trying to rush it can cause you to lose it all.

Realizing Value Takes Time

Moderna, the pioneer of mRNA medicines, was founded in 2008 and raised \$2.6 billion dollars in private capital and Pharma partnerships before going public in 2018. Twelve years after it was founded, the company is in mid-stage clinical development for multiple products but (with the possible exception of the accelerated SARS-CoV-2 vaccine) still years from its first approval.

Alnylam, the pioneer of siRNA medicines, was founded in 2002 and went public in 2004, but didn't get its first drugs approved until 2018 and 2019. It took sixteen years to get to its first product and seventeen years before becoming a true multi-product platform! In the early years, the company pursued viral diseases, neurodegeneration, and even agriculture before finally developing the capabilities to deliver siRNA to the liver and focusing on hepatic diseases.

Regeneron took even longer. Regeneron was founded in 1988, went public in 1991, and didn't get its first drugs approved until 2008 and 2011. The company also went through three distinct platforms during this time, starting with neurotrophic factors for regeneration (hence the company's name), then receptor-antibody fusions (the source of the first two approved drugs), and finally chimeric mice to create human antibodies for genetically defined diseases. The antibody drugs that now make up Regeneron's multi-product platform didn't hit the market until thirty years after the company was founded.

To stay afloat for a decade or more, you need to be able to convince people of the future potential of your MPP. Aligning your products with platform value propositions and *selling* that connection is how you buy the time. Persuading people that your product candidates inform and support the multi-product nature of your platform is how you raise money at increasing valuations, whether from public markets, private investors, or Pharma partners.

Beware the Quick-Win Schemes

In a classic episode of *The Simpsons*, Homer reflects on a miracle elixir he's brewing in his bathtub: "After years of disappointments with get-rich-quick schemes, I know I'm going to get rich with this scheme. And quick!"

Funny, yes, but also a warning. The road to realizing value is long and uncertain. You can be efficient and avoid pitfalls, but there are no shortcuts.

Certain common schemes promise a quick win in product development. These schemes tempt us with the allure of certainty and imminent clinical success. Tangible milestones,

a describable clinical path, and linear progress beckon like oases as you emerge from a desert of hypotheticals and optionality.

Don't be fooled by the mirage. Quick-win schemes are risky. Quick-win products come at the expense of your platform. They distract you from building the lasting value creation engine in your MPP. And the stink of failure, once upon you, is hard to scrub off. We've seen the schemes fail (and failed with them!) enough times to be confident in warning you against them.

The Minimum Viable Product The MVP Scheme suggests that a minimal version of your product, missing the costlier and time-consuming bells and whistles, is a suitable substitute for getting into patients early. This scheme falls apart, however, when the minimal viable product doesn't contain the features of your true product – in that case, you aren't actually testing the value propositions of the features of your platform. If the product doesn't validate the value propositions of the platform, its success won't inform subsequent products and therefore won't add value to your MPP. But its (likely) failure will be seen by all the world as a repudiation of the core premise of your MPP, and it will bring the entire platform down with it.

The Magic Asset The Magic Asset Scheme comes about when you find a molecule, often an existing drug with substantial clinical history, that you can repurpose for a new indication, with a fast path into the clinic that avoids costly optimization, manufacturing, and safety testing. The tangible milestones feel like real progress, and suddenly you see the finish line sharply defined. You can tell the world – your team, your investors, your friends – that you have a drug and a disease and a path forward.

Magic Assets often arise in New Technology or New Insight MPPs when you find a molecule that fits with the target or pathway you've just identified. Be careful not to overstate the connection between the Magic Asset and your platform. Was this a one-off? If so, it may not portend multiple products. Also, beware that we often round the Magic Asset up to "good enough" because it's the only thing available. If you've found one molecule, you're liable to convince yourself it's the best you'll ever find.

A risky consequence of a Magic Asset is that it decouples your product development from your platform development. By running out ahead with the Magic Asset, you suddenly find yourself off balance. The way-out-front program will drain resources and focus from building out the platform and prevent you from realizing your multi-product future.

The Can't Miss The Can't Miss Scheme seduces you with a perfect product that makes all the stars align for your platform. It seems too good to be true and irresistibly draws in all your energy. By luring you to accelerate too far ahead of your platform (in football, what you'd call out-kicking your coverage) and decoupling product development from platform investment, the Can't Miss product puts the MPP at risk.

A great example of a company that had a Can't Miss product is Seres Therapeutics, which is at the forefront of treating the human microbiome.^{xv} Flagship launched Seres in 2010 as the world was waking up to the notion that the bacteria in our guts constitute an organ of physiological consequence. Seres' first product was a collection of microbial spores, taken as a pill, that was designed to recolonize the gut of people suffering from recurrent *C. difficile* infection, a deadly disease afflicting hospital-bound patients in whom antibiotics have wiped out their natural flora.

The *C. diff* product was "Can't Miss." Here was a serious disease perfectly designed for a microbiome restoration therapy. Fecal transplants could restore healthy microbiomes and counteract the infection, an idea that provided scatologic titillation for the popular press. ("The Excrement Experiment" graced the pages of the *New Yorker* in 2014.)

Seres rode a wave of anticipation to a large IPO in 2015, the first ever for a microbiome company, but the very next year – in an unpleasant shock – the drug failed its Phase 2 trial. Patients receiving the spores did no better than those who received a placebo. The company's stock cratered, losing \$1B of its value, and the crash threatened to take the rest of the emerging field down with it.

It is worth noting here that clinical trials are not just milestones. They are also experiments. Like any experiment, they can be done well or poorly. They can fail because your product doesn't work or because you don't understand the biology, because you have the wrong controls or because you're measuring the wrong thing. The Seres scientists took a close look at the disappointing trial data to see what they could learn from their failure.

Around that time, lagging platform technologies finally caught up to the "can't miss" lead product and let the scientists better count the number of spores that recolonized the gut, suggesting that a higher dose might be needed. In a surprising twist, it also appeared that many of the *C. diff* patients may not have actually had the infection, because the standard nucleic acid test was not as accurate as a cellular assay that measured *C. diff* toxin.

Armed with these new platform-derived insights and fortunate that their principal investor, Flagship, was an MPP investor and had continued to invest in the company through the lull (whereas product-focused investors would have cut their losses and run), Seres returned to the clinic with a 10x higher dose of spores and a more accurate patient selection protocol based on detecting *C. diff* toxin. The Phase 3 trial, four years later, was a resounding success. The stock price rebounded in an instant, regaining \$2B in value. And Seres, now with multiple spore-based products in clinical development in ulcerative colitis, GI infections, and cancer, was back in the MPP game.

Striking a balance

Time is money. There are benefits to moving fast and taking advantage of the low-hanging fruits of specific indications or readily available molecules. But beware the

Quick-Win schemes that ask you to sacrifice the full future value of your MPP in exchange for a fast first product.

You should neither direct all your resources to one lead product at the expense of the platform (like a fire hydrant gushing in one direction) nor neglect your products in the quest for the perfect platform (like a rising tide imperceptibly lifting all of the boats in the harbor). With the former, you risk becoming a single-asset company and losing your MPP future regardless of whether the product succeeds or fails. With the latter, you risk running out of money before creating anything of tangible value.

Find the balance between the two approaches. Perhaps the proper balance between a fire hose and a rising tide is the Bellagio Fountains in Las Vegas, which have multiple discrete jets of water, orthogonal to each other, of varying intensities, changing over time. Together, the jets create a pattern that tells a story and captures the imagination of the viewer. That sounds about right as a way to build a portfolio of products and prosecute them strategically to create the most value in your MPP.

Key Takeaways

- Your first product must work *and* it must be valuable... But “valuable” means that it validates the value propositions of your platform!
- Let your product development choices guide your platform development.
- Develop multiple, orthogonal, products simultaneously, each with a strong Platform-Product-Market fit to maximize your optionality.
- Realizing value in an MPP takes time. Beware quick-win schemes and make sure your investors are with you for the long haul.

Chapter 7. Creating Value Through IP

Intellectual Property (IP) refers to the patents, trade secrets, and know-how that describe your company's unique compositions and processes. IP is also the only means by which you protect your inventions from competition and secure your share of the market. In an early-stage Multi-Product Platform, you are surrounded by better-resourced competitors that could (and often do!) try to seize upon your idea once you've shown it is viable. They have deeper pockets and could easily beat you to the market in a race. IP is how you fend them off.

Our major focus will be on patents, since that's the predominant form of IP for MPPs. Patents can be tricky business. The novice faces complex rules, arcane styles, and lots of jargon. As you become more advanced, you will begin to perceive the nuances threaded through the objective framework of the law, the arguments you can make, and boundaries you can push as you craft your filings. The expert integrates the full scope of IP – filing, licensing, prosecution, and enforcement – into the company strategy from Day One and uses IP to maximize value.

In this chapter, we'll explore how to create value through IP. We anticipate that you, our readers, will find yourselves all along the spectrum of experience, so we'll start at the beginning. We'll discuss the role IP plays in early-stage MPPs and cover the basics of patents: the rules of patentability, the parts of a patent, and the process of patenting an invention. Then we'll move into strategy. We'll cover how to get what you need from patents, how to maximally protect your MPP, and how to take advantage of time in your IP strategy. Finally, we'll discuss how to integrate IP into the company's experimental and scientific plans and how to incorporate IP into your value creation story.

Ch7.1 Why IP Matters in Early-Stage Biotech

Intellectual property is valuable in a start-up, but it looks different from IP in an established company. In Pharma, IP is literally money. The peak sales for a drug occur during the years it is patent-protected, since generic competitors enter the moment a patent expires. For blockbusters, each extra month under patent can mean hundreds of millions of dollars of revenue. As a result, Pharma patents tend to be very specific to the drug composition and are usually filed late in development, sometimes just before the drug hits the market, to ensure the longest time possible under patent.

In an early-stage MPP, IP plays a different role. You're a long way from marketing drugs, so your IP will lay the groundwork for specific compositions to follow. You may claim technical features of platform-derived products that will be shared by multiple future products. You may experiment with different ways of describing your inventions to see what gets traction at the patent office. Your early IP will project strength and provide a bubble of exclusivity – to ward off competitors, impress investors, and attract partners for whom you're the only game in town.

But beyond the optics, you must recognize that early IP is not just a paper exercise. It is real. The value is tangible. At some point in the future – sooner than you believe – you will make a major license or partnership deal. The partner will be paying you for the rights to your IP. The better your IP, the better the terms of the deal. And sometime in the not-too-distant future, you'll want to block a competitor who is infringing on your IP. The better your IP, the more likely you will be able to ensure market exclusivity. So take it seriously.

Get a Lawyer... But be your own best advocate

We are not your lawyers. Consult your lawyers before you make any decisions on legal matters. But there are several reasons to educate yourself as well.

First, your lawyers can only work with what you give them. They'll react to you, to the way you describe your technology, to the opportunities you perceive, and to distinctions you draw between your technology and others'. The more you know about how the patent system works, the better you'll be able to advocate for yourself and your company.

Second, patent law is a set of rules – objective, yes, but ever-evolving. And to describe how your particular invention fits into this web of rules, subject to some rules and not to others, is to make a subjective argument. To win the argument, you need to understand the patent system, its rules and boundaries, the levers at your disposal, and strategies for building value within it. You, the scientist-innovator-entrepreneur, know your technology better than anyone. You are the one who can design an experiment to get the data that supports the strongest possible claims. You are most motivated to create the most value for your company. IP is a big hammer, and you need to know how to wield it.

Ch7.2 Patents 101: The Rules of Patentability

A patent is a time-limited government-granted monopoly. If you invent something and then apply for and receive a patent on it, you can prevent others from making, using, or selling your invention for a period of time, typically twenty years. A patent gives you the right to exclude someone else from practicing your invention, but it doesn't affirmatively stop the other party from trying. It's up to you to police your patents and sue infringers when you deem it necessary.

The right to patent is based in Article I of the Constitution, a bargain made to accelerate technological progress in the young United States. The exclusive right to the profits of your invention for a limited period of time is the *quo*; the *quid* is that you must describe the invention in sufficient detail so that others can make it and build the next generation of innovations on top of it.

What makes something patentable?

The rules of patentability vary from country to country, but their framework is generally similar to that of the US. There are four simple rules. For an invention to be patentable, it must be (1) patentable subject matter, (2) novel, (3) non-obvious, and (4) described in sufficient detail. Let's look at each of these requirements in turn.

Patentable Subject Matter To be patentable, an invention must describe a new and useful process, machine, manufactured article, or composition of matter. The invention must have been “touched by a human hand,” which means you can't patent laws of nature, natural phenomena, or abstract ideas. Of particular relevance to biotech, you can't patent products of nature or processes that are inherent to the underlying biology.

Novel To be patentable, an invention must be new. It can't be something that exists already, is known already, or is described already in the prior art. (Prior art refers to any public information, of any form, that predates your patent filing.) Importantly for scientists, this means that any disclosure of your invention, even by *you* – publishing in a journal, giving a public talk with slides, presenting a poster at a symposium – prior to filing a patent renders the invention not new and therefore not patentable.

Non-Obvious To be patentable, an invention must be non-obvious given what is known already. Clearly, what is obvious to one person may be different from what is obvious to another, so the legal standard is a hypothetical “person having ordinary skill in the art” (with the cheeky acronym PHOSITA). If a PHOSITA would deem the invention obvious, then it is not patentable.

Sufficiently Described To be patentable, an invention must be described in sufficient detail to enable a PHOSITA to make and use it. Importantly, you don't have to physically build the invention or “reduce it to practice” in order to get a patent on it. You just have to describe it in sufficient detail (“full, clear, exact terms”) to prove that you know what it is.

The official requirements for patentability are set out in Title 35 of the US Code of Laws, which you can easily access online. Section 101 describes patentable subject matter. Section 102, the requirements for novelty. Section 103, non-obviousness. Section 112, enablement and written description. (Just in case, Section 105 covers inventions made in outer space.)

IP lawyers often refer to these requirements in shorthand, so you may hear phrases like “We got a 102 rejection” or “This suffices for 112.” The conversations may make you feel as though you’ve been dropped in the old story of the prisoners who have memorized all their jokes and amuse themselves by reciting them just by number. But if you know what the numbers mean, you can join the conversation.

The rules may be simple but their interpretation isn’t

In the joke’s punchline, when the new prisoner tries his hand and calls out “54,” he is met with silence and a few groans. His cellmate shrugs, “Some people just can’t tell a joke!”

Likewise, the rules of patentability may be simple to state, but their interpretation is not. Some of the fiercest debates in patent law history have played out in the courts over these four simple requirements.

Is a cancer-causing DNA sequence that you extract from a patient’s cell, amplify out of its genomic background, modify with new flanking sequences, and clone into a synthetic plasmid to use in a diagnostic test a product of nature (unpatentable) or a composition of matter touched by human hand (patentable)? (Unpatentable, according to the Supreme Court in *Association for Molecular Pathology v. Myriad Genetics* in 2013.^{xvi})

If a drug is metabolized in the body and you measure the metabolites to determine whether the dose needs to be adjusted, are you simply combining a natural phenomenon with abstract thought (unpatentable) or have you invented a non-obvious process for diagnosing and using the drug safely (patentable)? (Unpatentable, according to *Mayo v. Prometheus* in 2012.^{xvii})

If you have a patent on a humanized mouse for discovering antibodies where you’ve cloned human antibody variable region sequences into the mouse immunoglobulin locus (*Regeneron*), and someone comes along and puts the entire human antibody variable region (not just individual sequences) into the same locus (*Kymab*), is that an obvious extension for which you’ve provided enabling written description (you can block them), or would a PHOSITA consider it a non-obvious improvement (you can’t block them)? (Non-obvious and therefore not blocking, according to multiple international courts in *Regeneron v. Kymab*, 2020.^{xviii})

For all the formality of the law, the boundaries of patentability are not set in stone. Their application is debated with every new case, and new questions emerge alongside new technologies. The answers have big implications – financial, of course, for the ability to

market a product or service, but also because they can shape entire industries by setting precedents for what types of patents are valid.

You had better be ready. If you are successful, you will find yourself in court. It's the nature of capitalism that when there's money to be made, there will be competition for it. Whether you're enforcing your patents or invalidating a competitor's, you will be at the center of these arguments over patentability.

As you can see from the examples above, the rules are simple, but their interpretation is subject to debate. The words used to describe an invention matter. And science comes down on both sides, because both sides' arguments are firmly grounded in science and data. It matters all the more, then, that you be knowledgeable about patentability so that you can craft the strongest possible arguments and marshal the science in your favor.

Ch7.3 Patents 102: Parts and Process

All patents share a basic structure: a *specification* that describes the invention and a set of *claims* that specify what is covered by the patent, that is, the boundaries of the patent holder's right to exclude others. Patents also follow strict timelines as they are filed and prosecuted. To get the most value from IP, you need to know what's in the sections, how to read and write them effectively, and how to optimize for time.

In this section, we'll cover the basic parts of a patent and the process of patenting your invention.

Parts of a Patent

Specification The specification is the written description of the invention, per section 112, "in such full, clear, concise, and exact terms as to enable any person skilled in the art...to make and use the same." Alongside the description are examples, essentially any data and methods that you've developed in support of the invention.

We talk about "the invention," but the specification never describes just one thing. In fact, reading a specification can be tedious. You'll find laundry lists of diseases, tables of alternate chemical side chains, and seemingly absurd ranges for every value (e.g., "the particle diameter is 1000 nm, 750 nm, 500 nm, 250 nm, 100 nm, or less than 100 nm").

The madness has a method. Any claim you write must have literal support in the specification. If you want to be able to claim variants on your molecule to treat multiple diseases, they had better be listed in the spec. And if the examiner finds that 1000 nm particles are described in the prior art, you will need to have specified smaller diameters in order to get claims to those sizes.

Claims The claims list the specific compositions and methods that you can prevent others from making, using, or selling. Claims are typically nested, with an independent claim followed by several dependent claims that are variations on the first. Compared to the expansive specification, claims are concrete, but that doesn't mean they have to be narrow.

One of the most valuable forms of IP is a broad foundational claim that covers many specific instantiations of an invention. Genentech famously owned the process for making antibodies through its Cabilly patents (named for the inventor), whose independent claim read:

- (1) A process for producing an immunoglobulin (Ig) molecule comprising at least the variable domains (Fv) of the immunoglobulin heavy and light chains, in a single host cell, comprising the steps of:
 - (i) transforming said single host cell with a first DNA sequence encoding at least the Fv of the Ig heavy chain and a second DNA sequence encoding at least the Fv of the Ig light chain, and

(ii) independently expressing said first DNA sequence and said second DNA sequence so that said Ig heavy and light chains are produced as separate molecules in said transformed single host cell.

The claim is precise and concrete, and it also covers virtually any method of producing an antibody with recombinant technology. As long as you make your antibody in a cell to which you've given a heavy chain sequence and a light chain sequence, you're infringing the Cabilly patent. It doesn't matter what antibody you make, what cells you use, or what type of bioreactor you grow them in. For thirty years, anyone who wanted to make an antibody had to license Cabilly from Genentech, which brought in hundreds of millions of dollars per year in licensing fees.^{xix}

Granted Patents versus Patent Applications

Laypeople talk about "filing a patent," but technically, when you "file a patent," what you've actually filed is a *patent application*. A patent application is an opening salvo, your proposal to the Patent Office about what you've invented and how to claim it. The examiner assesses the patentability of your invention, what's known in the prior art, and whether your specification provides sufficient written description and then works with you to come to an acceptable (usually narrower) set of claims. Only then do you get a *granted patent* with rights that you can enforce against others.

Though the specifications are the same between an application and a granted patent, the claims of an application are usually much broader. But remember that the claims have no force behind them until the patent is granted, if it ever is. When you analyze patents for competitive intelligence or business strategy, make sure you know whether you're reading a patent application or a granted patent.

Getting your patent granted does not grant you immunity from future lawsuits. There are nearly 10,000 patent examiners. It is entirely regular for two patents with seemingly overlapping claims to be granted by separate examiners. It is entirely regular for a competitor to look at your granted patent and say "I don't think that claim is patentable as written" or "I don't think that claim infringes on my claim." Such disputes are resolved in court.

Timelines of Patent Applications

Patents follow strict timelines as they are filed and prosecuted. Here is a typical path.

At $t=0$ you file a priority document, often a US *Provisional Patent Application*. The date you file is called your priority date. In a dispute, the priority date determines who invented something first and separates knowledge that existed before the invention (prior art) from knowledge that came after it. You never know who else is out there inventing, so don't delay in setting your priority date. Also make sure you file your priority document before you disclose your invention – say, by publishing a paper or giving a talk. For our academic readers, be sure your technology development office is aware of your inventions and upcoming disclosures and is prepared to file on your behalf.

At *t+12 months*, the provisional patent application expires unless you choose to *convert* it to a standard non-provisional application. The conversion date starts the clock on your twenty years of exclusivity (should a claim eventually be granted). The conversion date is your last chance to modify the specification or to add new data that you've generated since the provisional application. Remember, any claims you may want to derive from this application at any point in the future must have literal support in the specification, so don't leave out anything important!

The conversion date is when you set the specific claims to prosecute and what countries to prosecute them in. The domestic path runs through the US Patent and Trademark Office (USPTO). The Patent Cooperation Treaty (PCT) covers most of the rest of the world, including the EU, Canada, Japan, China, and Australia. Each country costs money, so if you have limited resources, you may want to focus just on major markets – but remember you need patent protection in a market to drive sales or partner the drug, so balance today's costs with future value.

At *t+18 months*, the patent application *publishes*. At this point, others will be able to read your patent application and learn what you're doing. In PCT countries, the examiner's first search report of the prior art will also publish, so competitors can analyze your prospects for getting claims granted. From a competitive intelligence standpoint, the publication date marks a turning point if you had been stealthy until now.

Beyond *t+18 months*, the patent prosecution process continues. It can take anywhere from several months to multiple years, if claims are ever granted. You can file additional sets of claims from the same specification – continuations of the original filing – that will keep the original priority date but start a new twenty-year clock.

Ch7.4 Know what you need from a patent

The first piece of patent strategy is to know what you need from a patent. How you read a patent, how you understand it, and how you decide to engage with it depend entirely on what you need from the patent.

Patentability vs. Freedom to Operate

It is common to need to distinguish between asking a question of Patentability and asking a question of Freedom to Operate (FTO). This distinction can befuddle even the most experienced biotech entrepreneurs.

In a patentability scenario, you wonder: Given the patents in front of me, is my idea patentable? Often you're getting ready to file or are choosing how to craft your claims. You've met the criteria of patentable subject matter and written description, and now you're wondering whether your invention is novel and non-obvious given what's in the prior art. Remember that the prior art includes anything disclosed in writing – papers, granted patents, patent applications, abandoned patents. Therefore, as you read competitors' patents for patentability, you need to focus on their specifications, or their laundry lists of diseases and molecules, to see what impinges on your invention.

Because specifications tend to be written broadly, you'll often find a number of phrases that touch on your idea. It's up to you to figure out how to describe your invention so that its novelty and non-obviousness stand out. Be creative with language. Are there differences between the ways the competitors describe their molecule or process and what you've built? Is there a step in a process that makes it new? Can you carve around the specific examples described by the competitor to claim everything *but*? Is there an element of your method that would surprise a PHOSITA? In our experience, it is always possible to describe an invention to make it patentable given the prior art, but you have to be willing to delve deeply into the features, data, and language of your invention.

In a freedom to operate (FTO) scenario, you wonder: Given the patents in front of me, do I have the right to practice my invention or perform a set of activities? In contrast to patentability, where you focus on competitors' specifications, when you assess FTO your primary concern is the *claims* in competitors' patents. What do they have the right to enforce against you? It doesn't much matter what they disclose in their specifications (though be aware that future claims can issue from the same spec). Also, a current blocking claim might expire before you reach the market, leaving you in the clear.

It may seem counter-intuitive to say that you can have a granted claim but not be free to practice it. Remember, though, that a patent doesn't give you the right to affirmatively practice an invention. It only gives you the right to *exclude* someone who infringes your claim.

Let's take the Cabilly patents as an example. Imagine you invent a way to culture platypus cells and produce proteins from them. The invention is patentable. But if you

use platypus cells to make antibodies, you'll need a license to Cabilly. You don't have FTO.

Imagine you invent a new DNA sequence that increases the secreted yield of protein when put upstream of a gene. The invention is patentable. But if you put the sequence in front of heavy and light chain genes to make antibodies in a cell, you're infringing on Cabilly. You don't have FTO.

If you don't have FTO, you will either need to license the blocking patent or sue to invalidate it. (We won't cover invalidation here, since it usually requires a legal budget far beyond what a small start-up can afford, but suffice it to say you would try to make an argument that the competitor's granted claim is not patentable as written and therefore should be thrown out.)

Getting what you need from a License to IP

A license gives you the rights to use a piece of IP that you don't currently own. When you enter a licensing agreement as a start-up, you want to make sure you have the rights to get the value you need from the IP. These are the questions to ask yourself:

- *Is this the right IP?* Does this IP give you FTO? You don't want to pay for a license to IP that still leaves you unable to practice your desired activities. If you're licensing an application that isn't yet granted, does it support the type of claims you would want to eventually get? Make sure you've analyzed the IP landscape ahead of time to avoid unfortunate surprises down the road.
- 2. *Do you need exclusive or non-exclusive rights?* The license can be either exclusive, meaning only to you, or non-exclusive, meaning others have the right to license the IP as well. A non-exclusive license is sufficient for FTO. An exclusive license will also let you block others from practicing the invention. You'll pay more for exclusive rights. Decide which you need.
- *Do you have the right field?* The license will specify the field in which you have the rights to the IP. Make sure the field is broad enough to capture all the uses you envision today and in the future: a specific disease, all human therapeutics, therapeutics and diagnostics, or all human and animal health applications. The same patent can be licensed to multiple parties exclusively as long as the fields don't overlap. You don't want to find yourself with a cancer drug that turns out to be useful in treating autoimmune disease, but without the rights to use the drug in that application.
- 4. *Can you operate like a business?* As the licensee, you want to be able to operate as a business and use the patents accordingly. This means the right to enforce the patents in court so that you can prevent others from infringing and the right to sublicense the patent to future partners or affiliates. This stuff is boring, but very important!

For most early-stage MPPs, the licensor is a university technology transfer office. For many entrepreneurs spinning an invention out of an academic lab, this transaction is the necessary first step of their entrepreneurial journey. A typical license agreement will list the IP to be licensed; the field of use; the exclusivity; a common set of financial terms, including up-front, milestone, and royalty payments; and a set of diligence requirements to prove that you're developing products from the IP.

Different academic institutions develop their own expectations for fees and royalties, and venture firms that do dozens of deals a year have their own sets of comparable transactions. Our experience is that there is a large zone of potential agreement (ZOPA) in the negotiation because both sides are motivated to see the company succeed with the IP – the company for obvious reasons, and the university because the IP will bring in revenue and good press about the utility of the research at the institution.

Technology transfer offices are afraid that a licensed invention will sit undeveloped, hidden from competitors or used as an IP cudgel rather than transformed into useful products. Hence diligence requirements in the agreement. As the entrepreneur, try to anticipate how busy your life will be and don't sign off on overly onerous requirements. A simple amount spent per year developing the technology or a development timeline on par with your product development plans is a good way to go.

When not to worry about IP

Despite the importance of IP, there are a few times in the early days of your MPP when you shouldn't worry about it.

First, don't read IP during your exploration. When envisioning new technologies or potential uses of your ideas, the patent literature will induce despair: it will seem that everything has been thought of already. Specifications are written in extensive and broad language, but it's (almost) always possible to find a path to patentability for your invention. Don't let your naive perception of the prior art inhibit your creativity.

Second, don't worry too much about infringing patents during your early experimental work. You're off the hook because of the biotechnology Safe Harbor provision of the Hatch-Waxman Act of 1984, which lets generic drug manufacturers test their molecules against branded originals. Safe Harbor essentially lets you make and use on-patent material during preclinical development, as long as the work is eventually "for submission to a regulatory agency."^{xx} Legal experts we've spoken to note that there is a J-shaped liability curve with respect to Safe Harbor, meaning the bigger you are and the closer you are to market (the greater the competitive threat you pose), the more your actions will be scrutinized. But in the early days of an MPP, don't let the patent literature unduly curtail your experimentation.

Ch7.5 Protecting your MPP

Now that we've covered the basics of patenting, it's time to dig into strategy for protecting your MPP with IP.

File broad claims before narrow claims

Simply put, you should file broad claims before you file narrow claims. Patent law uses the language of biology here: a broad claim is a “genus” that encompasses within it individual “species.” The order of operations matters. You can patent the species of a disclosed genus, but once a species is known, you can't go back and claim the whole genus.

For example (ignoring the clear violation of patentable subject matter), you could patent a dog (e.g., a mammal with four legs and a tail that licks its owner's hand and is loyal) and then subsequently claim the species Chihuahua, husky, and labradoodle. (Pop quiz time: The species claims would be patentable but would not have FTO!) But if the Chihuahua is already described in the prior art, it's too late to get a patent to the genus *dog*.

You can't claim a genus if the species already exists, so make sure your early claims are genus-level claims. Push the boundaries of your early patent filings to make sure they encompass the broadest genus of your invention. Choose your words carefully to anticipate the future and claim the largest defensible territory.

Protect your MPP with multiple layers of IP

A strong patent portfolio is never just a single patent. Instead, protect your MPP with a web of patents with multiple overlapping claims that provide multiple layers of support and protection from multiple angles of attack. Like tightly woven fabric that is impenetrable to sharp blades, the layers of your IP portfolio will guard the body of your MPP from slings and arrows. Armed with claims that describe different aspects of your platform and describe your platform in different ways, you will have multiple shields against complex prior art and more ways to challenge infringers.

In our MPPs, the base layer is often comprised of foundational platform patents that cover underlying technologies, followed by patents that cover families of related products and topped with product-specific filings. Platform patents themselves can be layered too. To return to your F.A.B. framework (Ch3.2 **Features, Advantages, and Benefits**), you should strive for sets of claims that cover different specific features of your platform. The advantages of those features provide your novelty in relation to the prior art; the benefits provide your utility.

In New Modality MPPs, you can often describe your future compositions early on with sufficient clarity to meet the written description requirement for patentability, well before the molecule is ready for the clinic, and thus get product-specific claims granted early in the life of the company. But early IP is not the only IP. As you build your technologies

and develop products, you will continue to layer specific IP on top of this foundation. Later IP will include precise molecular compositions and manufacturing processes with claims that are not as broad but are deeply specific to your platform, which makes them an essential component of your portfolio.

Different types of claims have different relative value

Depending on the nature of your invention, you may want to protect a composition of matter, a process, or a method. These types of claims have different relative value based on how easy they are to get, how broad they can be, and how readily you can police them.

A *Composition of Matter* claim describes a thing like a molecule, a protein, a nucleic acid sequence, a cell, or a bioreactor. Composition claims are the strongest type of claim because they are specific and definite, and it's easy to determine whether a competitor's composition infringes.

For example, an early foundational CAR-T patent claims:

“A nucleic acid polymer encoding a chimeric T cell receptor comprising: (a) a zeta chain portion comprising the intracellular domain of human CD3 ζ chain, (b) a costimulatory signaling region... [corresponding to human CD28]; and (c) a binding element that specifically interacts with a selected target.”

The claim is simple, definite, and genus-level. It covers any synthetic receptor with three parts: a binding domain (to any target you want), a CD28 costimulatory sequence, and the intracellular domain of CD3. It describes the chimeric antigen receptor sequence – not the cell that carries it, the method of expressing it, or even its specific target. Because of its simplicity, it is easy to police the claim if another CAR-T infringes. (It happened in 2019: a \$750M judgment was awarded in favor of Juno against Kite.^{xxi})

A *Process* claim describes the steps used to make a composition. Process patents can be very strong if they cover an essential process that many people use, like the Cabilly patents for making antibodies. Because process patents have several steps, though, it can be easier to work around them than around composition claims.

For example, you can patent an improvement to a known process as long as you have at least one step that is different. (Generally, you can add a new non-obvious step, but you can't simply remove one of the existing steps, since you'd be guilty of claiming a genus when the species is already described.) To have FTO around a patented process, you need to show that your process skips at least one of the patented steps. The broadest, most valuable process claims list the minimum number of essential steps.

A *Method* claim describes how the invention is used. Method of use claims are generally less valuable than composition or process claims because they rely on having the rights to the underlying composition or process, but they can be useful added layers of protection.

A specific flavor of method claim describes a way of discovering a molecule. If your MPP is based on a new technology for identifying targets or a new insight into the relevant pathways, you might be tempted to patent the steps you take to come to your discoveries. The danger here is that method of discovery claims are very difficult to police. People rarely disclose their early discovery work publicly, which makes it nearly impossible to determine (much less prove) that someone is infringing on your patent. Furthermore, since no product is being sold yet, there won't be much in the way of monetary damages.

If you can't know whether someone is using your secret recipe, you might as well not divulge it. In some cases, keeping methods secret is a better choice than patenting them. The recipe for Coca-Cola is a trade secret. The copper-tin-silver alloy in Zildjian cymbals is a trade secret. Trade secrets make sense for information that is valuable to competitors, not known outside the company, and difficult for others to reverse-engineer. (You can sue someone for stealing your trade secret, but you have no recourse if the information leaks out or is independently discovered.) While we never entrust the protection of our MPPs to trade secrets alone, it is important to know that patents are not your only tools of self-defense.

Ch7.6 Filing Strategy

As you file the various pieces of IP in your portfolio, you can build value in the process. Here are some ideas for doing so.

Be Thoughtfully Expansive in your Specifications

You want to be expansive in your specifications. As we've discussed, you can file multiple sets of claims from the same specification. That means you can tap into this early filing to get an early priority date for later patents, extending the clock while keeping your place in line – as long as there is literal written support for the new claim in the old specification. Furthermore, you want to be certain that the specification has support for genus-level claims (that, is, that it's not so narrowly written that you prematurely describe just one species). Try to anticipate the future that you will wish you had specified, and specify it today.

Writing expansively also matters because your specification enters the prior art and can make life difficult for competitors who may try to piggy-back on your invention. The more thorough your description of adjacencies to your invention, the higher the bar for novelty will be for your competitors.

But you need to be *thoughtfully* expansive. Prior art cuts both ways. Be careful not to prior art your future self! Know what additional filings are in your plan across your platform, and make sure they aren't anticipated by an excessively broad first specification.

Use time to your advantage

We've covered patent timelines in Ch7.3 Patents 102: Parts and Process. You can take advantage of patent timelines to move quickly and maximize your value.

First, recognize that you don't have to have the invention in hand to file. You can file a *prophetic* provisional patent application that describes the invention before you have any data. You then have twelve months (the provisional application *conversion date*) to generate the data to support the invention. You might choose this path if you want to grab the earliest possible priority date. You may not know who your competitors are yet, but they're out there. We've been involved in enough cases where priority is determined by a matter of days or weeks that the idea of waiting until the full data package is in hand to file makes our pulse race.

If you don't generate the necessary data to support your invention by the conversion date, you can abandon the prophetic filing. An application that is abandoned before the conversion date just disappears. It never publishes and never enters the prior art. You may then re-file your prophetic application to restart the twelve-month clock with a new priority date, though you'll incur the costs of IP churn.

For applications that are enabled with data, another way to move quickly is to ask for an accelerated review (Track One) of your application at the USPTO, where, for an extra fee, you can guarantee a final disposition within about twelve months. Getting a claim granted early doesn't change anything about future prosecution: it doesn't affect the priority date of the invention, should someone sue you for infringement; it doesn't affect a court's decision about patentability, should someone sue to invalidate your claim.

But getting an early claim can be important for an early company. Knowing early what types of claims the patent office will grant can help you plan your future filing strategy. The optics of a granted claim can give you a real boost with investors and lend credibility to your overall IP strategy. Plus, you can begin to drive value with the patents, either by enforcing them against infringers or by attracting partners for whom you're now the only option to access the technology.

Ch7.7 Integrating IP and Value

You have an important role in all of this. The master practitioner knows how the legal system works, how to read and write patents, and how to be a good partner to the IP lawyers. Even more importantly, the master integrates IP strategy into the broader company plan, knows how to use data to get the broadest possible claims, and designs experiments to complement IP needs. Only when you're able to do these things can you create the most valuable patents, ensure that you're granted the strongest possible claims, and capture the most value from your IP estate.

Argue with Data

A recurring theme in our discussion of IP is that it's all an argument, and you can win it with data.

For example, you'll file an application and the examiner will come up with a list of references in the prior art that render your claim not patentable. The examiner will cite patents that describe something similar, or a set of papers that together make your invention obvious, or an amalgamation of facts that cloud the case for patentability. You will always get rejections. So what do you do?

You can choose to accept the rejections – to narrow your scope, discard your claims, go back to the drawing board and try again – or you can fight back!

And you fight back with data. You can argue that the references do not in fact describe what you're talking about, through a close reading of their data and yours. You can argue that it's not obvious to bring this collection of facts together, for example by finding art that points away from the invention. If people thought X and you showed Y, then Y couldn't have been obvious to a PHOSITA! Were the results surprising? Use negative experiments to prove that inventive experimentation was needed – that a result is not inherent to the biology. Find ways of characterizing your compositions of matter such that they're empirically and demonstrably different from what the examiner cites as prior art.

You have agency in this argument. You should – you must! – be prepared to do battle for your IP, for the broadest possible wording of your claims, and for the strongest possible interpretation of your results.

Incorporate IP into your Experimental Plan

By now we've said many times that you should center all your activities on value (Ch3.4 Orienting Around Your Value Propositions) and that the experiments you choose to do should be pulled into existence by the value propositions you need to demonstrate (Ch5.1 You have to do the RIGHT experiments). This is one more example of these principles. IP is a powerful driver of value, so it is *absolutely* worthwhile to design experiments with the goal of getting the most valuable IP.

What does this process look like? Perhaps you've filed a prophetic provisional application to reserve an early priority date, and now you have twelve months to generate the data you need to enable your claims. Make sure you include these experiments in your plan. But it's not enough to *do* them. You have to do them with an eye on the IP value you'll build.

You want to get the broadest possible genus-level claims. So make sure your experiments are designed to generate broadly applicable results that prove the genus and not just a single species. If you want to claim a cell that expresses multiple therapeutic proteins, think about how to express the largest number of distinct proteins in a single cell. If you want to claim that any binding domain will work in your chimeric antigen receptor, make CARs with several different binding domains. If you want to claim that your pathway is relevant in multiple diseases, make sure you run your assays in multiple model systems. If you want to claim that your process doesn't depend on a single parameter (or, conversely, if you want to hammer home that a single invariant step is the most important one), prove that this is the case. Design your experiments to get the data to support the broadest possible claims.

Anticipate the prior art you're likely to face. Design experiments to support your future argument about why your invention is patentable over the prior art or has FTO around existing claims. Know how the prior art describes its compositions and do experiments specifically to measure where yours are different. Characterize your platform features with an eye toward competing technologies so that you can prove novelty to the examiner.

And when data rolls in, when surprising results arise, when you prove that you're able to do something that you had only previously hypothesized – recognize what you have, advocate for it to your legal team, and prepare to file. You have no time to waste. If you can invite your IP lawyers to lab meetings, do so. Emerging data will trigger you to plan your next experiments, adjust your resourcing, and hone your story, so make sure your IP portfolio and strategy keep pace with the data.

Write Responsibly, Cautiously, and Minimally

So much of patents hinges on language. Do you have literal support for your claim in the specification? Do you describe how to practice your invention sufficiently well for a PHOSITA to practice it? Does your claim steer clear of the prior art?

Careless language can just as easily destroy your patent. When you succeed and are sued by those hoping to invalidate your patent, you will be subpoenaed and required to turn over all relevant communications: every email, document, and text message related to the case. If in any of those communications you've written anything in which you suggest that your patent may not be valid, or may be superseded by another, or may have different inventors, or may not have freedom to operate – no matter how naive your assertion, no matter how factually incorrect per your current understanding and arguments – the fact that you have written it will be a severe blow to your case.

Never write anything about your own or another's patent. That means no opinion about any patent in an email, no commentary in any slide, no appraisal in any document. Even when directly communicating with your IP lawyers under attorney-client privilege, express your opinions of your and others' patents in person or over the phone. You don't want an email saying "Hey, check out this patent – I worry it might block us!" coming to light during a trial. Instead, use innocuous phrases: "See this patent. Let's discuss. Please call me." Your lawyers can help you determine how your patent stacks up against others and find the best arguments in your favor. Have those discussions in confidence and in person, not in writing.

Similarly, don't write opinions about your experiments. As we just discussed, experiments are an essential part of your patent argument. Written comments that call into question your experimental data, suggesting that results are wrong or noisy or irreproducible, could invalidate your patent. Address concerns about the experiments in person or over the phone. Make sure the data in your patents is unassailable – it is illegal to misrepresent data to the patent office – but don't leave a discoverable paper trail of unformed opinions as you build to certainty.

You must protect your MPP in order to realize its potential. Design your experiments and technology development with future patent filings in mind. Behave and speak responsibly with future litigation in mind. Argue for the broadest and strongest protections you can get. Become a master of IP and capture the maximum value in your MPP.

Key Takeaways

- You must know how IP works so that you can advocate for yourself and protect your innovations as strongly as possible. IP is valuable in a start-up, but it looks different from IP in an established company
- A strong IP portfolio for a Multi-Product Platform will have many layers and different types of claims that cover the platform, the multiple product families, and individual products themselves. There is value to getting early claims granted.
- The rules of patents (patentability and FTO) are simple but their interpretation is not. Incorporate your future IP arguments back into the experiments you plan and the data you generate so that you can make the strongest possible case in your favor.
- Don't sabotage yourself. Communicate responsibly.

Chapter 8 Selling a Valuable Story

If there's one lesson that should be apparent in this book so far, it's that value is subjective. Over and over we've seen that the future value of an early MPP is a function of your assumptions, present decisions, and future actions. Value can be quantified, but it is not a fixed quantity. It is conditioned by your initial conditions but is not pre-determined by them. It is describable, but it's up to you to describe it. Value is in the eye of the beholder.

Value is objective in the same sense that a microscope whose lens you control is objective. You control what is in focus; you command attention where you want it; you magnify what matters and blur what doesn't for maximum effect.

This control is the act of selling. Selling is how you convey the value inherent in your company. Selling determines what people see: a transformative rocket ship MPP, a mixed bag of risky assets, or a science fair project. Selling is how you amass the resources you'll need to make your company vision real. Selling is essential if you want to be successful.

Among scientists, selling gets a bad rap as a shameful tool used when the data doesn't speak for itself. But data never speaks for itself. It needs context; its implications must be extrapolated. This is especially true when you are building a future, where, by definition, the data is scant or incomplete. And you, as the scientist-entrepreneur, must provide the context.

At any bookstore you will find shelves upon shelves of business books about selling, many of them good but aimed at readers with careers in traditional sales. In this chapter, we'll explain how selling manifests in building innovative MPPs and how those of us reared in the sciences can adapt to doing it well. We'll define what we mean by selling and explore what makes an effective sales pitch for an early-stage MPP. We'll tap into the psychology of persuasion to help you present well when you're selling and offer our tips for getting good at it.

Ch8.1 An Introduction to Selling

Let's begin by defining our terms. When we say "sell," we mean to champion your idea, to evangelize for the most valuable version of your company, to advocate for its brightest future. To sell your MPP is to embody the vision and potential of the multi-product future that is made possible by your transformative platform. If you, the scientist-entrepreneur at the heart of the venture, can't sell this vision, then who can? No one else is as well-positioned as you to make the strong case for the future value of your MPP; no one else has as clear-eyed a view of its potential.

Defining Selling

Selling is a transactional conversation between you and another party. It's transactional because you're trying to get something out of it, though it doesn't always result in a physical or financial transaction. Three things happen when you sell: you inform, you persuade, and you compel the other party to action.

When you sell, you must first inform the other party of who you are, what problem you aim to solve, what solution you've built, and what data you have. Information is both overabundant and scarce. You can't possibly convey all the details of your MPP in one sitting, so you get to choose what information is necessary, how important it is, and how to structure it as an argument. At the same time, because you've just started, there are big gaps in what you know, including hypotheses that are not yet known not to work, which leaves room for different interpretations of what is possible and what is likely to occur.

When you sell, you must then persuade the other party to care about the outcome. You must convince them that the problem is real and the solution impactful, that they desperately need what you are building and that you can, in fact, build it. From the initial remove, you have to bring them to your view of the world so that they "get it" and it becomes their view too. For a transformative MPP, this can be a big leap. Sometimes the mere description of your platform – whether it is a new modality to treat disease, a new tool to discover and make drugs, or a new insight to tease open biology – seems fanciful. It may not have even entered your listener's realm of the imaginable until your words brought it into existence. And yet you must persuade.

Finally, when you sell, you must compel the other party to act on their conviction. They may act by giving you money, giving you time, or choosing to work with you. Beyond just changing their perspective to yours, you must motivate them to participate in creating your vision. Mind you, you're not begging them to participate. Instead, you're making it impossible for them to miss out on the opportunity. Compelling is not driven by logic alone; you must also play on your listener's emotions and tap into the limbic brain.

You can think of a selling interaction as occurring in your value "potential energy" landscape. You're sitting at State A and you want to get to a more valuable State B. You would have gone there already but for an activation barrier (technological, financial, or temporal) that keeps you at your starting point. If someone can help you get to State B,

you must sell them on helping you. You must champion the goal of getting to B and convince the listener that you're the best person to do it. You explain the two states, their relative values, and the activation energy barrier between them. You persuade them that State B is worth reaching and that the activation barrier is surmountable. And ultimately, you compel them with an emotional kick to the pants, exciting them sufficiently such that they hop with you over the barrier.

You're Always Selling

Entrepreneurs typically think of selling as "pitching" – what you do in a conference room of investors, at a sponsored networking event, or in the metaphorical elevator. But the truth is that selling is both more common and more nuanced than pitching.

When you advocate for building an MPP based on your exploration to a colleague, a partner, or yourself, you're selling the potential value of a set of nascent ideas and the value of your creativity and entrepreneurial vigor to realize it. When you recruit scientists to join your as-yet-unnamed prototype company, or executives to leave their comfortable Pharma jobs and join you, you are selling the potential of the company, the excitement of the journey, and their role in bringing it to fruition. When you argue to your board of directors that your plan for developing multiple simultaneous products is the right one, you are selling your version of the future of the company and your ability to lead the organization. And when you present the company to future investors or partners, you are selling the potential of your technology, the future it unlocks, and the value you've created and will create. The audience can be anyone and everyone. You are always selling.

But it's not just entrepreneurs who sell. The best executives excel at selling – inwardly to motivate their teams to tackle seemingly impossible problems, and outwardly to investors who need to realize that value has been created and will be created. The best professors in academia excel at selling, inwardly to recruit and inspire students and outwardly to the academic community, scientific journals, granting agencies, and the press as they explain the impact of their discoveries. Even the best scientists excel at selling. They tell you just what you need to know about the data they generate, persuade you of its implications, and compel you to support their plan for what should be done next.

Selling is such an essential skill that we've taken to asking for it in interviews with our C-level executives (not just CEOs, but also CSOs, CTOs, and CFOs). Not asking *about* it, but asking for it directly: "Sell me something." Can the candidate triage the appropriate information, persuade us to care, and compel us to act? Selling is an essential skill that we've learned we need in the leaders of our early MPPs, but it's a skill that is nearly impossible to assess from a résumé. To believe it, we must see it.

Ch8.2 Selling your MPP

The first thing to keep in mind when selling is that, fundamentally, you're making an argument. You're making an argument that your MPP is on a path to a valuable future and therefore is valuable today. You're making this argument with data, but it's not an argument *of* data. The data serves to support your value propositions: those So Whats that make someone care, those benefits that your platform provides to a user as a function of its features and advantages.

To sell effectively, begin your argument with your So Whats, and then strategically unfurl your data. If you've oriented your experiments around your value propositions (Ch3.4 Orienting Around Your Value Propositions), your data should already be answering the So Whats, proving that key breakthroughs are possible, and resolving the dominant skepticisms you anticipate. If you've followed the tenets of scientific marketing (Ch5.2 Scientific Marketing), your experiments have been constructed backward from this moment, so that you know which results will best support your most compelling narrative.

It's hard to overstate the importance of this backwards-in thinking to your success. Start with the advertisement for your MPP that you want to have at the end of the day. Imagine the data that goes on the billboard *first*, and build your plans and experiments from there. Your job of selling will be immeasurably easier if you design your experiments appropriately and intentionally in the service of your argument.

Arguments specific to an MPP

You want to argue for the maximally valuable version of your MPP. Remember that an MPP has value in its platform and in the multiple products that come from it. These products include not just the products that exist today, but also the families of related future products that share common platform-derived value propositions. Your future products, by virtue of their shared feature-advantage-benefit triads, are more likely to work once the first product validates them.

One common pitfall we encounter is a heavy focus on the technological advances of the platform that fails to explain why anyone should care about it. As scientists, we instinctively gravitate to the technology. We get the Wow! factor in the data and want to see it front and center. But don't fall into the trap of reciting figure legends! The reason someone should care about your technology – the answer to So What, the benefits that the platform's features and advantages provide – cannot be overlooked, implied, or assumed. You must be explicit and emphatic about the benefits in your sales presentation. If you aren't, your audience will miss them.

A second pitfall we often see, especially with executives who come from careers in traditional drug development, is an overemphasis on the first products in development as if they were the full portfolio. The desire to latch onto something concrete leads these sellers to convey the snapshot of nascent products as if it represents the total future value of the MPP. But the full value of the opportunity includes the products you *do*

develop and those analogous products that you *could* develop. These analogous products share common features with the first products and therefore are highly plausible from where you stand today.

Your proof-of-concept data is much more valuable in an MPP-validating context than in an asset-centric context. You could sell a half-full cup of water as a future full cup of water, or you could sell it as proof that you've created reusable portable vessels for drinking (a New Modality MPP), as evidence of a process for rendering seawater potable (a New Technology MPP), or as a universal solvent for flavorings, electrolytes, and nutrients to improve human performance (a New Insight MPP). The early data validates the broader truths about your platform and offers a multi-product future. Keep the focus squarely there.

In an early MPP, the future matters. Your technologies may be great, but they're not yet fully (nor even partially) realized. When you sell, therefore, you have to emphasize the future potential and not just the concrete present. And you have to be able to do this even when the data is limited. We once interviewed a candidate who, when asked how she sold, said straightforwardly, "I present the data!" But what if you don't have all the data yet? Whether you have multiple products in development, early proof-of-concept data from the lab, or even just exploratory "what if" hunches (backed by observational anecdotes and reverse-engineered literature that shows that a particular destination would be valuable) – you never have all the data in hand. You can't just lay out the results and hope they tell the full story. Know what arguments to make, use what data you have to support them, and step up to tell the not-yet-built part of the story, because you're the only one who can build it.

Ch8.3 Crafting an effective narrative

This section is about crafting an effective narrative for selling your MPP, but we could just as easily call it Tapping into Emotions.

Think of any luxury good: a Ferrari, a Prada bag, a Patek Philippe watch. They're more expensive than their conventional counterparts not because they're made from more expensive components or with more expensive craftsmanship – though they often are – but because they're not priced based on their cost at all! They're priced based on value. Luxury goods appeal to our emotions. They compel us with legacy and seduce us with prestige. As the advertisements say, "You never actually own a Patek Philippe. You merely look after it for the next generation." People buy Ferraris for the mystique of the brand and the emotional thrum of the engine at ignition, not for their practicality.

If you want to induce people to desire your MPP, you have to shift them from a cost-based system to a value-based system. You have to move beyond their rational brain and tickle their limbic brain. The rational brain cares about the components of your MPP – the technology, the team, the IP, the data – but the emotional brain responds to

squishier concepts like novelty, breadth, scarcity, uniqueness, and the promise of transformation. To craft an effective narrative, logic is not enough. You must also leverage emotional suasion. Treat your MPP like the Ferrari it is.

Here are two frames that have helped us craft effective emotion-tapping narratives.

Generate Desire

One important emotional lever is desire. To compel someone to act, you must make them desperately want what you have. One way to generate desire for your MPP is to combine scarcity and inevitability.

What does the junction of scarcity and inevitability look like? Scarcity means saying, credibly and convincingly, that you're the only one able to bring forth a transformation and that this is the *only chance* the audience will get to participate. Your MPP is a rocket ship taking off with one seat left, and the countdown has begun. They have one chance to come aboard. Sellers of luxury goods do this all the time, driving demand through real or perceived scarcity. The fewer people own a good, the more prestigious it appears and the stronger the desire. FOMO is a natural human emotional response. Use it to your advantage to drive up demand and drive up value.

You control the scarcity of your MPP with the language you use to describe it. Don't let your MPP be sullied by comparisons to other companies or technologies that make you seem merely derivative. It's always possible to describe something, even something deeply technological, transformative, and brand-new, using language that compares it to what existed before. That's fine if you want to explain. It's a terrible idea if you want to sell. Develop your own language that exists on a separate plane. Emphasize your exceptional novelty and differentiation, the astounding benefits only you can provide, and the tremendous value only you can unlock.

You control the inevitability of your MPP via the way you describe the not-yet-existing future. The more details you give about the future that you will build, the more compelled your audience will be to act. It doesn't matter whether you're selling an investment, recruiting an executive, or doing anything in between. People want to feel that they're making a difference and taking part in a radical transformation, but they're also risk-averse. You can soothe their fears by making your future seem inevitable. Speak confidently and specifically about the world your technology enables, help them inhabit the opportunities that appear just beyond their horizon, and convince them that the data you've generated makes that future inevitable. Make the future achingly real. The nearer at hand it seems, the more compelling it will be.

Journey through the moment of discovery

A second important emotional lever is the journey. Humans have always made sense of the world through stories, and we attach strong conclusions to narrative arcs. When you sell, therefore, think of leading your audience on a journey. A particularly powerful journey is the one that takes your listeners through your moment of discovery. This is

the tale of realizing (with data) that your hunches were right, of seeing clarity emerge from confusion, and of proving that your invention works.

Every journey needs a starting point. The trope in marketing literature is to “meet the customer where they are and take them where you want to go.” Your MPP may require leaps of imagination, but you must begin the journey in a recognizable place for the audience. Use terms they know. Set the context with language they understand. Define problems they’re familiar with. If your MPP is New Technology for computational drug discovery, for example, and your audience is a room of Pharma drug hunters, you may start the journey with the challenges of traditional drug discovery. If instead your audience comes from the world of tech, you might start with the difficulty of encoding complex data in a machine-learnable format. There is no one right way to start; your choice will require knowing your audience.

But once your journey begins, like any good epic, it will need conflict and a climax before it resolves to a memorable lesson. In your case, the take-home lesson is the So What of your value propositions and the value your MPP unlocks. The conflict in your tale – the challenges and uncertainties you faced, the open questions you sought to answer, and the many reasons you were likely to fail – builds to the climax of a sudden realization. Create the suspense and then hit them with the data showing that, amazingly, you might be on to something.

Recapturing the magic of discovery can be difficult. Once you’ve seen your breakthrough, it becomes obvious, and it takes real effort to put yourself in the shoes of someone (the past you, for instance) for whom the discovery doesn’t yet exist. What facts did you assume? What premises would have seemed absurd? What results would have been most astounding? Furthermore, after immersing yourself in your platform for months, you can become inured to the real extent of its novelty. Force yourself back into a “before-times” mindset, because that’s where your audience is today.

Key phrases can be your friends as you craft the narrative of your journey. “It turns out...” is one of our favorites because it locks in the novelty of discovery and the surprise of your observation in a single climax. You pose a problem, and then resolve it with a simple phrase that inherently makes the solution seem counterintuitive – hard to stumble upon – and therefore more valuable. (It turns out that the phrase has a similar effect on relatively mundane topics too!) Find the phrases that work for you and anchor your narration to them.

Invite your audience to accompany you on the path to discovery. Even better, take them to the brink and let them propel themselves over the threshold. Aristotle alluded to this technique 2300 years ago in *On Rhetoric*, when he noted the power of enthymemes: “A is to B, B is to C, therefore A is to...” We instinctively complete the truncated argument. The conclusion may be evident, but when we are induced to reach it for ourselves, the message lands more forcefully.

Ch8.4 Perfecting your Presentation

Selling is much more than presenting data. But to sell well, you have to present well. To present well is to explain the data clearly, convey what you need to convey concisely, and project calm in a high-stakes situation. Here are some suggestions for presenting effectively when you're selling.

Attention spans are short so get to the point!

We've discussed the psychology of persuasion, but to present well, you also have to understand the psychology of attention. In a nutshell: we don't have much attention; attention is hard to capture; and it's nearly impossible to retain.

If you've ever been in front of a room of VCs addictively checking their Blackberries, iPhones, or smart watches (depending on your generation), you know that attention spans are short. But that's not the worst of your problems.

Ed O'Brien and his colleagues at the University of Chicago Booth School of Business have studied how we use information, especially new information, to make decisions.^{xxii} The evidence shows that people make up their minds right away. Any new information isn't dispassionately compared to a prior belief, but rather *reacted* to emotionally in real time as it arises. "We won't need to see later information if we already love or hate the very first piece," they write.

This is a daunting conclusion when you're selling. It means you have to get to the point right away. It also means that the first impressions you make matter the most! If the audience doesn't care, they won't do the work to understand what you're telling them. You need to make them care.

How do you hook the audience from the get-go? By leading with the So What. Hit the So What before you get to the What and the How. This is the inverse of most scientific talks or presentations, where you begin with a broad foundation of What, build slowly through the How, and end with a flourish at the great reveal. To sell effectively, you must invert that pyramid. No one cares about the What or the How until they know they care about the So What. So make them care.

Establish Credibility

To get the audience members to suspend their disbelief as you paint a picture of an unbuilt future, you must establish credibility. Some entrepreneurs can show their credibility through their experience – achievements that speak to their trustworthiness. But those of you who are more junior, or entering a new field for the first time, or speaking to audiences that are unfamiliar with your exploits should think explicitly about how to establish your credibility.

One path to credibility is through scientific authority. Scientific authority means speaking knowledgeably about the science. This means knowing not just your specific technology

or platform, but also the field you're transforming – and knowing it backward and forward, inside and out, historically and to the cutting edge. Beyond your pitch, be ready to draw analogies to past events in the field and to reference the common guideposts a knowledgeable audience will expect. Do your homework.

A second path to credibility is through the projection of competence, which you can dial up with the tone of your voice. In his negotiating treatise *Never Split the Difference*, based on years of front-line experience as an FBI hostage negotiator, Christopher Voss describes the “Late-night FM DJ voice,” a lowered-octave downward-inflected cadence that says “I’m in control. You’re going to listen to me and everything will be okay.” Strive for this tone as you speak confidently about your science, your ambitions, and the future you will build.

Be in tune with your audience

Presenting is a conversation. Your audience receives your words, and it also reacts to them. You need to be attuned to the non-verbal cues your audience is giving you and attend to them in real time.

Connect with your audience members, but don’t let them set the tone. When you’re presenting, you’re in control. There is deep literature on the power of mirroring emotions to drive empathy. But instead of mirroring your audience’s emotions, try modeling the emotions you want them to mirror. Infect your audience with your enthusiasm. Transfer to them your surprise and delight at a new discovery. Enact for them your excitement as you recount the aha! moment when the realization hits that your platform opens entirely new realms of product opportunities. Perfect your cadence. Control your delivery. Manage your breath. Pause, expectantly, before landing the knockout punch. You are a puppeteer, and the audience members are your puppets.

Finally, remember that audiences are diverse and dynamic. The story that works for one crowd won’t work for another. One audience’s transformative breakthrough is another’s ho-hum feature. Different people will care about different elements of your platform, respond to different components of your data, and foresee different insurmountable challenges to your technology.

Because your audiences are diverse and dynamic, you must be too. Have multiple storylines ready for different audiences. Keep an eye on your audience to see what resonates and judge whether they’re engaging. If you’re “on,” keep going. If you’re losing them, don’t be an automaton. If their eyes are glazing over at the data, you’re likely going too deep. In that case, pull back, jump to the So What, and watch them re-engage.

Ch8.5 How to get good at Selling

How do you get good at selling? As with any skill, you work at it. Tap into the three Ps – preparation, practice, and performance – and along the way, strive to overcome any self-imposed limitations that hold you back.

Prepare

Selling is an argument, and the argument has to be airtight. It can have no holes. Don't take it lightly or assume you can "wing it."

We often advise people to start by writing the argument out in prose. You will usually present in PowerPoint, but not always. Recall that selling doesn't just happen in the boardroom. You have to be prepared to sell in any conversation, even when slides are not handy. Furthermore, PowerPoint is a medium that prioritizes visual information. When you prepare slides, your attention is drawn to the slide layout, crowded text, color schemes... none of which contribute to building a strong argument, and all of which make it easy to gloss over holes in the argument.

Craft the argument in prose. Prose doesn't tolerate slipshod logic. Instead, it shines a devastating spotlight on inconsistency and meandering. Prose forces you to sharpen your rhetoric; it pushes you to identify the pillars of your argument, crystallizes the key phrases that anchor your narration, and makes explicit the implications of your data. (Perhaps this is why all internal investment decisions in VC firms are communicated as written memos.) Once you've prepared the argument in prose, it becomes trivially easy to convert it to PowerPoint. Regardless of what your slides say, you'll know what to say.

Practice

Practice your presentation. Rehearse your facts. Train your Late-Night FM DJ voice. Find a team to help you, advise you, and provide feedback. Find low-stakes engagements where you can practice selling. This might mean describing the company in an interview with an entry-level scientist, or talking about it over drinks with a group of entrepreneurial peers, or having a low-stakes fact-finding conversation with experts in your field.

As you practice, remember what you're seeing. Observe your audience as though you were an anthropologist. Notice what produces a spark in one person's eye, and try to create the same effect on the next person. Maybe it's how you told a part of the story, or your cadence when describing the moment of discovery. Maybe it's the open-mouthed surprise when you described an experimental result, or the excitement when you articulated a certain set of implications. Maybe it's the way the other person described the forever-unsolved challenges in their field that your platform solves. Identify the breakthrough moments when you connect with someone; remember the words and argument that got you there and incorporate them into your presentation.

Though these conversations, you'll begin to learn what resonates with audiences. Often the resonances cluster by audience phenotype – the molecular biologists love one element; the drug discoverers another; the marketing and sales folks another. Learn these resonances and file them away, ready to deploy on new audiences as you find them.

In familiar settings, force yourself to build a fourth wall between yourself and the audience. Use every chance you get to practice selling. With friends, don't fall into the easy trap of describing the sales pitch – “Now I'll say this... then I'll say that...” Instead, sell as though they're the real audience. Practice being “in the zone.”

Perform

Though your pitch may be practiced, it doesn't have to be rote. Selling is, after all, a performance. Think of any creative performative endeavor: not only is the on-stage moment preceded by years of practice, but the performance itself is given, in nearly identical form, night after night. And yet for every audience, the performance must feel fresh.

Rock bands rehearse their arena shows, including the between-song banter, and yet when the lights go down, every audience member feels that Bono is speaking directly to them. Comedians workshop their material for months, fine-tuning jokes to nail the cadence, the tone, the word choice, and the delivery. But when they stand up at the microphone, the punchline lands fresh every time. Politicians hit their stump speeches – phrases used over and over again at campaign stops, interview shows, and debates – with enthusiasm because there's always someone in the audience who is hearing the speech for the first time. Dancers and figure skaters know every leg, arm, and head movement of their performance, and yet when the music starts, they create inspired art in the moment.

Clive Swift, a British actor, once described his work in the theater as a sport: “On stage you must be ready to move like a tennis player on his toes. Your concentration must be keen, your reflexes sharp; your body and mind are in top gear, the chase is on. Acting is energy.” You, too, are a performer. Workshop your material to find what works best on your audiences. Know the key signature, the chords, and the count. And when the lights go on – and they could go on at any moment – be ready to perform.

Overcome the trap of “authenticity”

Many scientists we know struggle with selling. All this talk about context and future implications and what-ifs can feel deeply inauthentic. It's not part of our training, and it's certainly not how we are taught to communicate our results. It can make us feel like impostors, as though we're going against our “true” selves.

Herminia Ibarra, who has studied organizational behavior at Harvard, INSEAD, and London Business School, describes this phenomenon as the Authenticity Paradox.^{xxiii} When we do things that run counter to our natural inclinations, we tend to latch onto “authenticity” as an excuse for sticking with what is comfortable. But any change that

puts us in new situations – advancing in our careers, taking charge in unfamiliar circumstances, shouldering new responsibilities – requires that we move beyond our comfort zones. The skills that the “authentic” you developed in one arena need to be updated if you are to be successful in a new arena.

At the beginning, using these updated skills often feels superficial, unnatural, and calculating instead of genuine and spontaneous. Some of our suggestions about selling may have made you uncomfortable for the same reasons. But trying something new is not an insult to the “true” you. It’s an experiment in which you’re testing and figuring out your new self. It’s okay to be inconsistent, to prototype new styles. Try to experience the process of selling as a playful chance to break out of your habitual patterns of thought and action. With time and repetition, the new self will begin to feel more comfortable.

Find the style that works for you. Who knows – maybe you’ll discover a new “authentic” version of yourself who is a comfortable, effective seller of your MPP.

Key Takeaways

- You're always selling: informing, persuading, and compelling to action. You must be the champion of your MPP because no one else will be. It doesn't matter how much value you create if you can't convey it effectively.
- Selling is an argument with data, but it's not an argument *of* data. Start from your argument and work backwards to design your experiments and efforts accordingly.
- Value is deeply emotional. Generate desire and take the audience through your moments of discovery. Don't get locked into the concrete present but emphasize the future.
- Craft an effective narrative that leads with "So What" because attention spans are short.
- Selling is a performance. When the lights go down you must be ready. Prepare, practice, and overcome the trap of "authenticity".

Chapter 9 Building Value Through People

You can't create a successful company alone. Entrepreneurship is fundamentally a human enterprise done best by teams. It requires many minds thinking in concert and many hands pulling together. It requires people with diverse and specific expertise in science, technology, business, law, and leadership. It requires that these diverse and specific skills come together productively and efficiently in a culture that encourages individual excellence and collective achievement.

To succeed, you must know how to recognize and cultivate the value in people.

This is a particular challenge if you intend to innovate, because the process of innovation is full of contradictions. For example, you have to strive to realize a futuristic vision while adhering to the rigorous norms of science. You explore many divergent paths but somehow move forward. You thrive on creativity but also on discipline.

Importantly, and contrary to all human nature, to innovate well you must act at both poles of every contradiction simultaneously. There is no happy medium where you're moderately creative and mildly disciplined. There's no optimal average where you're less visionary and also somewhat lax with data. You have to sustain seemingly contradictory behaviors and emotions at the same time. It's like a heavy barbell that you can only lift if the weights are balanced. And it's hard.

To maximize the value in your people, you must help your team navigate the tension in these dichotomies. If you don't, the pull from these opposing forces may break the company. Imbalances can cause interpersonal conflicts. Unbalanced cultures will become toxic and unproductive. Unbalanced teams, no matter how highly qualified, won't function together. Succumbing to the tension can overwhelm you and leave you unmoored and unhappy.

In this chapter, we will explore how the dichotomies of innovation manifest in cultures, among individuals, and within you. We will offer tips for how to thrive within the dichotomies so that your company can survive and innovate. We'll address how to build a culture of innovation, how to construct an innovative team, and how to maximize your own value along the way.

Ch9.1 Building a Culture of Innovation

Your innovative MPP needs a culture that is high-powered, dynamic, and motivating. But you don't get a culture merely by painting slogans on a wall. Culture emerges from the aggregate behaviors of individuals and sets the guiding norms for future conduct. As leaders, we influence culture through our behaviors and attitudes, which others on the team emulate; our practice shapes what others perceive as proper.

To build innovative companies, we need to create cultures that withstand, and in fact embrace and thrive on, the dichotomies of innovation. Three of these dichotomies are especially important. Innovative cultures must be both collective and unilateral (in how decisions get made), free and structured (in what gets worked on), and supportive and accountable (in how we deal with failure).

Let's discuss each dichotomy in turn to explore how our actions as leaders can help create innovative cultures.

Collective and Unilateral

Innovation thrives in cultures that are both collective and unilateral.

A collective culture is one where the voices of all of the individuals on the team are welcomed and heard. Collective cultures are sometimes structurally flat, with few layers of hierarchy, but they are always behaviorally flat: the most junior RA is in the same room as the senior executive, and information is shared openly. Collective cultures promote innovation. People feel empowered when their contributions are listened to, and thus they take more independent initiative. People feel valued when they see their suggestions adopted and realize that their work has a real impact on the company. When information is shared openly, there is room for serendipity. When challenges are posed frankly, people are motivated to offer suggestions, and the diversity of their perspectives tends to uncover more and better solutions for the company.

At the same time, an innovative company must be able to act decisively, which means someone has to make decisions unilaterally. As the leader, you face dozens of choices in an ever-changing landscape against the backdrop of intense competition. There is no time for endless debate, no time to wait for certainty. When you have no roadmap, you must make decisions quickly, try something and see where you end up, and make the next decision from there.

The need to act unilaterally runs headlong into the desire for a collective culture. The tension manifests when you, the leader, make a decision that doesn't go someone's way – and that means essentially always, since every decision implies alternatives, closes doors, or de-prioritizes certain projects or activities. How, then, can you balance the innovative benefits of a listen-to-all-voices collective culture and the need to act decisively?

One part of the answer comes from you, the leader. You must model openness in your assertiveness – not just by requesting input from the team, but also after the fact, when you explain your actions. Open the window into your decision-making process – the choices you’re making and why you’re making them – and acknowledge the risks and uncertainties in the decisions. Your openness to the team will reinforce the benefits of the collective culture. Sharing with scientists how you understand and optimize for value will help them generate data that is optimized for value. Welcoming them into the rationale will strengthen the sense of shared mission and collective bonding.

A second part of the answer comes from the team. You should establish a code of “disagree and commit.” Jeff Bezos described this philosophy in his 2016 annual letter to Amazon shareholders: when you cannot reach consensus, which happens often when no one knows the answer for sure (and all the time when you’re innovating!), you can acknowledge the disagreement, but everyone must pledge to commit fully and vigorously to making the chosen path as successful as possible.^{xxiv}

Openness from you, the leader, about the rationale for your decisions combined with a pledge to “disagree and commit” when consensus can’t be reached lets you maintain an innovative collective culture that acts unilaterally and decisively when necessary.

Free and Structured

Innovation thrives in cultures where work is both free and structured.

A culture of freedom is one where people have the right, the encouragement, and even the mandate to pursue hunches. A free culture nurtures creativity and curiosity, which are essential in an MPP. You are trying to create a wholly new future with wholly new technology – the only path forward is a creative one. Your testable hypotheses from the exploration value landscape leave lots of wiggle room; there is no linear checklist to diligently follow. Instead, there will be some chaos, a whole lot of open-endedness, and the need for a culture that can harness entropy to produce innovation.

But in a start-up, freedom runs headlong into deadlines. You don’t have infinite resources. If the money runs out, the show is over. If you don’t create value by the time your runway runs out, you’ll be left begging for scraps. You need to have structure around your programs. You can’t allow unfettered academic experimentation. You don’t have the time!

This tension is not unique to a start-up – the “productivity” of R&D is a long-running concern in the management literature^{xxv} – but it is particularly acute in the early laboratory phase of an MPP. When you’re just starting out, the world of opportunities is big, and you must navigate a hazy outlook with many paths to (potential) future value. How do you encourage freedom, creativity, and exploration while also reaching your milestones on deadline?

One component of the solution is to communicate the company’s value objectives clearly to the team. Everyone in the company should know what the value propositions

of the platform are and what results create value for the company. (The cause is helped immensely if you've already oriented everything you do around a framework of value.) When you are clear about value, you can be clear about the purpose of every experiment – why it matters and how it fits into the larger picture. Is the goal to find new targets? To evaluate a bunch of ways of making the modality to see which is best? To set up a rapid prototyping platform? Sometimes the work is exploratory; other times it is not. If you choose to end a project or refocus your resources elsewhere, you can explain the decision as an assessment of where and when you're likely to create value.

Furthermore, when you frame the discussion of value, you can make it clear that while the destination is specific, the path to get there is ambiguous. The goals and timelines are fixed, but you leave room for people to be creative and flexible because you don't yet know how you'll get where you're going. Cultures are much more forgiving of tactical changes when the strategic goal is constant. Teams in early start-ups are resilient, but you can cushion the inevitable whiplash by keeping everyone focused on the long-term goal of value creation.

A second component of the balance between freedom and structure is knowing your team and matching individuals to the projects that best suit them. Know who is comfortable flitting from idea to idea, nimbly testing hypotheses, and who prefers stability. Know who works well on deadlines and who tenses up. Manage your team to their strengths. Be clear from the get-go, especially with scientists in their first industry job, that no project will be like an academic experience, and no project is inviolable or guaranteed to go on forever. Celebrate successes collectively and make it abundantly clear (in word and deed) that a definitive and quick closing of a low-value door is as valuable to the company as the validation of a high-value path.

Supportive and Accountable

Innovative cultures must be both supportive and accountable.

To be supportive in a start-up is practically fiat. Those of us who work in start-ups have chosen an uncertain and storm-battered path instead of a steady and comforting one. We've chosen to try to create something new rather than slot ourselves into an existing structure. We get through the challenges collectively, with many hands at our backs, forging bonds through the crucible of late nights and long hours. Supportive cultures celebrate this undertaking. We encourage one another to take risks and celebrate after failures, recognizing that if none of us sticks out our neck to try something new, we will never build anything new.

But if you want a culture of support, you must also create a culture of accountability. Celebrating failure only works if you simultaneously demand excellence. Gary Pisano, a scholar of innovation at Harvard, highlights this dichotomy in his piece "The Hard Truths of Innovative Cultures."^{xxvi} According to his findings, innovative cultures tolerate failure but are intolerant of incompetence. Failing because experiments are poorly planned or sloppily executed is different from failing because you industriously tried something bold and learned something important from the attempt. Celebrating the former lowers the

standards for everyone and erodes future innovation. Celebrating the latter – the learning from the failure and the willingness to try – enables innovation.

This dichotomy often manifests itself as interpersonal conflict between the “Jerk” and the “Nice Brigade.” The blunt truth-teller disturbs the warm, supportive ethos with pointed criticism – but what if we need the bluntness to succeed? How should innovative cultures communicate? How openly should we say what we think to one another? How do we maintain rigor and accountability without discouraging risk-taking?

Yet again, one component of the solution is for you, the leader, to model the behavior you seek. Set the precedent that we can disagree vigorously yet without contempt. Show that we trust each other to be grown-ups, to not take criticism personally, and to have the honest debates that we need to reach the best plans and therefore the best outcome. Model this behavior in your interactions with the team – giving honest and direct criticism of plans when needed, and actively celebrating the results even when the outcome of a well-designed and executed experiment is negative.

A second component of the solution is hiring the right people. Don’t fixate on “niceness” as an attribute or prioritize “getting along.” Small biases in hiring can quickly lead to a culture that silences direct dissent and open communication. An innovative culture needs honesty. Accountability and creativity can come across as abrasiveness, but a culture of directness need not be toxic. Without accountability, debate, and disagreement, you risk failing to hold people to the high standards of excellence that innovation requires.

Ch9.2 Constructing an Innovative Team

If culture is the collective norms that emerge from individual behaviors, then to build an innovative MPP, you must hire innovative individuals who thrive in the messiness of innovation. Roles in start-ups are rarely siloed; they often require a blend of skills that are usually attributed to single positions in a more established company. As a result, individuals in your company may inhabit multiple personas. Here we'll consider three common start-up personas – the executive, the scientist, and the entrepreneur – and the specific dichotomies they face. The executive must be a visionary optimist *and* a pragmatic pessimist; the scientist must be both specialized *and* polyvalent; and the entrepreneur must be simultaneously convinced *and* questioning.

The Executive: Visionary Optimist and Pragmatic Pessimist

When you take on the persona of the executive, you are responsible for the success of the company. To helm an MPP requires that you deal with many of the contradictions of innovation, perhaps the biggest of which is the tension between visionary optimism and pragmatic pessimism.

You must be a visionary optimist, able to depict and illustrate the unbuilt future the MPP is striving toward and to describe it with specificity, confidence, and optimism. Your team will look to you for reassurance, and your visionary optimism will make them believe this wild goal is achievable. It will motivate your team to do the experiment one more time, to stay late, and to put in the effort. Beyond your team, visionary optimism is essential to selling. Projecting and embodying the future world in which the MPP has succeeded is the only way to convince investors to invest in you and partners to join you.

But living in the future doesn't erase doubts. And you must be attuned to your doubts, because they reflect instincts you've honed over years. Doubts may point to real failings that must be addressed for the success to materialize. Listening to doubts – being a pragmatic pessimist – is the counterpoint to visionary optimism. You must be pessimistic about the things that could go wrong, which means you must get deep in the weeds of the science and technology of the MPP and of the value landscape on which the MPP rests. If something isn't working well, you must recognize it and get cracking on solutions. Maybe the answer is to pull the emergency cord and set everyone to surmounting the biggest hurdle. Maybe you should declare a path closed and pursue an orthogonal value hypothesis. Perhaps the answer is to carry on as usual, confident that the team will overcome the hurdles with time and a steady, optimistic, guiding hand.

From hour to hour, you will need to shift between visionary optimism and pragmatic pessimism. Sometimes you must even inhabit them at the same time in the same conversation – perhaps to direct a scientific plan while simultaneously keeping everyone motivated and upbeat. It is not easy, but it is essential.

The Scientist: Specialized and Polyvalent

For the scientist persona, especially in the earliest days of an MPP, the most important dichotomy is between specialization and polyvalence.

You must be specialized to get hired, unless you have a specific skill set that fits with what the company needs to do. The MPP must build a technology platform at and beyond the frontier of what's currently the state of the art, which requires people with deep and cutting-edge expertise. The expertise may be in a field of science, a technology, or an assay. It may be in drug development or in manufacturing. Having a specialized skill set is essential.

At the same time, you must be polyvalent. You must know about areas beyond your expertise. Moreover, you have to be fearless enough to jump into those new areas, away from your comfort zone, and be effective in them. An early start-up job isn't a position in an established company, with established programs and clearly delineated job descriptions. The uncertainty of an MPP, especially in the early days when we're still experimenting with the value landscape and prototyping ideas, begs for polyvalence. None of us know where things will go; only the results will tell us. You have to be willing to push the science in whatever direction value creation demands.

If you're considering your first job in an early MPP, you're likely already specialized. Now ask yourself whether you're polyvalent. Are you nimble in picking up new topics, assays, and tools? Are you fearless? Are you energized by trying new things and undeterred by failure because you're motivated by what you'll learn? Are you curious about business, intellectual property, manufacturing, and operations? There's no shame in answering no. But if your answer is no, it's best to steer clear of the start-up.

Polyvalence extends beyond science. An early start-up is a great opportunity to see and participate in the beyond-science elements of a company. Lab meeting is a time to share with highest-level management not just the results of your experiment but also the implications of the results – what they suggest the company should do next. If you're interested in making the leap from the lab bench to strategy, business development, or executive leadership, an early start-up job is a tremendous opportunity. You will learn, first-hand and under pressure, how to understand and communicate value. This will likely be a new skill, especially if you're fresh from your academic training. But when there are few people in the room, you get to be one of those people. You'll do things that, according to your résumé, you have no business doing. Some of our most exceptional scientists have advanced rapidly from individual contributors to very senior roles managing large teams in just a few years. The opportunities will arise if you are both specialized and polyvalent. Seize them.

The Entrepreneur: Convinced and Questioning

As the entrepreneur, you have your own dichotomies to deal with. You play many roles – you're an executive, a scientist, a salesperson, and so on. In addition to these specific roles, you're ultimately responsible for initiating the innovation, writing the first notions of

the future on the whiteboard, and figuring out how to make it real. In this process of origination, you must be both convinced and questioning.

Convinced here means having conviction. You must persevere against all odds, especially the skepticism and discouragement of people who think they know better than you how the world works. You must persevere because things won't always go well and certainly won't be smooth. It's easy to tip over the edge and be consumed by doubt and negativity. But you must be convinced of the future you envision, convinced that you can build it, and convinced when you act.

At the same time, you must be skeptical and agile, willing to question your dogmas and to revisit your assumptions. You must anticipate the unanticipated, but also accept that full foreknowledge is impossible and that you will have to deal with eventualities as they arise. Preparation helps – it's less terrifying to turn away from a path if you've already mapped the alternatives – but there is no guarantee that every path will look as you imagined.

I've found that this mind-bending duality feels more manageable when I think of solving a tough crossword puzzle. When you face an empty grid, if you think you might have an answer to a clue, you must have the conviction to write it in (in ink!). Only then will the rest of the puzzle materialize. And yet you must be ready at any moment to overwrite the original input as new information forces you to rethink your first answer.

Have convictions, but don't become enamored of them. Be convinced and keep questioning.

Ch9.3 Maximizing Value in Yourself

At the heart of all of this is you. The challenges of creating innovative organizations, the excitement of building new technologies, the thrill of making new medicines – all of it rests on your ability to innovate. In this work, you'll face two big dichotomies: between the need to do it all and to be effective, and between the need to be authentic and to evolve.

We've discussed how to maximize the value in every other aspect of your MPP. Now it's time to maximize the value in yourself.

Do it All and Be Effective

When you're running the show, it's clichéd to say that you wear many hats. But, boy, do you ever! Some days you're the idea generator: you read papers, have freewheeling conversations, and brainstorm on a whiteboard. Other days you're the chief scientist, reviewing data and planning experiments. In the morning you're the facilities operator; over lunch you're pitching the company to investors; by mid-afternoon you're the head of talent acquisition; and by evening you're filing IP. With apologies to the Lion King, there is more to do than can ever be done. All you can do – what you must do, because no one else will – is pick up a shovel and dig. But if you aren't careful, you'll end up overwhelmed and ineffective.

You need to know what's worth working on. You may be the best person for any job, but you can't let yourself become the best person for *every* job. You have to prioritize your time, a finite resource, and set aside less important tasks. You have to know what will create the most value and commit to working primarily on that. Be intentional with your time. Attention management is the secret to navigating a world of excess choices. Be thoughtful about what you choose to spend your time on.

This dichotomy often shows up in time management. You're falling behind, feeling swamped... How will you finish everything on your plate?

First, take charge of your to-do list. Array your tasks, both immediate and long-term, on a matrix of importance (*x-axis*) and urgency (*y-axis*), as popularized by Stephen Covey in *Seven Habits of Highly Effective People*. When we try to Do It All, we usually find ourselves in perpetual crisis mode, tackling urgent issue after urgent issue, regardless of importance. The important items that aren't currently urgent – the ones, arguably, that we should be spending the most time on – don't come up until they become urgent, at which point we can't do them justice. Organize your to-do list and make sure you make time for important goals before they become urgent.

But this prioritization will still likely leave you with an overload. So next, delegate and drop. Tasks that can be done by others? Delegate them. Tasks that are neither important enough nor urgent enough? Drop them. And don't take on new tasks blindly. As William Oncken, an organizational theorist, brilliantly described it, tasks are like monkeys on your back, weighing you down with responsibility and sapping your energy

and time.^{xxvii} Every interaction is a chance for someone to toss their monkey onto your back. Protect yourself! Don't accept new monkeys.

As you begin to wrest control of your calendar, make time for attentive, contemplative thinking. Prepare your days and plan for the work you need to do. Check in periodically with your assessment of the value of your planned work in case it has changed given new developments in your MPP. If you have a framework for what's worth doing, you're in control. If you've reflected first, you won't simply react when the next thing comes up. If you are intentional, you can immerse yourself more fully in what you do, knowing that you're making good use of your time. Find the modality of reflection that works for you – journaling, meditation, running, Outlook calendar gazing – and make the time to put your world in order.

Be intentional. You will still end up working on all sorts of things, but you'll be doing so effectively and without drowning in excess.

Be Authentic and Evolve

The company you build will reflect your values. Therefore, it's important to know yourself. At the highest level, nurture your guiding principles: do things the right way, for the right reasons, and treat others with respect. More immediately, learn what makes you run smoothly. Identify the moments that put you in a state of “flow” and figure out why they engage you so fully. Learn which activities energize you and which exhaust you. Find the times of day and the settings when you are most effective. Incorporate these insights about your maximal productivity as you plan your workdays and engineer your work environment.^{xxviii}

At the same time, as we discussed in Ch8.5 How to get good at Selling, growth requires evolution. As you build your MPP, no matter how many times you've done it before, you will face situations in which, to be successful, you'll need new skills that may feel unnatural or inauthentic at first. (If you don't, you're likely not pushing yourself to the fullest – as Ellen Johnson Sirleaf said, “if your dreams don't scare you, they aren't big enough”!) The solution is to untether yourself from the anchor of authenticity and force yourself to behave “inauthentically,” to try on new selves, to experiment in the new role until it stops feeling unnatural and you begin to fit your new self into the new role.

Who are you? What is your “true” nature? These are weighty questions to wrestle with. One way to tiptoe into the balance of authenticity and evolution is to play with the story you tell about yourself. When you meet someone new, how do you introduce yourself? It's so simple and such a common occurrence – every call, every meeting begins with a round of introductions – and such a good testing ground for your new self.

You probably have a stock introduction. Perhaps it's chronological, starting with your college degree or graduate school research. You can probably recite it from memory. It may feel pretty stale. So try on a new story. This time, make it thematic. Pick highlights that hang together as a theme. List not just the “whats” of your past, but also the “so whats.” What are the implications of your achievements? What motivates you, and why

are you doing what you are doing? What impression do you want to make; which is the most relevant *you* for the conversation? Consider the different elements of your experiences – how might you describe them in new ways? Out of infinite narratives, which one do you want to tell?

Work backward from the value *in yourself* that you want to convey through your story. Try on new stories until you find the one that basically fits, that works for this purpose, that has a nice ring to it, and that you can grow into. Keep telling it and tweaking it, and it will begin to feel natural. Evolve your way to a new authenticity.

If you can do this for your story in a low-stakes environment, you can do it for other things when the stakes are higher. You can sell your company story with conviction to investors. You can describe your value creation plan to your early scientists in the prototyping days. You can craft a compelling MPP vision from the mess of an exploration. And you'll feel good doing it, too.

The dichotomies of innovation are not easy to inhabit, but you must do so if you want to succeed while creating something new. You must know yourself and how to manage the tensions of living at two poles simultaneously. You must select people who are capable of handling the tension and support them as they navigate the inevitable bumps. You must build a culture that makes the most out of everyone's effort. If you do all this, you will maximize the value your team brings to your MPP.

Key Takeaways

- You can't create a successful company alone. Entrepreneurship is fundamentally a human enterprise done best by teams.
- Innovation is full of contradictions, but to innovate well you must accept and embody both poles of the contradictions simultaneously. There is no "happy medium."
- To innovate you must build a culture that thrives in these dichotomies, help your team through these dichotomies, and learn how to navigate them yourself.

References

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- ⁱⁱ Analysis from Bruce Booth, a.k.a. LifeSciVC at <https://lifescivc.com/2020/04/booming-vc-backed-biopharma-strong-market-despite-pandemic/>
- ⁱⁱⁱ Advanced Materials 2004, DOI: 10.1002/adma.200400767
- ^{iv} A Novel Amino Lipid Series for mRNA Delivery: Improved Endosomal Escape and Sustained Pharmacology and Safety in Non-human Primates. Sabnis et al., *Mol Ther*. 2018 Jun 6; 26(6): 1509–1519. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5986714/>
- ^v Big hat-tip to Seven Holtzman, biotech executive and veteran of many early stage companies, who discusses this and other nuances of fundraising in his blog post “Early-Stage Biotech Value Creation: The Roles of Equity and Partnerships,” August 15, 2018. http://leadershipandbiotechnology.blogspot.com/2018/08/early-stage-biotech-value-creation_15.html
- ^{vi} *Between the World and Me*, Ta-Nehisi Coates (2015), p. 34. Ta-Nehisi is describing his question-heavy investigative writing process, not the creation of new biotech innovations, but the approach to forging a new understanding from a swirl of emerging facts and opinions is highly analogous.
- ^{vii} <https://cdixon.org/2013/08/04/the-idea-maze>
- ^{viii} The studies appear every five years or so, consistently finding in favor of older entrepreneurs. Take, for example, the Global Entrepreneurship Monitor (2010) and the Kellogg School of Management (2018), covered in the lay press in Slate (<https://slate.com/business/2010/12/why-old-people-make-better-entrepreneurs-than-young-ones.html>) and Inc (<https://www.inc.com/jeff-haden/a-study-of-27-million-startups-found-ideal-age-to-start-a-business-and-its-much-older-than-you-think.html>).
- ^{ix} Jenny Odell interviewed on the Ezra Klein Show, May 23, 2019: <https://www.vox.com/ezra-klein-show-podcast/2019/5/23/18636332/jenny-odell-how-to-do-nothing>
- ^x A fascinating history of Chinese Hamster Ovary (CHO) cells from LSF Magazine: <http://biomanufacturing.org/uploads/files/547998065159985597-cho-history.pdf>
- ^{xi} The Nimbus journey is detailed in Bruce Booth’s LifeSciVC blog: <https://lifescivc.com/2019/05/a-decade-of-discovery/>
- ^{xii} Tom Wolfe. *The Right Stuff*. New York: Farrar, Strauss, and Giroux, 1979. The behind-the-scenes look at the response to Sputnik is on p. 57.
- ^{xiii} Steve Holtzman’s discussion of biotech start-ups is here: [Early-Stage Biotech Value Creation: The Roles of Equity and Partnerships](http://leadershipandbiotechnology.blogspot.com/2018/08/early-stage-biotech-value-creation_15.html)
- ^{xiv} The fascinating tale reminds is a great pick-me-up amidst the chaos of a start-up... the early days of Genentech could describe any start-up today! Sally Smith Hughes. *Genentech: The Beginning of Biotech*. Chicago: University of Chicago Press, 2011.
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- ^{xvi} Myriad: Assoc. for Molecular Pathology v. Myriad Genetics, Inc. (2013) https://www.supremecourt.gov/opinions/12pdf/12-398_1b7d.pdf

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<https://www.supremecourt.gov/opinions/11pdf/10-1150.pdf>

^{xviii} Regeneron v Kymab (news articles): [Kymab holds off Regeneron in patent fight on home turf | FierceBiotech](#) and [Kymab wins UK court case against Regeneron's patent claims/](#)

^{xix} The Cabilly patent estate includes the three patents: US4816567, US6331415, and US7923221 (the claim in the text is from “Cabilly II). A nice summary of the legal history of enforcement and licensing of these patents can be found in Storz, U. “The Cabilly Patents: Status quo and relevance for antibody companies” MAbs 2012 Mar-Apr; 4(2): 274–280.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3361663/>

^{xx} For more on Safe Harbor: <http://www.tiplj.org/wp-content/uploads/Volumes/v12/v12p23.pdf>

^{xxi} This particular CAR-T patent is *US7,466,190, by Sadelain, Brentjens, and Maher at MSKCC*. Ashley Jean Yeager published a nice review of the litigation around CAR-T technology in GenEngNews in 2017: <https://www.genengnews.com/insights/car-t-in-the-courts/>

^{xxii} Ed O'Brien, Harvard Business Review, March 7, 2019 (<https://hbr.org/2019/03/we-use-less-information-to-make-decisions-than-we-think>); also O'Brien & Klein, Proc. Nat. Acad. Sci. 2018 Dec 26;115(52):13222-13227. <https://pubmed.ncbi.nlm.nih.gov/30530692/>

^{xxiii} Herminia Ibarra, Harvard Business Review, Jan-Feb 2015. <https://hbr.org/2015/01/the-authenticity-paradox>

^{xxiv} Jeff Bezos' 2016 letter can be found here: <https://blog.aboutamazon.com/company-news/2016-letter-to-shareholders>

^{xxv} The science historian Steven Shapin covers this long-running debate, dating back to the earliest industrialization of science in the 1920s at General Electric, Westinghouse, Estman Kodak, AT&T Bell Labs, and others, in his fascinating book *The Scientific Life: A Moral History of a Late Modern Vocation*. Chicago: University of Chicago Press, 2008, pp. 132-145.

^{xxvi} <https://hbr.org/2019/01/the-hard-truth-about-innovative-cultures>

^{xxvii} For more on managerial monkeys and tips for deflecting them, see [Management Time: Who's Got the Monkey?](#)

^{xxviii} Bill Burnett and Dave Evans' book, *Designing Your Life* (New York: Alfred A Knopf, 2016), has a wonderful set of exercises for knowing oneself. The Harvard Business Review's *Guide to Being More Productive* (Cambridge: Harvard Business Review Press, 2017) is full of interesting variants on finding your optimal place, time, and environment for work.